



CHAPTER -1

NUTRITION AND DIET

- National Institute of Nutrition (NIN) is located at Hyderabad
- Nutrition is process by which food is taken in and used by body.
- Nutrition may also be defined as the science concerned with the study of food nutrients relation to health conditions.

Composition of Nutritional Food:

- Food can be classified into:
 - 1) Energy giving Food (Carbohydrates & fats)
 - 2) Body Building Food (Proteins)
 - 3) Protective Food (Minerals & vitamins)
- **Macro-Nutrients:** They are also called as proximate principles of food.

Ex:

- 1) Carbohydrates
- 2) Proteins
- 3) Fats.

These are energy Producers.

- **Micronutrients:** They are also called as protective principles of food.

Ex:

- 1) Vitamins
- 2) Minerals.

- Vitamins and minerals are mainly metabolic regulators.
- Minerals and vitamins are in the form of small molecules and many of them need no digestion.

(1) CARBOHYDRATES

- Carbohydrates are compounds of carbon, hydrogen and Oxygen in the ratio of 1:2:1 respectively.
- It is starchy and sugar element of diet, fuel of human body.
- One gram of carbohydrates provides 4.2k calories of Energy.
- It is manufactured by green plants through the process of photosynthesis.
- In mammals excess carbohydrates are stored in the form of Glycogen in Liver Part.
- Excess glucose-is not stored as glycogen and is converted into fat and stored in fat cells (adipose tissues).

Sources of Carbohydrates

Wheat (More carbohydrates grain) Rice
Potato (More carbohydrate vegetable) Banana
Maize
Bajra
Soya Beans

- Chemically Carbohydrates are of three main categories:

- (1) Monosaccharides
- (2) Disaccharides
- (3) Polysaccharides

(1) Monosaccharides

- It is the simplest form and human body can derive energy only in the form of Monosaccharide.
- These are easily soluble in water.

Ex:

- Glucose (Dextrose or grape sugar)- It is basic unit in simple sugar. It is simplest carbohydrate useful as an instant source of energy. It is also called Reduced Sugar. All athletics must take this type of carbohydrates before entering into the play ground.
- Galactose (soluble milk sugar).
- Fructose (fruit sugar) - It is sweeter among natural sugars. It is also called Honey sugar.

(2) Disaccharides:

It is formed by hydrolysis of two monosaccharides molecules. Commonly there are three Disaccharides

- **Sucrose (cane sugar).** It is also called Table sugar.
- At polar regions sugar is produced by
- Beetroot and Potato
- **Lactose (milk sugar).** It is found in Milk. Lactose found more in human milk than of animal milks.
- **Maltose (malt sugar).** It forms during germination of starchy seeds. It is used in making of alcohol

(3) Polysaccharides:

- It is complex form and built in the form of chains of monosaccharide.
- It is difficult to digest the polysaccharides easily. Human can digest all forms of carbohydrates except cellulose.

**Ex:**

starch
glycogen
cellulose

- Starch is common in plants. Found more in Rice, Wheat, Potato, Banana etc.
- Glycogen is common in animals. It is also called Animal starch.
- Cellulose is found more in Cotton fibers natural material. It is the largest natural polymer in the world.

FATS

- It consists of carbon, Hydrogen, Oxygen elements.
- Fat is a concentrated source of energy and supplies per unit weight more than double the energy, furnished by either proteins or carbohydrates.
- One gram of fat provides 9.3k calories of energy.
- Fatty acids are of two types viz. saturated fatty acids and unsaturated fatty acids.

(1) Saturated Fats:

- These are produced within the body
- No need to take from outside.
- These are harmful to humans
- They increase cholesterol content in the body.
- These are maximum in the form of oils

Ex:

- Palmitic acid
- Stearic acid.
- Acetic Acid (Saturated fat in grapes juice)
- Capric Acid (saturated fat in castor)

(2) Unsaturated Fats: Are those which have one or more double bonds between the carbons of the chain. Unsaturated fatty acids are found in fish oil and vegetable oil.

- These are not produced within the body.
- These are responsible for free passing of blood circulation so that these are good cholesterol.
- These are available in animal based products

Ex: Milk, Fish, Eggs, Meat

- Salmon, Mackerel, Tuna fishes consists of "Omega-3- Fatty-Acids".
- Fats are mixture of lipids (triglycerides).

Lipid mixture (fat) is solid at 20° C.

- Fats are stored as future food.
- Fats are esters of fatty acids with- glycerols.

The esters are called monodi or triglycerides.

- Human body is able to synthesise most of the fats in enough quantity except three-polyunsaturated fats. They are- Linoleic, Linolenic and Arachidonic acids. These essential fatty acids must be provided in the diet from outside.
- Fats are found in adipose tissues. The fat cell of adipose tissue can store upto 95% of their volume of triglycerides (fats), and hence adipose tissues are often called as the Fat Depot of the body.
- On an average an adult should get 20-30% of energy from fats.

Sources

Butter and Ghee,

Coconut oil,

Hydrogenated vegetable oil,

Vanaspati oil, Meat contain high proportion of saturated fatty acids and greatly increase cholesterol level.

- Cholesterol is steroid which reinforces the structure of the cell membrane in animal cells.

Cholesterol is known to cause arteriosclerosis (thickening of wall of arteries), high blood pressure and heart diseases.

- Fats that are generally liquid at room temperature are called oils.

Fats are stored beneath the skin at adipose cells.

- Fats are used to convert sun rays into Vitamin D.

Fats regulate normal body temperature.

PROTEINS

Proteins contain carbon, hydrogen, nitrogen and oxygen and less commonly sulphur, phosphorus, iodine and iron.

One gram of protein provides 4.2k Calories of energy

Proteins are the basis of many animal body structures

(Ex: muscles, skin, hair)

It is the primary food - stuff of the animal, the chief substances of the cells of the body. Proteins provide base for manufacturing enzymes, hormones, antibodies, etc.

Sources

Spirulina Algae- 60-70% (Highest source of proteins)

Soya Beans - 45-50%

Meat - 50%

Mushrooms - 21-35%

Fish - 15-25%

Ground Nut - 35-40%

Eggs - 40-50%

Milk - 15-20%

Pulses - 40-50% Proteins regulate metabolism. They make 12- 15% of diet calories.





Animal proteins are mostly first class proteins and plant proteins are second class proteins.

Casein is a protein which forms the part of food for young animal. Best source of casein is milk.

Protein is essential for growth and replacement of cells. Thus, proteins are essential for body growth and functioning.

Below 14 years children growth proteins are essential.

Haemoglobin is a Protein

Proteins are made of 24 amino-acids which are the only source of nitrogen in the body. Amino-acids are called Building Blocks of Protein.

- Excess of amino acids cannot be stored in the body.
- Nitrogen is essential for replacement of body cells and for growth.
- Amino Acids represent the basic structure of proteins.
- Their deficiency of protein into human body may cause Kwashiorkor which means red haired and Marasmus in infants.

Use of proteins

- (1) Cell Growth & development
- (2) Enzymes, Hormones are made
- (3) Antibodies are made
- (4) Haemoglobin is made

MINERALS

- Vitamins and minerals are called micronutrients while protein, carbohydrates and fats are called macronutrients.
- Minerals are inorganic substances. At least 29 elements are found in our body.
- Minerals have no energy value but imbalance of mineral level in the body may cause diseases.
- Minerals maintain the body metabolism and preserve the physical shape of the body cell.

Major Minerals

- (1) Calcium
- (2) Chlorine
- (3) Phosphorus
- (4) Magnesium
- (5) Potassium
- (6) Sodium
- (7) Sulphur
- (8) Trace Minerals

Minor Minerals

- (1) Chromium
- (2) Chlorine
- (3) Copper

- (4) Magnesium
- (5) Iodine
- (6) Sodium
- (7) Manganese
- (8) Molybdenum
- (9) Zinc

Calcium

- Calcium (Ca^{++}) helps in blood coagulation or blood clotting.
- It helps for Neuro muscular function, cardiac function and actions of many enzymes and hormones.
- Calcium plays an important role in sustaining intestinal peristalsis and growth of body tissues.
- It helps for formation of Vitamin D.
- Calcium is needed for normal muscle and nerve function.
- Calcium and phosphorus are deposited in bones and teeth to give them strength and rigidity.
- Calcium is the raw material for bone and teeth formation.
- Calcium deficiency causes -Rickets and Osteomalacia (Bones become weak).

Sources

- (1) Milk
- (2) Eggs
- (3) Green leafy vegetables
- (4) Cauliflower
- (5) Cabbage

Phosphorus

- Phosphorus helps in maintaining constant composition of the body fluids.
- It is essential for the formation of bone and teeth.
- It also maintains blood pH
- Phosphorus enters into many compounds such as nucleic acids and phospholipids, many coenzymes and high energy compounds like ATP.

Sources

- (1) Milk
- (2) Eggs
- (3) Vegetables

Magnesium

- Magnesium is required as a catalyst for many intracellular enzymatic reactions.
- Low magnesium concentration causes increased irritability of nervous system and peripheral vasodilation (widening of blood vessels)

Sources

- (1) Green leafy vegetables
- (2) Meat



**Chlorine (Cl)**

- It is the principal mineral in extra cellular fluid.
- It is essential for maintaining Osmotic pressure of body fluids.

Sources

- (1) Common salt
- (2) Milk

Potassium (K1)

- It is the principal cation inside the cell (inside cytoplasm of cell).
- Potassium deficiency causes – Hypokalemia (Vomiting and acute diarrhea, rise in heart-beat rate, kidney damage, weakness and paralysis of muscles).

Sources

- (1) Meat
- (2) Vegetables

Sodium (Na1)

- It is the principal mineral cation in the extra cellular fluid (fluid outside the cell).
- Sodium and potassium are essential for propagation of nerve impulse and for water balance in the body.
- Sodium deficiency causes – Hyponatremia (Dehydration, low blood pressure, loss of body weight).

Sources

- (1) Common salt
- (2) Milk

Iron

- Iron is required for haemoglobin and cytochrome synthesis.
- Iron gives red colour to RBC.
- It helps for formation of Vitamin B6.
- Deficiency of hemoglobin in RBCs. Persons look pale loses appetite and fatigue easily. This is called Anemia.
- Iron is essential for both transportation of oxygen to tissues and for operation of oxidative systems within the tissue cells.
- Iron is the raw material for RBC while Iodine for thyroid glands functioning.

Sources

- (1) Meat
- (2) Eggs
- (3) Green leafy vegetables
- (4) Jaggery
- (5) Dry Fruits

Fluorine

- Fluorine maintains normal dental enamel and prevents dental cavities.
- If fluorine is less than 1 mg/L in water, it causes dental cavities.
- If fluorine is greater than 1.5 mg/L in water it causes fluorosis which is represented by mottled teeth and enlarged bones.

Sources

- (1) Drinking water
- (2) Milk

Cobalt

- Cobalt helps in Erythropoiesis and III activities of some enzymes.

Zinc

- Zinc is important for digestion of proteins in the alimentary canal.
- Zinc helps in healthy functioning of tongue.
- Zinc helps in CO₂ transport.
- Zinc is a component of at least 70 enzymes like carbonic anhydrase and some peptidases.

Copper

- Copper helps in the utilisation of iron.
- Copper deficiency may produce anaemia.

Sulphur

- It is the component of hormones (Ex: insulin).
- It is necessary for normal metabolism.

Iodine

- Iodine is essential for the formation of thyroxine hormone which is secreted by thyroid gland.
- Deficiency of iodine causes Simple Goitre (Enlargement of thyroid due to low iodine content in drinking water).

Sources

- (1) Sea foods
- (2) Common salt

Mineral	Requirement
1. Chlorine	3500 mg
2. Sodium	3500 mg
3. Calcium	1200 mg
4. Phosphorus	1200 mg
5. Potassium	1000 mg
6. Magnesium	400 mg
7. Iron	18 mg
8. Iodine	0.15 mg.





CHAPTER -2

VITAMINS

- Vitamins are discovered by Hopkins.
- The name of vitamins put forth by Funk in 1912

- Vitamins are complex organic substances and are essential for growth, metabolism and full utilization of food.
- Vitamins are two types:
 - Fat soluble Vitamins - A, D, E, K
 - Water soluble vitamins - B, C
- Vitamins are not produced in the body except Vitamin D and K.
- They are organic components, taken in required quantities.
- They are micronutrients, Biological and Metabolic regulators.
- Many vitamins are portions of coenzymes.
- Only plants and symbiotic bacteria, present in intestine of most mammals manufacture vitamins.
- Liver generally stores fat soluble vitamins.
- Insufficient vitamins in diet lead to deficiency diseases.
- Vitamin B17 is a newly discovered vitamin with anti-cancer properties.
- Vitamins do not produce energy but essential for humans till death.

Vitamin - A (Retinal)

- Known as anti - infective vitamin, strengthen the integrity of skin, mucous membranes, epithelial cells.
- It is synthesized in liver from yellow and Red carotenoid pigments.
- Vitamin-A is necessary for the normal growth of a child.
- Essential for the formation of visual Rods and Cones of the retina of the eyes.
- Excess vitamin is stored in liver.

Sources

Carrot (more Vitamin)
Fish(Cod)
Milk
Egg yolks
Green Vegetables
Mango
Papaya
Banana

- Carrot appears orange colour due to presents of β - Kerotene
- Deficiency Disease:
 - (1) Night blindness (Nyctalopia),
 - (2) Xerophthalmia(dry eye/Lachrimal glands stop produce tears),
 - (3) Abnormal changes in various epithelial membranes,
 - (4) Retarded growth,
 - (5) Dermatitis (dry skin).

B-Complex Vitamins (B_1 , B_2 , B_3 , B_6 , B_7 , B_9 , B_{12})

All B vitamins help the body convert food (carbohydrates) into fuel (glucose), which is used to produce energy. These B vitamins, often referred to as B-Complex vitamins, also help the body use fats and protein. B-Complex vitamins are needed for healthy skin, hair, eyes, and liver. They also help the nervous system function properly. All the B vitamins are water-soluble, meaning that the body does not store them.

Vitamin - B_1 (Thiamine)

It is the first vitamin to be discovered by Eijkman.

Essential for healthy nerves and mucous membranes and for metabolic process.

It helps to form Cocarboxylase enzyme which involved in Citric Acid cycle.

Essential daily intake is around 1.5 mg. It is easily lost into heat.

Source:

Un polished grains,
Especially the outer seed coats
Meats
Yeast
Green vegetable
Liver etc.

- **Deficiency Causes:** Effect on growth of children like Beri-Beri (affect on peripheral nervous system)

Vitamin - B_2 (Riboflavin or Ariboflavin)

- It is essential for normal growth and healthy Comea.
- It is called yellow vitamin.
- It is called Anti Glositis Vitamin.
- Cow milk is yellow because of B_2 Vitamin

**Sources:**

Milk products

Egg

Liver

Cheese

Yeast

Green vegetables

Deficiency Disease:

- (1) Cataract
- (2) Defective skin patches around mouth and tongue
- (3) Sore mouth, skin turns into Red and rough
- (4) Slow metabolism and retarded growth.

Vitamin - B₃ (Nicotinic Acid / Niacin)

- For those who takes Maize as staple food effected with Pellagra.
- Essential for healthy digestive tracks.
- It is also called AntiPellagra Vitamin

Sources

Milk

Ground nut

Sweet potato

Yeast

Eggs

Deficiency Disease:

- (1) Pellagra
- (2) Dementia
- (3) Insomnia

Vitamin -B₅

Pantothenic acid is a vitamin, also known as vitamin B₅. It is widely found in both plants and animals

Sources:

Meat

Vegetables

Cereal grains

Legumes

Eggs

Milk

- Vitamin B₅ is commercially available as D-pantothenic acid, as well as dexpantenol and calcium pantothenate, which are chemicals made in the lab from D-pantothenic acid.

Pantothenic acid is frequently used in combination with other B vitamins in vitamin B complex formulations.

Vitamin - B₆ (Pyridoxine)

It is called Anti-anemia.

Essential for protein metabolism and blood formation

Deficiency is rare because it is found in most of the food.

In all delivery women it is a common deficiency Vitamin.

It is useful transmission process of synthesizing of Amino Acids and useful at the time of Radio Therapy.

Sources:

Yeast

Eggs

Cheese

Nuts

Cereals

Dry fruits

Deficiency Disease:

- (1) Mild anemia
- (2) Atrophied lymph tissue
- (3) Insufficient leucocytes and antibodies
- (4) Low resistance to infection.

Vitamin B₇

Chemical name Biotin.

Vitamin B₇ or Biotin is a natural nutritional supplement.

Sources

soy milk, egg yolks, liver, cheese, butter, poultry, cauliflower, bananas, watermelon, peas, brewer's yeast, nuts, beans, oat bran, salmon, and mackerel.

B₇ helps restore hair loss and keeps skin and nails looking healthy.

- In this way, Vitamin B₇ acts as an integral component in the skin care regime.
- It is used to treat the condition of hair loss or alopecia.
- This, in fact, is one of the most important and well-known commercial uses of Vitamin B₇.
- Vitamin B₇ helps naturally revitalize skin discoloration caused by pollution, bad health, and various other reasons.
- Vitamin B₇ helps in the growth and replication of cells.
- Vitamin B₇ provides numerous benefits including maintaining good metabolic activity, tissue maintenance, weight loss, heart problems, synthesis of vital components, and treating any kind of variations associated with blood sugar levels in the body.

Vitamin B₉

- Vitamin B₉, also called folate or folic acid, is one of 8 B- Vitamins.



- Folic acid is the synthetic form of B₉, found in supplements and fortified foods, while folate occurs naturally in foods.
- Folic acid is crucial for proper brain function and plays an important role in mental and emotional health.
- It aids in the production of DNA and RNA, the body's genetic material, and is especially important when cells and tissues are growing rapidly, such as in infancy, adolescence, and pregnancy. Folic acid also works closely with vitamin B₁₂ to help make red blood cells and help iron work properly in the body.
- Vitamin B₉ works with vitamins B₆ and B₁₂ and other nutrients to control blood levels of the amino acid homocysteine. High levels of homocysteine are associated with heart disease. However, researchers aren't sure whether homocysteine is a cause of heart disease or just a marker that indicates someone may have heart disease.
- It's fairly common to have low levels of folic acid. Alcoholism, inflammatory bowel disease, and celiac disease can cause folic acid deficiency. Also, certain medications may lower levels of folic acid in the body. Folic acid deficiency can cause poor growth, tongue inflammation, gingivitis, loss of appetite, shortness of breath, diarrhea, irritability, forgetfulness, and mental sluggishness.
- Pregnant women need more folic acid to lower the risk of neural tube birth defects, including cleft palate, spina bifida, and brain damage. Neural tube defects are birth defects caused by abnormal development of the neural tube, a structure that eventually gives rise to the brain and spinal cord. Since folic acid has been added to many grain foods in the U.S., such as bread and cereal, neural tube defects have decreased dramatically.

Sources:

Liver
Beans sprouts
Yeast
Dried herbs
Sunflower seeds
Dry roasted
Soya-beans
Spinach
Turnip

Vitamin B₁₂

(Cobalamin or Cyanocobalamin)

- Essential for the development of Red blood cells (RBC)
- It is cobalt containing vitamin.

Source:

Liver
Meat
Fish

- Only Vitamin not found in green vegetables.

Deficiency disease:

- (1) Pernicious anemia (acute form of anemia)
- (2) Failure of RBC maturation.

Vitamin-C (Ascorbic Acid)

- Most abundant vitamin found in nature and earliest known vitamin.
It is quickly susceptible to atmosphere.
- It is not found in animal based products (Egg, milk, Meat).
- Plants are rich in it but is easily lost due to heat and storage and excreted in Urine.
- Frequent Alcohols and smokers suffers the Vitamin C deficiency.

"Vitamin C" uses:

- (1) Formation of Collagen Proteins
- (2) To add broken bones
- (3) To heal Wounded parts
- (4) Anti-viral agent
- (5) Anti- Cancer agent
- (6) It acts as Anti-Oxidant

Sources:

Lemons
Oranges
Goose - Berry (More vitamin C)
Cabbage
Leafy vegetables

Deficiency Disease:

- (1) Scurvy disease (Spongy & Bleeding gums)
- (2) Retarded growth

Vitamin- D (Calciferol)

- It is a natural vitamin.
- Also known as Anti-Rickets or Sunshine vitamin.
- Essential for absorption of Calcium and Phosphates in the body so it regulates the bone and teeth formation.
- It is produced in skin when exposed to sunlight particularly if, exposed to Ultraviolet- B radiation. Hence, it synthesized in skin.
- Vitamin D is in 2 forms
 - (1) Ergocalciferol (D₂)
 - (2) Cholecalciferol (D₃)
- It is more needed by the children and nursing mother. It promoted Bone formation and strong skeleton. It regularize Calcium and Phosphorus levels in the blood and promoted.



- It can also be produced in the human body. It restrains the secretion of Parathyroid hormone from Parathyroid gland and also works on immune system in order to promote Anti-tumor activity.

Sources:

Sun rays
Animal oils
Butter
Milk
Fish
Cereal grains

Deficiency disease:

- (1) Liver & kidney disorders,
- (2) Bone diseases
- (3) Osteomalacia(in adults).
- (4) Rickets(in Children)

Vitamin - E (Tocopherol or Tocotrienol)

- Also called as Anti - sterility vitamin.
- Also called as beauty vitamin because it removes skin wrinkles and scares.
- It prevents oxidation of vitamin A and prevents the damage of membrane lipids as it inhibits peroxide formation.

Sources:

Fresh Fruits
Saffola nuts
Dry fruits
Unpolished grain
Animal and
Vegetable oil
Wheat

Deficiency disease:

- (1) Sterility or infertility (impotence),
- (2) Muscular atrophy.

Vitamin- K

Napthoquinone / Phylloquinone)

- 'K' stands for Koagulation or Coagulation and (clotting) commonly called as Anti Heamorrhagic vitamin.
- Essential for blood clotting and manufacturing of protein Prothrombin in liver.
- It is manufactured in colon by Bacteria.
- Can be absorbed in body only in presence of bile and can be synthesized by intestinal bacteria.

Sources:

Leafy vegetable
Cabbage
Spinach
Green vegetables
Tomatoes

Deficiency disease:

- (1) Uncontrolled bleeding.
(Haemiphilia)

WATER

- An average man contains about 45 liters of water which is 70% of the body weight.
- Water is absolutely necessary for the digestion and absorption of the foods taken in.
- It is the carrier of transporting medium of the nutrients and body substances. It also regulates the body temperature.
Sea water - 7.8 pH
Rain water - 5.6 pH
Acid rain - 2.4 pH
Pure water - 7 pH
- The only living organism does not take water entire life time - Silver fish.



CHAPTER-3

SKELETON SYSTEM

This system comprises the networks of bones, joints, and muscles. It provides the proper defined shape to the body and protects vital organs from harm. It is also the major ground of the RBC and WBC formation and works as the storehouse of calcium and phosphorus

Types of skeleton systems

- (1) Exoskeleton system
- (2) Endo skeleton system
- **Exoskeletons**, the hard material is formed mainly on the outside of the body and is called an exoskeleton. Insects such as beetles or dragonflies and crustaceans like crabs lobsters have a hard covering to their bodies called as cuticle.

Function of Skeleton System

- (1) Movement facilitation
- (2) Protection
- (3) Mineral storage (calcium phosphorous)
- (4) Support
- (5) Haematopoiesis (birth place for red blood cells)
- Endo skeleton system (Bones, Joints and muscles).

BONES

- Study of bones is called Osteology.
- At the time of birth the total number of bones are 350 while in adults it is 206.
- Bone is a hard connective tissue in which the ground substance is very hard and contains calcium salts.
- More than 112 of bones are made of water remaining 2/3 are of calcium, phosphorus, capillary blood vessels, fibrous tissues.
- In children, bones are more elastic due to more fibrous tissues and less mineral salts. But in old aged persons the bones are brittle due to the high level of mineral salts and less fibrous tissues.
- Bones are formed by calcium and phosphorous (in the form of calcium phosphate, calcium carbonate).

Types of bones

- (1) Irregular bones
Ex: Vertebrae, facial bones
- (2) Short bones
Ex: wrist, Ankle

- (3) Flat bones
Ex: Cranial, sternum, ribs, scapulas
- (4) Long bones,
Ex: leg bones, toes, arms, fingers.
- Location of bones in the body.

Head	22
Ear	06
Shoulder Girdle	04
Hands	60
Face	14
Vertebrae	26
Ribs	24
Legs	60
Sternum	01
Pelvic Girdle	02

- Head: cranium(8)+face(14) = 22 bones
- Study of cranium is called cranillogy.
- Ear: study of ear is called "Otology"
- Bones 6 (3+3)
 - (1) Malleus,
 - (2) Incus
 - (3) Stapes (smallest bone)
- **Shoulder Girdle: 4(2+2)**
 - (1) Collar Bone (clavicle)
 - (2) Shoulder bone (scapula)
- **Hands: 60 (30+30)**

1	Humerus (upper arm)	02
2	Radius	02
3	Ulna (fore arm)	02
4	Wrist (Carpals)	16
5	Metacarpals	10
6	Phalanges	28
	Total	60

- **Vertebrae column (back bone): 26**
In children there is a 33
- **Legs: 60 (30+30)**

1	Thigh bone (Femur) Longest bone is a 33 body	02
2	Tibia	02
3	Fibula	02
4	Knee cap (Patella)	02
5	Phalanges	28
6	Tarsal	14
7	Metatarsals	10
	Total	60



BONE DISEASES

(1) Arthritis

Arthritis caused by the inflammation of the joints.

Types of Arthritis:

(a) Rheumatoid arthritis

It is diagnosed by the presence of rheumatoid factor which is a type of immunoglobulin (Ig M). It is the primary symptom of inflammation of synovial membrane.

(b) Osteoarthritis

It is a degenerative joint disease characterized by the degeneration of the articular cartilage and proliferation of new bones.

(c) Gout arthritis or gout

It is caused due to excessive formation of uric acid Crystals.

(d) Osteoporosis

Mostly elder men and women are affected by it. It is an age-dependent systemic disorder characterized by low bone mass, micro architectural deterioration of the bone, increased fragility and proneness or susceptibility to fracture.

(2) Fluorosis

(3) Rickets

(4) Osteomalacia

(5) Spondylitis

(6) Cartilage

- It is tough connecting tissues, found in bone joints.
- It is more elastic and provide easy movement
- Cartilages are covered by a membranes Synovial Membrane. For easy movement of the joints synovial membranes are lubricated by Synovial fluid. Lack of synovial fluid causes pain in joints which occurs mostly in old age persons.

(7) Ligaments

- Study of ligaments - Syndesmology.
- These are tough bands of fibrous connecting tissues, usually found in bone joints. They are non - elastic in nature.
- They hold some important organs in the body like uterus.

(8) Tendons

- They are tough bands of fibrous connecting tissues which connects a bone with muscles.
- They are also non - elastic in nature.

JOINTS

- The study of joints are "Orthology"

Joints are the place of articulation between two or more bones or between a bone and a cartilage.

Type of Joints:

S. No.	Joint	Features
1.	Fixed or Immovable or Fibrous Joints:	There is no space between the bones and sutures present between the skull bones and the articulation of the roots of teeth with sockets of maxillae and mandible. Ex:- Upper Jaw (maxilla), Skull Bones.
2.	Imperfect immovable / Cartilaginous Joints	In such joints, the opposing surfaces are connected by fibrocartilage. Joints between adjacent vertebrae are examples of such joints.
3.	Perfect Movable or Synovial Joints	In such joints a space between the bones is present, called synovial cavity. This cavity remains filled with a viscous and slippery synovial fluid. The examples of synovial joints are
	(i) Ball and socket joints	Ex: Shoulders, and hip joints These joints permit movement in three planes.
	(ii) Hinge joints	Ex: elbow, ankle These joints movement only in single plane



(iii)	Pivot joints	Ex: neck and ulna These joints movement is monaxial
(iv)	Gliding joints	Ex: sternum, clavicle, scapula These joints only side to side and back and fourth movements are permitted. Movement is not around an axis
(v)	Ellipsoidal joints.	Ellipsoid joints are similar to a ball and socket joint. They allow the same type of movement to a lesser magnitude. Ex: Wrist

MUSCLES

- The study of muscles is called "Sarcology".
- Our muscles provide movement to the body and help in maintaining the body-heat. The total number of muscles in human beings are around 636.
- Human muscle constitute nearly 40-50% of total body weight.
- Muscle tissue is composed of 75% water, 20% protein and 5% non-protein organic material and minerals.
- Muscles convert chemical energy (glucose) into mechanical energy.
- The muscle itself is made up of bundles of hair-thin muscle fiber, which contain even thinner microscopic myofibrils. In turn, each muscle fibril contain bundles of long chain like substances. These are muscle protein act in and myosin.
- A muscle is a body part designed to get shorten or contract.
- The muscles that act together to produce a movement are called synergists.
- The record of electrical activity of muscle is called as Electro-Myogram (EMG) (Record of heart is known as Electro-Cardiogram (ECG)).
- On the basis of the presence or absence of a pigment called myoglobin, the skeletal muscles are classified into red and white muscles.

Variation between Red and White Muscles

- The Diameter of Red muscle is Smaller, white muscle is Bigger.
- Mitochondria number is more in Red muscle, Less in White muscle
- Blood Capillaries are More in red muscle, Less in white muscle
- Sarcoplasmic reticulum is less in red muscle, More in white muscle

Classification of Muscles:

Muscles are classified into three.

1. Skeletal Muscles or voluntary muscles

- These are striated muscle attached to the bones by tendons and help in the movement of the parts of skeleton.

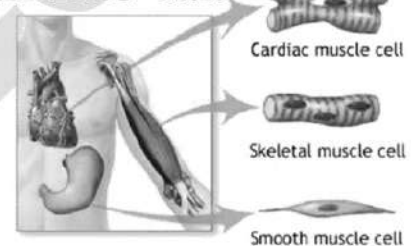
- Skeleton muscles are responsible for the movements to facilitate locomotion as these muscles are controlled by conscious mind
- They can move on their free will. Hence they are also known as voluntary muscles.

Types of Muscle

SKELETAL – striated & voluntary

Smooth – involuntary

CARDIAC - heart



2. Cardiac Muscles

- These are also striated muscles and occur exclusively in heart, and these are involuntary muscles.

3. Smooth Muscles or involuntary Muscles

- They are found inside the wall of the hollow internal organs, like alimentary canal, reproductive track, blood vessels, etc.
- Cardiac muscles and smooth muscles are known as involuntary muscles, as their action is not under the control of will power.
- Ex: intestines, stomach, fallopian tube muscles
- Muscles that act most powerfully during any given movements are called prime movers.

Muscles that act in opposition to each other are called antagonists.

- The largest muscle is the gutters maximus in the buttock.
- The smallest' muscle is the stapedius, deep inside the ear. It is just a few millimeters long and thinner than cotton thread.
- A muscle can only pull on a bone not push it.
- Each muscle is linked by nerves to the brain.



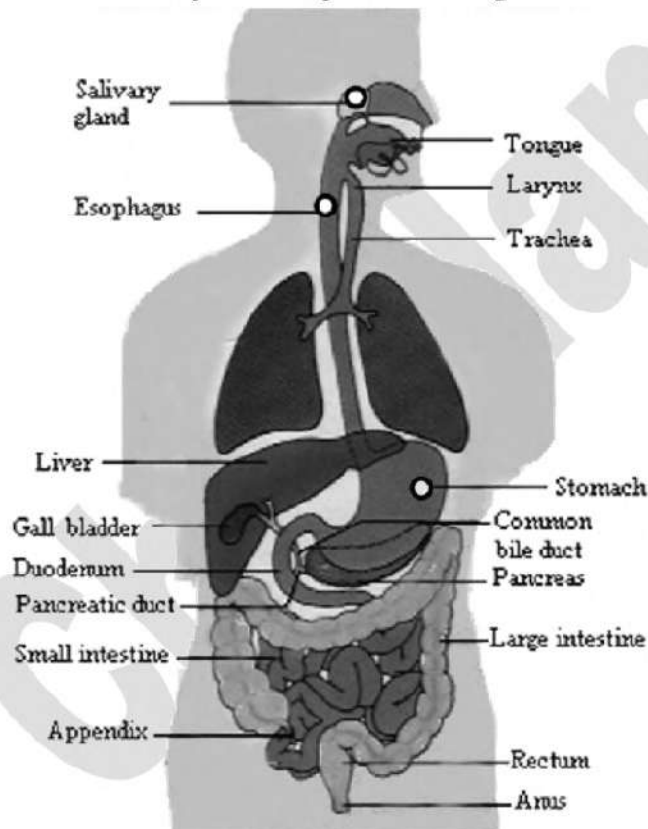
CHAPTER 4

DIGESTIVE SYSTEM

- Food is taken into the Alimentary canal through the mouth and is propelled along by the movements of the muscles on its wall. Glands located in its wall (gastric glands) and in associated organs like liver and pancreas secrete enzymes into the lumen of the canal.

CANAL

- It starts from the mouth and passing through oesophagus, stomach and small intestine finally ends into large intestine. Absorption of digested food is mainly taken in small intestine.
- Undigested and unabsorbed material then pass down into the colon and rectum which absorb the excess water and with the help of purifying bacteria from faeces. Finally the faeces passes out through the anus.



- It is a coiled, muscular, continuous tube running through the ventral body cavity extending from the mouth to the anus. It is about 30 feet long. It consists of many specialized sections.

Alimentary systems consists of,

- 1) Mouth
- 2) Buccal cavity
- 3) Pharynx
- 4) Oesophagus
- 5) Stomach
- 6) Small Intestine
 - a. Duodenum
 - b. Jejunum
 - c. Ileum
- 7) Large Intestine
 - a. Caecum
 - b. Appendix
 - c. Colon
- 8) Rectum
- 9) Anus

Mouth

- Taking food through the mouth Ingestion.
- Amoeba takes food through Pseudopodia.
- It has 3 pairs of salivary glands (parotids, submaxillary (sub mandibular) and sublingual glands) which secretes saliva.
- Daily secretion of saliva in the mouth is 1 to 1.5 liter.
- Saliva is acidic in nature.
- Saliva lubricate the food after mixing in it.
- It is also aid in digestion of food as it has ptyalin or amylase enzyme which acts on cooked starch changing them into sugar maltose.
- Oral Cavity or Buccal cavity
- Mouth leads into oral cavity.
- The oral cavity leads into a short pharynx which serves as a common passage for food and air.
- The mouth leads into oral cavity that has two lateral walls (cheeks), an anterior roof (hard palate), a posterior roof (soft palate) and a pair of Jaw (upper maxilla and lower mandible).

Pharynx

- The oral cavity leads into a short Pharynx.
- The pharynx serves as a common passage for food and air.
- The oesophagus and the trachea (wind pipe) open into the pharynx.



Oesophagus

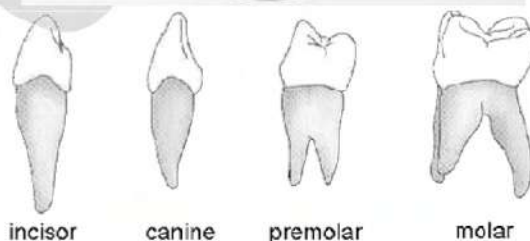
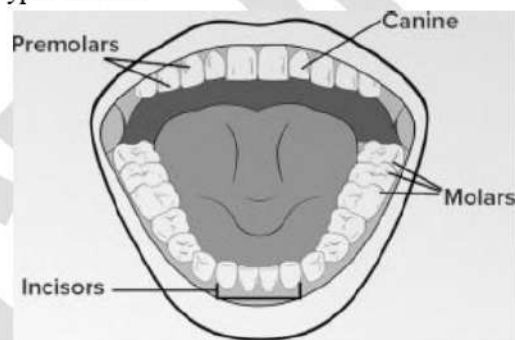
- It is a long, narrow, muscular and tubular structure.
- It runs downwards through the neck behind the trachea, passes through the diaphragm and opens in to stomach in the abdomen.
- It conducts the food to stomach rhythmic contraction is peristalsis.

TEETH

- The study of teeth is called "Odontology"
- Teeth are hard yellow substance surrounding the pulp.
- Parts of teeth
 - 1) crown
 - 2) neck
 - 3) root
- Teeth are formed by calcium phosphorous.
- Teeth are maximum covered by dentine.
- A small shining, whitish and thin layer on the crown is called enamel.
- Enamel is the hardest substance in the body.
- Enamel protects teeth from sensitive heat and cold.
- As the milk teeth are replaced by permanent teeth, human teeth are also called as Diphyodont.
- In lower vertebrates, the teeth are replaced continuously, the old worn out ones falling out and their places being taken by new ones throughout life. This condition is Polyphyodont.
- Cats, Dogs have Heterodont teeth
- Whereas Reptiles and Fish have Homodont teeth, they replace their teeth frequently (Crocodile 40 times in its life time).

Teeth in human

- Types of teeth.



- 1) Incisors
- 2) Canines
- 3) Premolars
- 4) Molar

- Milk teeth are of yellowish colour while the permanent teeth are white. Permanent teeth appear between 9 to 20 years of age.
- First, milk teeth to appear is incisor while the first permanent teeth to appear is 1st molar. Lower teeth comes earlier than the Upper teeth.
- Total milk teeth 20.
- Milk teeth formula (child teeth) $\frac{2\ 1\ 2\ 0}{2\ 1\ 2\ 0}$
- Molars are not found in children
- Man like the Primates also has 32 teeth.
- Adult dental formula (permanent teeth) $\frac{2\ 1\ 2\ 3}{2\ 1\ 2\ 3}$
- 4 incisors
- 2 canine
- 4 premolars
- 6 molars
- So, both jaws have 8 incisors, 4 canine, 8 premolars, 12 molars (Total 32).
- Premolars and Molars are also called Cheek teeth.
- First molar is largest of all teeth while the third molar (wisdom teeth) is smallest and appears at last.

Teeth type	Functions
Incisors	Cutting and biting
Canines	Stabbing
Premolar	Crushing
Molar	Grinding

- Each teeth has 3 layers - Crown, Neck, Root.

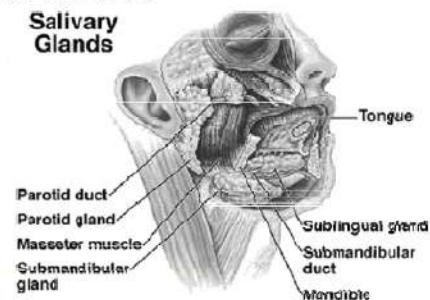
Terms related to teeth,

Teeth Diseases:

- 1) Gum disease (Periodontal)
- 2) Tooth Decay
- 3) Tooth sensitivity
- 4) Toothaches and dental emergencies
- 5) Fluorosis
- 6) Pyorrhea (formation of puss on the junction of tooth and gum).

SALIVARY GLANDS

Salivary Glands





- Human have 3 pairs of salivary glands:
 - Parotid glands in the cheeks
 - Submandibular (submaxillary)
 - Sublingual
- Both submandibular and sublingual opens into the floor of the mouth.
- Saliva is a mixture of water and electrolytes (Na^+ , K^+ , Cr , HCO_3), derived from the blood plasma, mucus and serous fluids, salivary amylase or ptyalin (enzyme) and lysozyme (antibacterial agent), all are secreted by the salivary glands.
- Salivary pH value 6.5 (acidic)

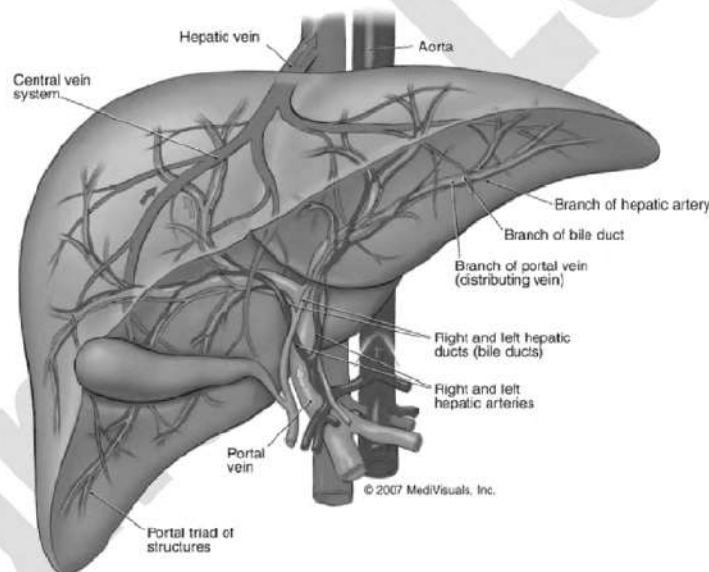
Stomach

- It has strong muscular wall with mucus.
- The stomach is a large muscular C-shaped sac and located on left side in the upper position of the abdominal cavity.

LIVER

- Food is stored in the 3 to 4 hrs
- The muscle of the stomach churn the food and mix it with water and gastric juice to form a thick, acidic fluid called chyme and then it pass to Duodenum.
- Food mixed with gastric juice in the stomach is called chyme.
- Gastric juice is produced in the stomach.
- Gastric juice consists of,
 - **Hydro Chloric Acid:** It activates enzymes in the stomach.
 - **Renin (Enzyme):** It is enzyme found in stomach. It is absence in an adult.
It is more active in children. This enzyme works on Caseinogen (Milk products) and converts into paracaseinate products (curd like products).
 - **Lipase (enzyme):** It acts on fats and converts into fatty acids and glycerol.

Internal Anatomy of Liver



- The study of liver is called Haepatology.
- It is the largest organ in the body.
- It is a regenerative power organ.
- It produces "Bile Juice". Which contain "no enzymes". But it acts as enzymes. i.e. it "promotes the action of the enzymes.
- The bile salts play a very important role in the digestion of fats.
- Bile is a watery greenish fluid mixture containing bile pigments, bile salts, cholesterol and phospholipids.
- It is alkaline in nature and hence neutralizes the acidic nature of the food.

- Bile pigments are the products of dead erythrocytes.
- Bile juice acts on fats & converts into fatty adds and glycerol.
- Bile juice is stored in "Gall Bladder".

Duties of Liver

- R.B.C birth place & Death place.
- Urea (waste material) is stored in liver.
- A, D, E, K, B_{12} vitamins are generated.
- Controls (or) Stabilization the toxins in the body.
- Storage of Glycogen Excess glucose converts into glycogen in the liver.
- Heparin anti blood clotting agent also generated in the liver.



- Diseases for liver are,
(1) Hepatitis, (2) Sirosis
- First country developed Vaccine for the "Hepatitis -B" in the world is "INDIA". i.e., Sanvac -1997.
- Sirosis → caused due to excess consumption of alcohol.

Gall Bladder

- In vertebrates the gallbladder (cholecyst, gall bladder, biliary vesicle) is a small organ that aids mainly in fat digestion and concentrates bile produced by the liver.
- In humans, the loss of the gallbladder is, in most cases, easily tolerated.
- The surgical removal of the gallbladder is called a cholecystectomy
- When food containing fat enters the digestive tract,
- It stimulates the secretion of cholecystokinin (CCK). In response to CCK, the adult human gallbladder, which stores about 50 milliliters (1.7 U.S. fl OZ; 1.8 imp fl oz) of bile, releases its contents into the duodenum.
- The bile, originally produced in the liver, emulsifies fats in partly digested food.
- During storage in the gallbladder, bile becomes more concentrated which increases its potency and intensifies its effect on fats.

Bile Juice

- Bile or gall is a bitter-tasting, dark green to yellowish brown fluid, produced by the liver of most vertebrates, that aids the digestion of lipids in the small intestine.
- In many species, bile is stored in the gallbladder and, when the organism eats, is discharged into the duodenum.
- Bile is 85% water, 10% bile salts, 3% mucus and pigments, 1 % fats, and 0.7% inorganic salts.

Small Intestine

- It produces succus entericus
- It is a over 7 m (23 feet 4 inches) long track.
- It is divided into 3 structural parts
 - Duodenum
 - Jejunum
 - Ileum

Duodenum

- It is 'U' shaped organ.
- It is the first and largest part of the small intestine.
- Duodenum is most responsible for the breakdown of food in the small intestine, using enzymes.
- The villi of the duodenum have a leafy-looking appearance, a histologically identifiable structure.

- Brunner's glands, which secrete mucus, are found in the duodenum only.
- The duodenum wall is composed of a very thin layer of cells that form the muscularis mucosae.
- The duodenum is almost entirely retroperitoneal. It has three parts and each parts has its own significance.
- The duodenum also regulates the rate of emptying of the stomach via hormonal pathways.

Jejunum

- It is a middle part of the small intestine.
- Jejunum is specialized for the absorption, by enterocytes, of small nutrient particles which have been previously digested by enzymes in the duodenum. Once absorbed, nutrients (with the exception of fat, which goes to the lymph) pass from the enterocytes into the enterohepatic circulation and enter the liver via the hepatic portal vein, where the blood is processed

Ileum

- It is a small part of the small Intestine.
- Ileum is mainly to absorb vitamin B12 and bile salts and whatever products of digestion that were not absorbed by the jejunum.
- The wall itself is made up of folds, each of which has many tiny finger-like projections known as villi on its surface.
- In tum, the epithelial cells that line these villi possess even larger numbers of microvilli. Therefore the ileum has an extremely large surface area both for the absorption (attachment) of enzyme molecules and for the absorption of products of digestion.
- The DNES (diffuse neuroendocrine system) cells of the ileum secrete various hormones (gastrin, secretin, cholecystokinin) into the blood.
- Cells in the lining of the ileum secrete the protease and carbohydrase enzymes responsible for the final stages of protein and carbohydrate digestion into the lumen of the intestine. These enzymes are present in the cytoplasm of the epithelial cells.

Villi

- The villi contain large numbers of capillaries that take the amino acids and glucose produced by digestion to the hepatic portal vein and the liver.
- Lacteals are small lymph vessels, and are present in villi.

Enzymes in small intestine

- 1) Peptidase
- 2) Lipase
- 3) Maltase
- 4) Lactase
- 5) Sucrase





Large Intestine

- The large intestine is the last part of the digestive system in vertebrate animals.
- No enzymes are produced.
- Its function is to absorb water from the remaining indigestible food matter, and then to pass useless waste material from the body.
- This article is primarily about the human gut, though the information about its processes are directly applicable to most mammals.
- The large intestine consists of the,
 - Cecum
 - Appendix
 - Colon
 - Rectum
 - Anal canal.

Cecum

- Cecum, also spelled caecum, pouch or large tube like structure in the lower abdominal cavity.
- It receives undigested food material from the small intestine and is considered the first region of the large intestine.
- It is separated from the ileum (the final portion of the small intestine) by the ileocecal valve (also called Bauhin valve), which limits the rate of food passage into the cecum and may help prevent material from returning to the small intestine.
- The main functions of the cecum are to absorb fluids and salts that remain after completion of intestinal digestion and absorption and to mix its contents with a lubricating substance, mucus.
- The internal wall of the cecum is composed of a thick mucous membrane through which water and salts are

DIGESTIVE HORMONES

There are 4 hormones that regulate the digestive system

Digestive Hormone	Function
Gastrin	➤ It is in stomach and stimulates the gastric glands to secrete pepsinogen and hydrochloric acid. ➤ Secretion of gastrin is stimulated by food arriving in stomach. ➤ The secretion is inhibited by low p^H.
Secretin	➤ It is in duodenum and signals the secretion of sodium bicarbonate in the pancreas and it stimulates the bile secretion in the liver. ➤ This hormone responds to the acidity of the chyme.
Cholecystokinin (CCK)	➤ It is in duodenum and stimulates the release of digestive enzymes in the pancreas and stimulates the emptying of bile in the gall bladder. ➤ This hormone is secreted in response to fat in chyme.
Gastric inhibitory peptide (GIP)	➤ It is in duodenum and decreases the stomach churning in turn! slowing the emptying in the stomach.

absorbed. Beneath this lining is a deep layer of muscle tissue that produces churning and kneading motions.

Appendix

- At the place where small and large intestine are joined, two organs are found namely, caecum and appendix, but both of them have lost their utility. Appendix, is a vestigial organ.
- Pancreatic juice work well in the presence of Succuss - entericus.

Colon

- Colon is thicker than small intestine and thinner than caecum.
- The colon is the last part of the digestive system in most vertebrates.
- It extracts water and salt from solid wastes before they are eliminated from the body and is the site in which flora-aided (largely bacterial) fermentation of unabsorbed material occurs.
- Unlike the small intestine, the colon does not play a major role in absorption of foods and nutrients. However, the colon does absorb water, sodium and some fat soluble vitamins

Rectum

- Rectum do not secrets any enzymes. Rectum stores digestive residue and absorbs water from residue.
- Rectum is the last part of the alimentary canal and is about 7-8 inches long.
- The rectum is the last stop before the feces is eliminated through the anal canal.
- Similar to the colon electrolytes are absorbed (sodium, potassium, chloride) and indigestible food ingredients are decomposed by anaerobic bacteria.
- The stool is thickened through water absorption and mixed with mucus.



Chandan Logics

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S.No.	Enzyme Name	Location and Digestive Juice	Enzyme Act on	Enzyme Changed into
1.	Ptyalin or amylase	Mouth-Salivary juice(acid)	Starch	Maltose (via dextrins)
2.	Pepsin	Stomach - Gastric Juice	Proteins	Peptodes
3.	Renin (absence in an adult)	Stomach - Gastric Juice	Caseinogen (Milk products)	Calcium paracaseinate
4.	Lipase	Stomach - Gastric Juice	Fatts	Fatty acids and glycerol
5.	Hydro Chloric Acid(HCL)	Stomach - Gastric Juice	It activates all enzymes in the stomach	-----
6.	Trypsin	Pancreas - Pancreatic juice	Proteins	Polypeptides
7.	Chymotrypsin	Pancreas - Pancreatic juice	Proteins	Polypeptides
8.	Bile juice (no enzymes found in bile juice)	Liver -Bile juice	Fatts	Fatty acid and glycerol
9.	Erepsin	Small intestines - succus entericus juice	Polypeptides	Amino acids
10.	Maltase	Small intestines - succus entericus juice	Maltase	Glucose
11.	Lactase	Small intestines - succus entericus juice	Lactase	Glucose
12.	Sucrase	Small intestines – succus entericus juice	Sucrose	Glucose

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CHAPTER-5

RESPIRATORY SYSTEM

Respiratory is the chemical breakdown of food in the living cells to release the energy, essential for life. The term 'Breathing' simply means the act of inhaling and exhaling air out of the lungs.

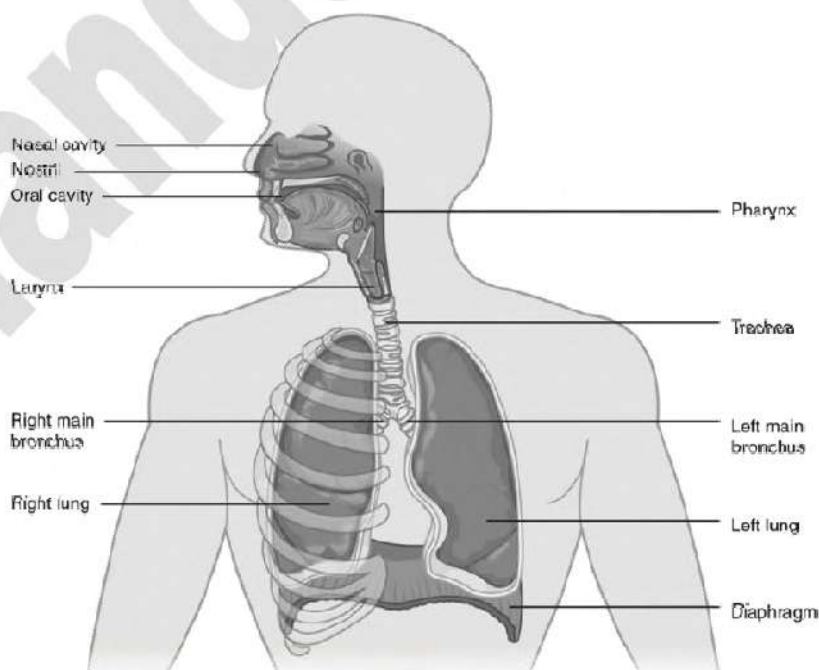
Elements in atmosphere	Inhaling Elements	Exhaling Elements
(1) Nitrogen (N ₂)	78.028%	78.028%
(2) Oxygen (O ₂)	21%	16%
(3) Carbon dioxide (CO ₂)	0.03%	4.4%

Oxidation:

- Haemoglobin absorbs oxygen and converts into oxyhaemoglobin in the body.
- Oxy haemoglobin is the combination of the red blood haemoglobin cells with the oxygen in the lungs. The oxygen is then transported all over the body by the blood and the process of blood oxygenation continues in the lungs. This process is called Oxidation.
- Respiration is done in all plants, animals and bacteria. These are called living organisms. But respiration cannot take in virus.
- Respiration takes place at mitochondria in the cell. It is called power house of the cell.
- The energy is made available the form of A TP (Adenosine tri-phosphate), referred to as the energy currency of the cell.

Catabolic Process:

- It is the set of metabolic pathways that breaks down molecules into smaller units to release energy and is related to wakefulness.
- In catabolism, large molecules such as polysaccharides, lipids, nucleic acids and proteins are broken down into smaller units such as monosaccharides, fatty acids, nucleotides, and amino acids, respectively.
- The human respiratory system consists of,
(1) External nostrils, (2) Nasal cavity (3) Nasopharynx (4) Larynx (voice box) (5) Trachea (6) Bronchi (7) Bronchiole (8) Lungs.





Nose

- The study of nose is called Rhinology
- The visible part of the human nose is the protruding part of the face that bears the nostrils.
- The shape of the nose is determined by the ethmoid bone and the nasal septum, which consists mostly of cartilage and which separates the nostrils.
- On average the nose of a male is larger than that of a female.
- The nose has an area of specialized cells which are responsible for smelling (part of the olfactory system).
- In Snakes and Dogs olfactory cell are well developed so that they have power of smelling 40 times greater than the humans.
- Another function of the nose is the conditioning of inhaled air, warming it and making it more humid. Hairs inside the nose prevent large particles from entering the lungs.

Nasopharynx

- The nasopharynx (nasal part of the pharynx) is the uppermost part of the pharynx.
- It extends from the base of the skull to the upper surface of the soft palate;
- It differs from the oral and laryngeal parts of the pharynx in that its cavity always remains patent (open).

Larynx

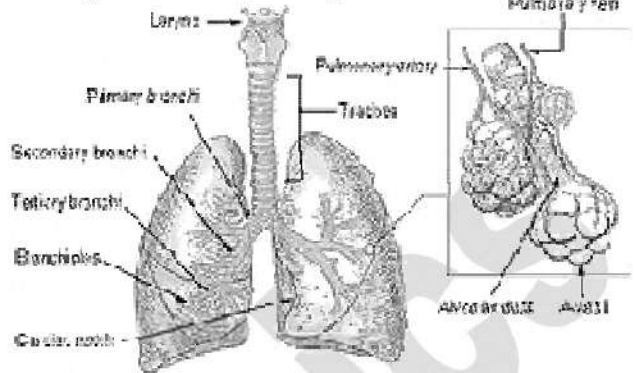
- Larynx is the connection between pharynx and the windpipe or trachea. Larynx is also known as voice box.
- Larynx is covered by pieces of cartilage forming a box. In males, the adam's apple is the protruding cartilage of the voice box.
- It does not contribute to the breathing movements.

Trachea

- Trachea is a tubular structure of about 12 cm in length and 2.5 cm in diameter.
- The trachea and bronchi are strengthened by rings of elastic cartilage. Internally trachea has a lining of mucous glands and cilia.
- It includes the long wind pipe or trachea running through the neck and the thoracic cavity where it divides into 2 bronchi which enter the lungs.

LUNGS

Branchi, Branchial Tree, and Lungs



- The study of lungs are called Pleurology.
- The lungs are a pair of spongy, air-filled organs located on either side of the chest (thorax).
- The trachea (windpipe) conducts inhaled air into the lungs through its tubular branches, called bronchi.
- The bronchi then divide into smaller and smaller branches (bronchioles), finally becoming microscopic.
- The bronchioles eventually end in clusters of microscopic air sacs called alveoli.
- In the alveoli, oxygen from the air is absorbed into the blood. Carbon dioxide, a waste product of metabolism, travels from the blood to the alveoli, where it can be exhaled. Between the alveoli is a thin layer of cells called the interstitium, which contains blood vessels and cells that help support the alveoli.
- The lungs are covered by a thin tissue layer called the pleura. The same kind of thin tissue lines the inside of the chest cavity -- also called pleura.
- A thin layer of fluid acts as a lubricant allowing the lungs to slip smoothly as they expand and contract with each breath.
- Lungs are paired structure present in the thoracic or pleural cavity.
- Human's right lung has 3 lobes while the left lungs has only 2 lobes.
- Lungs tissues are elastic, porous and spongy, so float in water.
- Lungs are situated in the thoracic cavity. It is a spongy mass of tissue with greatly increased internal surface due to innumerable alveoli.
- Diaphragm is not the part of respiratory system but help in expansion and contraction of the lungs.
- During inhalation, diaphragm contracts and lungs increases while during exhalation, the diaphragm relaxes and the lungs are compressed.



Breathing

- The cycle of normal breathing (ventilation of lungs) consists of three phases: Inspiration, Expiration and Pause.
- Inverse Breathing consists of three phases Inspiration, Pause and Expiration.
- Inspiration is an active process. It is the result of muscle contraction.
- Expiration is a passive process. It is the result of muscle relaxation.
- Breathing rate is lowest while we are sleeping.
- A person with fever may breathe faster than normal.
- Breathing occurs involuntarily but its rate is controlled by the respiratory centre of brain.
- Under normal conditions rate of breathing is 15-18 times per minute.
- New-born child's normal Breathing Rate is 5-11 per minute.
- 5 years old child's normal Breathing Rate is 26 per minute
- 25 years old person's normal Breathing Rate is 16 per minute
- 50years old person's normal Breathing Rate is 18 per minute

Aerobic Respiration

- It is done in the presence of the oxygen (O_2).
- Here glucose combining with oxygen is break down into CO_2 H_2O and energy.
- This is the complete respiration and provides 24 hours more energy than anaerobic respiration.
- It can be done all plants and animals.

Anaerobic Respiration

- It is done in absence of the Oxygen (O_2)
- It takes place without oxygen and very little amount of energy is released.
- Glucose is broken down in intermediary products like alcohol or lactic acid and not into final products.

Lungs Diseases

(1) Asthma

- Asthma is characterized by recurrent episodes of wheezing, shortness of breath, chest tightness, and coughing. Sputum may be produced from the lung by coughing but is often hard to bring up.

Causes

- (1) Genetic
- (2) Allergy
- (3) Environmental problems

(2) Bronchitis:

- Bronchitis is an inflammation of the mucous membranes of the bronchi (the larger and medium-sized airways that carry airflow from the trachea into the more distal parts of the lung parenchyma).
- The people who are working in cotton and cement industries are mostly suffer with Bronchitis.

- (3) Pneumonia
- (4) Bird flu
- (5) Swine flu
- (6) Diphtheria
- (7) Pertussis
- (8) T.B
- (9) SARS (Severe Acute Respiratory Syndrome)

Animals	Respiratory organ
Sponges	Plasma membrane
Flatworms	Plasma membrane
Coelenterates	Plasma membrane
Scorpio	Book lungs
Fish	Gills
Crab	Gills
Frog, leech, Earthworm	Skin
Snakes, Birds, mammals	lungs

CHAPTER -6

BLOOD CIRCULATORY SYSTEM

The circulatory system is an organ system that permits blood and lymph circulation to transport if nutrients (such as amino acids and electrolytes), oxygen, carbon dioxide, hormones, blood cells, etc. to and from cells in the body to nourish it and help to fight diseases, stabilize body temperature and pH, and to maintain homeostasis.

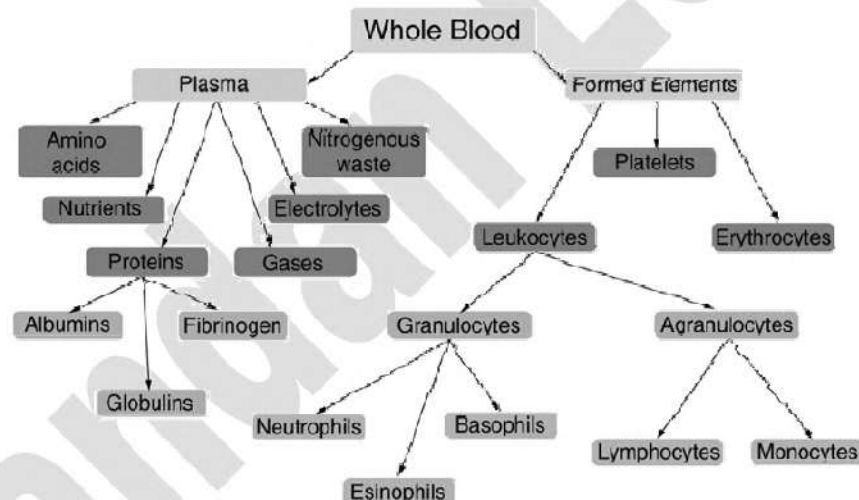
Animal of the body require nourishment and oxygen and metabolic wastes need to be removed from the body. This is carried out by blood through circulatory system.

➤ Blood circulatory system was discovered by Willam Harvy. (He was called father of blood circulation)

BLOOD

- The study of blood is called Haematology.
- Blood is slightly alkaline in nature (pH 7.4).
- Blood is in Red colour, sticky and salty fluid.
- It transports gases, hormones and food substances. Hence, blood is used as transport vehicle for any compound within the body.

COMPOSITION OF BLOOD



- Blood maintains water balance of body, pH of body fluids and maintain body temperature by thermoregulation. It also maintains and heat dispersion.
- It removes waste products from body cells.
- In adult man normally 4-5 liters of blood is found.
- It prevents haemorrhage by formation of blood clot.

(1) Red Blood Cells (RBC)/Erythrocytes

- They are numerous in blood they contain haemoglobin. [The red, oxygen carrying pigment] Mature RBC are dead cells.
- The entire volume of RBC is filled with Haemoglobin. In the foetus, RBC are mainly formed in the liver and spleen but in adult, RBC formed from bone marrow. RBC have an average life span of 120 days.
- RBC died in spleen (It is called grave yard).
- Old and damaged RBC are phagocytosed (eaten away by macrophages).
- The haemoglobin catabolished into the yellow pigment called Bilurubin which is excreted in bile.



- The skin and mucous membranes of eyes assume a yellowish colour, if all the bilirubin cannot excreted from the body. This condition is called Jaundice.
- The excess RBC is stored in the spleen which serves as blood -bank.
- Hilly people have higher number of RBC than the plain - men.
- RBC count in male is (10%) greater than female.

Haemoglobin:

- There are small bags of haemoglobin, a reddish coloured protein containing iron.
- The organs which produce blood corpuscles are spoken haemopoieses.
- Haemoglobin has good affinity for oxygen but affinity to CO is 200 times more than that of O₂.
- The normal quantity of haemoglobin present in blood is 12-15gm in every 100ml of blood. A decrease in this quantity is called anaemia.
- Since haemoglobin is responsible for red colour in blood, anaemia results in reduced redness of skin and mucous membrane.

Diseases Associated With RBCs

Diseases Name	Symptoms
Anaemia	➤ Deficiency of iron, folic acid or proteins which are required for manufacturing of RBCs. ➤ Abnormal loss of blood especially in women due to excessive heavy menses.
Polycythaemia or Erythaemia	➤ RBC count goes abnormally high. ➤ Blood becomes thicker and moves slowly. ➤ Skin becomes bluish.
Thalassemia	➤ It is a hereditary disease. Body is not able to make enough haemoglobin. ➤ It is common in India in sindhis and punjabis. ➤ Remedy of thalassemia is regular blood trans-fusion. ➤ Probably the disease came to India with Alexander's army.

(2) White Blood Cells (WBC) / Leucocytes

- WBC are non pigmented, colourless blood cells. They are larger in size than RBC but far lesser in number.
- The number of WBCs in blood is in the range of 7000-1 0000/mm³. Ratio of RBC to WBC count is 500: 1.
- WBCs are nucleated and they do not have haemoglobin.
- WBC are of 2 types:
 - (a) Granulocytes
 - (b) Agranulocytes

a) Granulocytes

They are the cells containing granules, and a polymorphic nucleus in the cytoplasm. These are of three types viz. neutrophils, eosinophils and basophils.

Cell	Characters
Neutrophils	Neutrophils do not take colour when exposed to acidic, as well as basic dyes. Its nucleus has two to seven lobes. Neutrophils are mainly responsible for protection against infection. This constitutes 62% of total leucocytes.
Eosinophils	Eosinophils are characterised by a bilobed nucleus. The cell contains numerous coarse granules that are stained bright red with eosin, an acid dye. The proportion of eosinophils is less than 3% They play an important role in detoxification.
Basophils	Basophils are found in least proportion (i.e. 0.5 to 1 %). The nucleus has two to three lobes. Basophils are significant in allergic reactions.



**b) Agranulocytes**

They are leucocytes that lack granule and consist of two types of cells lymphocytes and monocytes.

Cell	Characters
Lymphocytes	Lymphocytes have a single large nucleus with pale blue cytoplasm. They have a proportion of about 30%. An important function of the lymphocytes is the manufacture of serum globulin, and they play a key role in immunological reactions. This count falls in HIV patients.
Monocytes	Monocytes are large cells with kidney-shaped nucleus. These constitute 5-6% of total WBC.

Disease associated with WBCs**Leukemia (Blood Cancer)**

- It is more common in children.
- This is caused because of cancer of red bone marrow.
- In case of leukemia, abnormal increase of WBC and decrease of RBC takes place.
- Bone transplantation and radiation therapy is only remedies for leukemia.

(3) Platelets or Thrombocytes

- They are non-nucleated, colourless, smallest of all blood components, tiny circular or oval discs like.
- They are derived from giant cells of megakaryocytes of bone marrow.
- These are non pigmented like WBCs.
- Average life of platelets is 10 days. These are smaller in count.
- These are manufactured in red bone marrow.
- Platelets disintegrate and produce Thrombokinase thromboplastin, vital in blood clotting.
- Ratio between WBC and Thrombocytes is 1 : 44.
- These are essential for blood clotting. Vitamin K is essential for production of prothrombin.
- Their number in blood ranges between 2,50,000 to 4,00,000/mm³. They play an important role in blood coagulation.

Plasma

- Plasma is the liquid part of blood and extracellular fluid of blood.
- Albumins, globulins and fibrinogen are the important types of proteins present in the plasma.
- Albumins provides the colloid osmotic pressure in plasma.
- Fibrinogen is essential for clotting of blood.
Thus, these proteins act as enzymes, coagulating factors and antibodies.
- Plasma is a faint yellow coloured viscous fluid which is alkaline in nature. Salinity of plasma is about 1/3rd of the salinity of sea water.

- Plasma constitute about 55% of total blood volume.
- Some other proteins (globulins) present in plasma aid in body defence mechanism.

Blood Clotting

- The blood contains an inactive enzyme known as Prothrombin which is kept in check by another substance in blood called Antiprothrombin or Heparin.
- Thus, the blood flowing normally in blood vessels does not coagulate due to the presence of heparin.
- Clotting depends upon the presence of Prothrombin, Thromboplastin and calcium ions and absence of any of these will prevent coagulation.
- 3 minutes is the normal time for blood to clot.
However, average coagulation time varies from 2 to 6 minute depending upon the amount of blood.
- Plasma protein fibrinogen forms mesh over the wound & within a few minutes clot becomes firm.
- When injury is caused, thromboplastin is secreted by injured platelets.

Blood Groups & Transfusion

- Blood groups were discovered by Land Stiner in 1902.
- June 14th world Blood donation day.
- The four types of blood groups are
O → Lack of antigens it is called universal donar.
A → Largest blood group in the world.
B → Least blood group in the world.
AB → It is the universal acceptor(lack of antibodies)

Blood Group	Can Accept	Can Donate
A	From A & O	To A & AB
B	From B & O	To B & AB
AB	From A, B, AB and O	To AB Only
O	From O only	To A, B, AB, & O



The Rh Factor

- Rh in factor is first 2 letters of the monkey Rhesus in which first of all it was discovered.
- Rh is an antigen present in the RBC of a large majority of people.
- Such people who have this antigen are called Rh positive (Rh+). In India about 97% people are Rh positive blood, they will develop antibodies.

LYMPHS

- The lymphatic system is a major component of the immune system.
- It is a complex network of lymphoid organs, lymph nodes, lymph ducts, lymph tissues, lymph capillaries and lymph vessels that produce and transport lymph fluid from tissues to the circulatory system.
- The lymphatic system has three interrelated functions:
- Removal of excess fluids from body tissues.
- Production of immune cells (such as lymphocytes, monocytes and antibody producing cells called plasma cells).

Lymph originates as blood plasma that leaks from the capillaries of the Circulatory system, becoming interstitial fluid, and filling the space between individual cells of tissue.

- Plasma is forced out of the capillaries by oncotic pressure gradients, and as it mixes with the interstitial fluid, the volume of fluid accumulates slowly. Most of the fluid is returned to the capillaries by hydrostatic pressure gradients.
- The proportion of interstitial fluid that is returned to the circulatory system by osmosis is about 90% of the former plasma, with about 10% accumulating as overflow.
- The excess interstitial fluid is collected by the lymphatic system by diffusion into lymph capillaries, and is processed by lymph nodes prior to being returned to the circulatory system.
- The excessive volume of lymph fluid causes swelling called dropsy or oedema. The movement of Lymph is mainly due to squeezing action of the surrounding muscles.
- Lymph organs include the bone marrow, lymph nodes, spleen and thymus.

B-lymphocytes

- Bone marrow contains tissue that produces lymphocytes. B-Lymphocytes (B-cells) mature in the bone marrow.

- B-cells make deadly proteins called antibodies that surround the invaders and macrophages, which swallow them up and destroy them.

T-lymphocytes

- The thymus secretes a hormone, thymosin that causes pre-T-cells to mature (in the thymus) into T-cells.
- T-cells are the cells which identify invaders.
- T-lymphocytes (T-cells) mature in the thymus gland. Other blood cells such as monocytes and leucocytes are produced in the bone marrow.
- T-Lymphocytes and B-Lymphocytes collectively known as primary lymphoid organs and lymph nodes, spleen and mucosa associated lymphoid tissues such as tonsils are part of secondary lymphoid organs.
- Lymph nodes are areas of concentrated lymphocytes and macrophages along the lymphatic veins. The spleen is similar to the lymph node except that it is larger and filled with blood.
- The spleen serves as a reservoir for blood, and filters or purifies the blood and lymph fluid that flows through it. If the spleen is damaged or removed, the individual is more susceptible to infections.

HEART

- The study of heart is called cardiology.
- First heart transplantation was done by "Christian Bernadshaw" (South Africa 1967)
- First heart transplantation in India was done by Dr. VenuGopal (1984).
- Heart weight 300 grms.
- Heart is a hollow, muscular, somewhat conical four-chambered force pump, enclosed a fibrous bag.
- Wall of the heart is made up of cardiac muscle fibre, connective tissue and tiny body vessels.
- The heart of mammal, conical in shape, is completely enclosed in a membrane called 'Pericardium'.

Heart of mammal is 4-chambered

Animal	Heart Chambers
Reptiles	4 chambers
Amphibian	3 chambers
Fish	2 chambers
Earth worm	16 chambers
Cockroach	13 chambers



In human heart

Upper chambers are called Auricles -2

Lower chambers Ventricles -2

Blood circulatory system in the heart:

Body parts (Except lungs) → Right Auricle



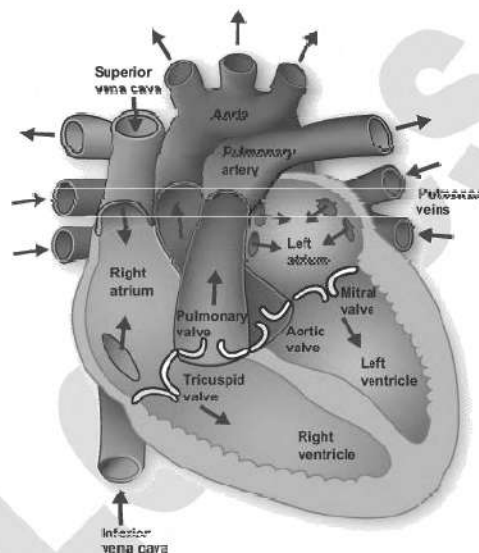
Left Auricle ← Lungs ← Right ventricle



Left Ventricle → Aorta → Body parts

- The 2 auricles are separated by an inter-auricular septum and the 2 ventricles by 2 ventricular septum. The right auricle receives venous blood from all parts of the body except the lungs. Left auricle receives oxygenated blood from the lungs through Pulmonary veins.
- Right auricle leads into the right ventricle by an opening guarded by Tricuspid valve this valve allows the flow of blood in one direction only i.e. from auricle to ventricle. Similarly, the left auricle which receives oxygenated blood coming from lungs, opens into left ventricle by an opening guarded by a valve called Bicuspid valve or Mitral valve.
- The valves between auricles and ventricles are membranous in mammals which in frog, they are muscular.
- For the pumping action of Heart, the wave of contraction originates from a 'Pacemaker' or 'Contraction centre' (Located at right auricle) called the 'Sinoauricular node'. Through the contraction originates in the node but its frequency is controlled by the nerves.
- Left heart is a term used to refer collectively to the left atrium and left ventricle of the heart. Sometimes aorta also included.
- Left atrium receives oxygenated pulmonic blood from the pulmonary veins. The blood is then pumped through the mitral valve into the left ventricle, which in turn pumps the blood through the aortic valve into the aorta.
- The left side of the heart is thicker than the right because it has to pump blood, from the left ventricle into the aorta and the body.
- Right heart is a term used to refer collectively to the right atrium and right ventricle of the heart. Occasionally, this term is intended to refer the right atrium, right ventricle, and the pulmonary trunk collectively.

- The right atrium receives deoxygenated system blood from the superior and inferior vena cava.
- The blood is then pumped through the tricuspid valve into the right ventricle, which in turn pumps the blood through the pulmonary valve into the pulmonary artery.



Auricles:

- Auricles are situated at broader end while the ventricles are situated at the lower conical end. Internally, the four chambers are separated by septa and valves.
- Auricles are much smaller than the ventricles and are relatively thin walled.
- Right auricle receives the blood from the venae cavae (the great veins returning the deoxygenated blood from all parts of the body) and empties it into the right ventricle through an opening guarded by the right atrio-ventricular (tricuspid) valve.
- Thus, the right and left atria receive blood from different body parts.
- Auricles contract first and then the ventricles.
- Right atrium receives deoxygenated blood from all parts of the body, except the lungs, through the superior and inferior vena cava.
- Pulmonary veins bring oxygenated blood to the left atrium from the lungs.
- Left auricle receives the four pulmonary veins- from the lungs and empties into the left ventricle via the left atrio-ventricular (bicuspid or mitral) valve.
- In humans the heart pumps blood into arteries.
- Arteries divide and sub-divide into narrower ducts called capillaries.



- Capillaries join together forming small veins called venules and these leads into larger and larger veins which carry blood back to the heart.
- The aorta is the longest and largest artery in the body.
- In human adult at rest, the heart beats at the rate of about 72 times per minute.

Ventricles

- Left ventricle, (the larger of the two ventricles) forms the apex of the heart. From the right ventricle arises a pulmonary trunk, which bifurcates to form right and left pulmonary arteries which supply deoxygenated blood to the lungs of the respective side. From aorta, the oxygenated blood is supplied to the coronary arteries and the systemic circulation of the body.
- Number of valves present inside the chambers of the heart regulate the direction of flow of blood. All heart valves prevent the backflow of blood. Atrio-ventricular valve separates the atria and ventricles.
- In a normal resting man, the rate of heart beat is 72 beats per minute. One heart beat (cardiac cycle) lasts $60/72 = 0.8$ second.
- Sequence of events that occur during one heart beat is called cardiac cycle.
- Complete heart beat consists of both systoles and diastoles of atria and ventricles.
- The contraction phase is called systole and the relaxation phase is called diastole.
- Rythmic closure and opening of the valves causes the sound of the heart beat.
- The first sound (lub) is of longer duration (0.16 - 0.90 sec) and a louder one, is created by the closure of the atrio-ventricular valves immediately after the start of the ventricle systole.
- The second sound (dub) is of shorter duration (0.10 sec) and is created by the closure of the semi-lunar valve at the end of the ventricular systole.
- There is a gap or pause between this second sound and the first sound of the next cardiac cycle. This pause is nearly twice as long as the pause between first and second sound of one cycle.
- Sympathetic nerves are accelerator nerves.
They increases heart beat by secreting adrenaline (epinephrine), where as stimulation of para sympathetic nervous system (vagus supply) decrease the rate of heart beat by secreting acetylcholine.
- An excess of calcium ions increase heart rate.
- The heart beat is somewhat faster in females.

- The heart beat is fastest at birth, moderately fast in youth, average in adulthood and below average in old age.
- The heart murmur is due to defective valves.

Blood Pressure

- It is measured by sphygmomanometer.
- Blood pressure means the arterial blood pressure.
- Blood pressure is usually taken in the left brachial artery.
- Normal systolic BP in a healthy adult man is 120 mm Hg. Normal diastolic BP in healthy adult man is 80 mm Hg. Thus the average blood pressure is expressed as 120/80 mm Hg.
- Taller person has larger heart. Taller the person lower the blood pressure.
- Male have lower blood pressure than female.
- Body's entire supply of blood is 4.7 litre, is pumped through the body every minute. In one day heart pumps about 7600 litres of blood.
- Pulse pressure is the difference between systolic and diastolic pressure.

HEART DISEASES

Congenital Heart Diseases

- It is present right from birth due to defective formation of heart in mother womb

Rheumatic Heart Disease:

- It is common in children e.g. arthritis follows sore throat.

Coronary Heart Disease

- It is the most harm among the heart diseases.
- Higher age group people are more affected.
- Severe chest occurs, in this disease.

Electro-cardiogram

- It is the process of recording of the electrical changes that accompany the cardiac cycle.

Electro-cardiograph

- It is the instrument used to record potential differences of heart muscles.

Stethoscope

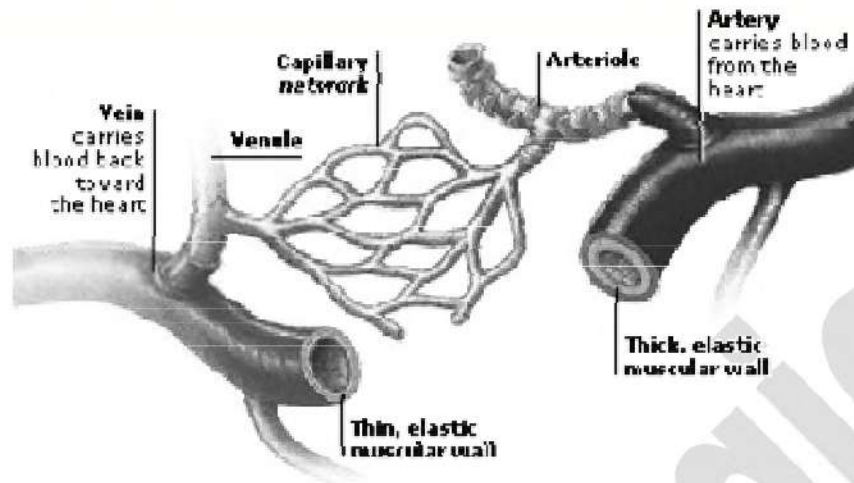
- It is the instrument used to hear heart sound. (PCG = Phono-cardiogram).

Electro-cardiogram

- Electro-cardiogram (ECG or EKG), is a graphic produced by an electro-cardiograph, which records the electrical activity of the heart over time.
- Understanding of the various waves and normal vectors of depolarization and depolarization yields important diagnostic information.



Blood Vessels



Arteries

- All arteries carry oxygenated blood away from heart except pulmonary artery which carries blood from right ventricle to lungs.
- Largest artery is aorta which carries blood from left ventricle to other parts of body.
- Arteries have capacity to contract and expand on their own. Therefore, they are also able to force the blood forward.
- Flow of blood in arteries is intermittent flow.
- Pulmonary artery & aorta are called great vessels.

Veins

- All veins except the pulmonary vein carry deoxygenated blood.
- All veins bring back blood to the heart.
- They cannot contract & expand on their own.
- Flow of blood is continuous in veins.
- They have thinner muscular wall of larger diameter.
- They are provided with valves to prevent backflow.
- Pulmonary vein carries oxygenated blood from lungs to left atrium.
- Largest vein is superior vena cava, which brings blood from head and arms.
- Inferior vena cava carries the blood from legs and trunk.

Capillaries

These are hair like thin blood vessels that connect the arteries and veins.

They are smallest blood vessels in the body.

A capillary has no muscular wall.



CHAPTER - 7

HUMAN GLANDULAR SYSTEM

Gland is an organ in animal's body which synthesises substances, such as hormones, which are either discharged into blood stream by endocrine glands or carried outside through duct by exocrine glands.

GLANDS

Glands are divided into two groups

(1) Endocrine Glands (or) Ductless Glands

Those glands which secrete their product directly onto a surface rather than via a duct
They produced hormones.

(2) Exocrine Glands (or) Ducted Glands

Are those glands which secrete their products via a duct. They produced enzymes

EXOCRINE GLANDS

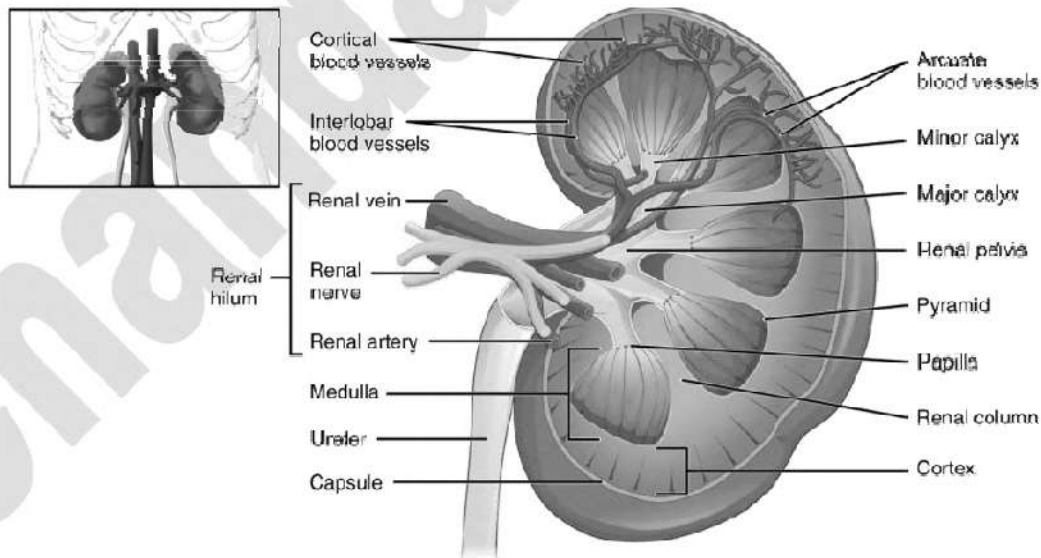
- Exocrine glands do not empty their secretions into blood.
- Their products are carried by ducts to the surface of the skin or other organs.
- They involve in large quantity of secretion
- Their Secretion is known as enzyme.
- Exocrine glands include sweat glands, mammary glands and many glands of the digestive system.

EXCRETORY SYSTEM

- The study of excretory system is Urology.

In plants all three waste products- water, oxygen and carbon dioxide leave the plant body by the process of gaseous diffusion through stomata or lenticels. Some plants produce insoluble solid waste products like tannin, alkaloids such as quinine or morphine. These waste products of human use are stored in leaves, fruits or bark cells and are removed when a leaf or fruit falls or when the bark peels off the plant.

Kidneys

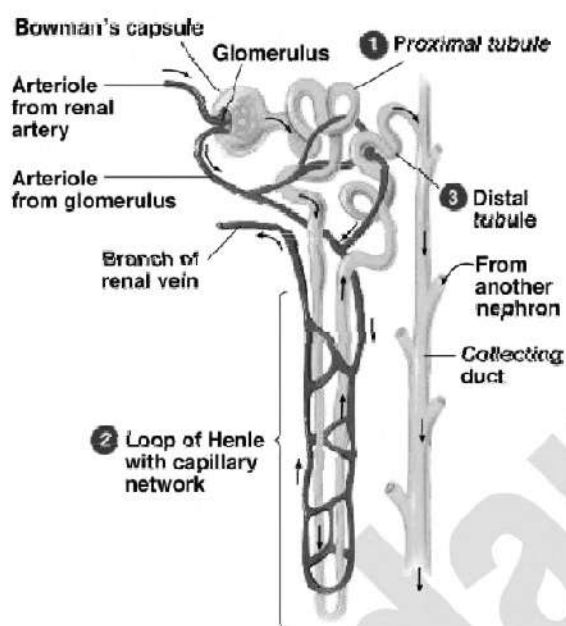


- The study of kidneys is called Nephrology.
- In human the main excretory organ is kidney.
- Kidney is a bean-shaped structure and measures about 10 cm in length, 5 cm min breadth and 3 cm in thickness. In adult, it weighs about 125-170gm.



- Normally every human being possesses two kidneys. However, one kidney can perform all functions.
- one located on each side against the dorsal inner surface of the lower back, at the level of 12th thoracic and 1st or 2nd lumbar vertebrae inside the abdominal cavity.
- Nephron The structural & functional unit of kidney is the nephron, an intricate epithelial tube about 3 cm long and 20-60 μ m in diameter. Each kidney contains about 1 to 2 million nephrons and each nephron can be divided into three main regions.

Proximal Nephron, Loop of Henle and Distal Nephron



- It secretes hormones called erythropoietin which stimulates production of RBCs and renin necessary for osmo-regulation.
- Kidneys remove a major part of waste product from body.
- Kidneys remove uric acid, urea, ammonia, NaCl, phosphates, hormones, etc. from blood.
- Kidneys helps in maintaining alkaline nature of blood by removing acidic products from blood.
- Kidney & sweat glands co-ordinate their activities.
- In summer sweat gland sweats more and role of kidneys is reduced. But in winters it is reversed.
- Diseases related to Kidney
- Besides kidney, there are some other organs which are excretory.
- Carbon dioxide is produced in the tissues through respiration and it is eliminated by lungs. The water is eliminated through the skin.

- The alimentary canal serves to remove certain products like bile pigments and bile products.
- A pair of kidney are situated on the dorsal wall of the peritoneal cavity outside the coelom.
- Each kidney is dark-red, bean-shaped, its outer side is convex and the inner concave side bears a notch, from where the ureter runs back.
- The right kidney is placed more anterior than the left kidney.
- Kidneys help to carry out 3 essential functions:
 1. The excretion of nitrogenous materials, excess water and undesirable salts.
 2. Osmo-regulation and
 3. Conservation of useful material such as glucose and mineral salts by selective reabsorption.
- Urine contains 95% water, 2% urea, 0.6% chlorides of N&K and some Uric acid.

Mucous Membrane

- It secretes sticky substance mucous which traps dust particles and pathogens.
- It also moistens the air which is breathed.

Human Urine

- It is a pale-yellow coloured, slightly acidic (average pH 6.0, range 4.5-8.2) watery fluid (specific gravity 1.015-1.025) with a characteristic smell.
- Human urine consists of:
 - Water (about 95%), Salts (about 2%) Urea (about 2.6%), Uric acid (about 0.3%) Traces of creatine, creatinine, ammonia etc.
- Due to presence urochrome urine appears yellow colour.
- Human adults produce around 1.5 litre of urine per day.

Kidney Stones:



- Stones are formed due to concentration of calcium Phosphate, calcium Oxylate, Uric acid crystals.



Occurrence of ketone bodies, glucose, albumin, blood cells, excess pigments, pus cells, calculi or casts (kidney stones) are some of the major abnormal constituents of urine.

- Abnormal Conditions of Urine

Hemodialysis

It is the process in which an artificial kidney is used to remove excess urea from a person suffering from abnormally high level of urea.

Animal	Excretory Organ
Amoeba	Contractile Vacuoles
Tape Worm	Flame Cells
Crabs	Green Glands
Birds, Fish, Reptiles, mammals	Kidneys

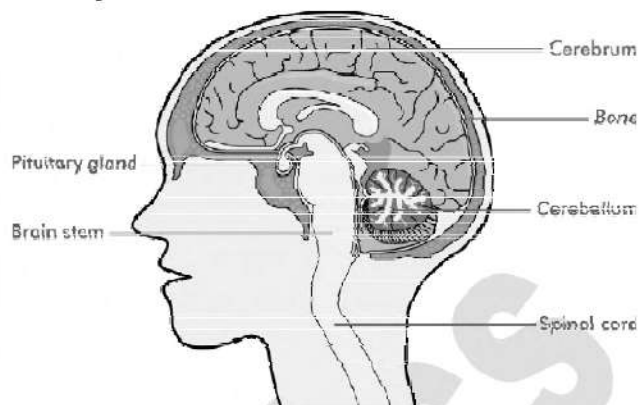
ENDOCRINE GLANDS

- Endocrine system is a group of ductless glands.
- It means that these glands pour their produce directly into the blood stream without it having to pass through a duct.
- Endocrine system achieves co-ordination and integration by transmitting information through chemical messenger's called hormones.
- Endocrine glands secretions are called as hormones.
- Hormones are known as Steroids.
- Hormones regulate metabolism, growth, secretion, digestion, excretion, reproduction. They are

synthesized in the body i.e directly discharged into blood.

- They do small quantity of secretion.
- They are easily soluble and diffusible.
- Many important functions of the endocrine system are under the control of nervous system. The endocrine and nervous system are often collectively, referred to as neuro-endocrine systems.
- Glands in the human beings are as follows,
 - 1) Pituitary Gland
 - 2) Thyroid Gland
 - 3) Para Thyroid Gland
 - 4) Adrenal Gland
 - 5) Ovary
 - 6) Pineal Glands
 - 7) Thymus
 - 8) Pancreas
 - 9) Testis

Pituitary Gland



- It is located at below and connects hypothalamus at the base of brain /skull.
- It is also known as Master Gland or king of the Glands.
- It produces 10 types of hormones.
- Hormones like Proteins (Peptides) / or Glycoproteins are secreted by Pituitary.
- It is slightly larger size in women than men.
- Pituitary gland has 2 parts-
- Anterior lobe - It produces a cluster of hormones.
- Posterior lobe - It stores 2 hormones secreted by hypothalamic neurons.

(1) Growth Hormone (GH) (or)

Somatotrophic hormone (STH):

- It acts directly on all tissues and influences the growth of long bones, muscles and viscera.
- Due to abnormal increase of growth hormone leads to Gigantism.
- Due to lower secretion suffer with dwarfism.

(2) Thyroid Stimulating Hormone (TSH)

- It influences the structure and secretory of the thyroid.

(3) Leutinising Hormone:

- It is responsible for secretion of Estrogen and Progesterone in females.

(4) Follicle Stimulating Hormone:

- It is a responsible for producing of Ova and sperm cells in male and female.

(5) Prolactin (Lactogenic- Hormone)

- After delivery this hormone is responsible for producing of the mother milk.

(6) Melanocyte Stimulating Hormone (M.S.H)

- This hormone is responsible for skin colour.

(7) Anti Diuretic Hormone (A.D.H):

- This hormone control B.P and Urine secretion.
- Due to deficiency of this hormone people suffer with diabetes, insipidus.



(8) Oxytocin (Birth Hormone)

- It is responsible for free delivery in females.

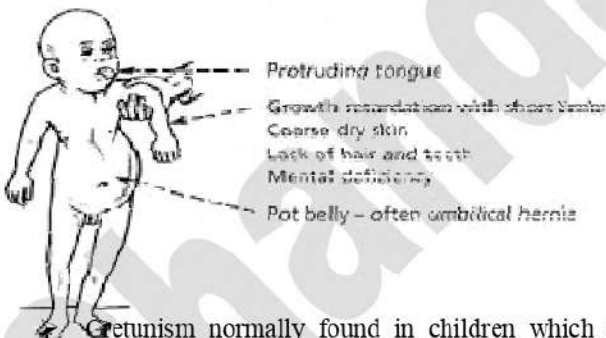
Pineal Gland

- Pineal gland is situated in mid brain. looks like pine cone.
- It secretes melatonin (hormone) which lightens the skin color in some animals.
- This is not important for human beings.
- More melatonin is produced during darkness.

Thyroid Gland

- It is largest Endocrine Gland.
- It is located at base of neck at lower extremity of larynx, in butterfly shape.
- Thyroid gland secretes two hormones (a) Triiodothyronine (T3) (b) Tetraiodothyronine (T4) - It is also known as Thyroxine which increases the rate of oxidation of glucose releasing energy, i.e. regulates basic metabolism.
- Thyroid gland regulates metabolism through stimulation of protein synthesis, regulate metabolic rate, influences heat production and helps in overcoming the effects of exposure to severe cold, essential for mental and physical development!
- It improves actions of neuro-transmitters adrenaline and non-adrenaline.
- It is only endocrine gland that stores secretory product in large quantity. Thyroid gland also secretes calcitonin which decreases the amount of calcium in blood.

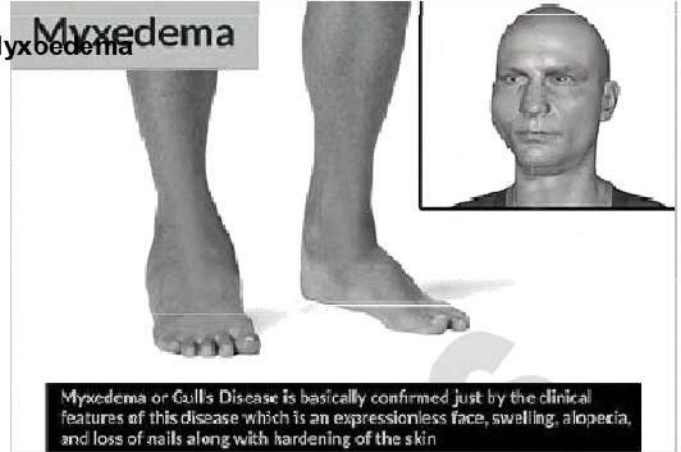
Thyroid Gland Disorder Cretinism:



Cretinism normally found in children which results due to hypothyroidism.

- It causes dwarfism because the skeleton fails to grow and mature.
- Patients are severely mentally retarded as the brain fails to develop fully.
- Other features include dry skin, thick tongue, prolonged neonatal jaundice, lethargy, respiratory problems and constipation.

Myxoedema



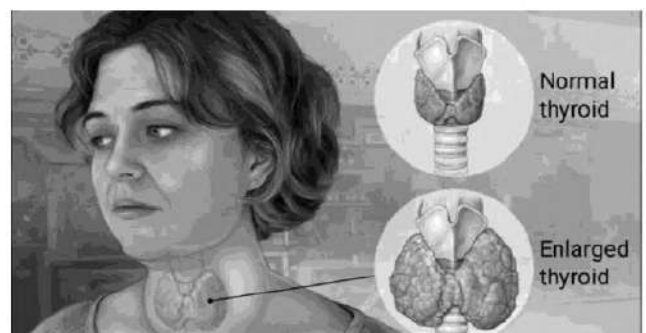
- Myxoedema normally found in adults which results due to hypothyroidism.
- The cardinal symptom of this disorder is edema (accumulation of interstitial fluid) that causes the facial tissues to swell and look fluffy.

Grave's Disease



- It is also known as exophthalmic goiter which results due to hyperthyroidism.
- It is an auto-immune disorder in which the person produces antibodies that mimic the action of TSH, but are not regulated by normal negative feedback controls.
- These patients have a peculiar edema behind the eyes called exophthalmos, which causes the eyes to protrude. This disorder, like myxoedema, also occurs more often in females.

Goitre





- It is also called as bronchocele, is a swelling in the neck just below Adam's apple or larynx due to an enlarged thyroid gland.
- They are in different ways:
 - A diffuse goitre is a goitre that has spread through all of the thyroid (and is contrasted with a "simple goitre", "single thyroid nodule" and "multinodular goitre").
 - Toxic goitre refers to goitre with hyper-thyroidism. These are derived from inflammation, neoplasm and some kinds of activating autoimmune disease (Grave's disease).
- Nontoxic goitre (associated with normal or low thyroid levels) refers to all other types (such as that caused by lithium or certain other autoimmune diseases).
- The most common cause for goitre is iodine deficiency (E0 1). This condition is commonly called endemic goitre.
- It is curable by mass food-supplementation with iodine (in the form of iodide or iodate).

Para Thyroids

- They are Four in number embedded within the substance of thyroid gland capsule.
- They are independent of Thyroid glands. They are 4 pea sized organs in human body.
- Parathyroids control the metabolism of Calcium.
- Parathyroids secrete Parathormone hormone. Parathormone raises calcium level and lowers the Phosphate level in the blood. Deficiency of Parathormone leads to Tetany (Restlessness).

Thymus Gland

- Thymus gland is located in upper anterior position of the chest cavity.
- It is also called the throne of immunity or training school of T-lymphocytes.
- It plays an important role in the development of the immune system, being primary site of the cell maturation.
- It produces lymphocytes which train WBCs in fighting against pathogen.
- Thymosin hormone is produced by thymus glands which has stimulating effect on the entire immune system.
- Decline and disappearance of thymus gland by late middle age may be the primary cause of ageing.

Adrenal Gland

- It is located at the top of each kidney, shaped like a tricorn hat.
- Adrenals also called as- Suprarenals and also 4S gland (because responsible for Sugar metabolism, Salt retaining, Sex hormone, Source of energy).
- It activates hypothalamus and also activates Pituitary gland. It activates us when there is a stress or emergency situation.
- Adrenal gland divided into-
 - **Outer Adrenal Cortex:** Bright yellow colour, derived from mesoderm. It secretes sex hormones of Male and Female
 - **Inner Adrenal Medulla:** Reddish brown colour derived from ectoderm. It contains chromaffin cells secrete 2 hormones like Adrenaline (Epinephrine) & Non-Adrenaline (Norepinephrine)
- **Glucocorticoids:** Regulate metabolism of Carbohydrates, Proteins and fats, help to overcome the stress associated with severe cold by elevating blood glucose due to conversion of Protein to glucose.

Pancreas

- It is positioned within the curve of Duodenum.
- It is a mixed gland (Heterocrine gland). consists of
- Exocrine portion - consists pancreatic acid (maximum portion of gland).
- Endocrine portion - consists clusters of cells known as Islets of Langerhans. They secrete hormones which regulate glucose level in blood.
- Islets lie scattered between the pancreatic acini.
- There are 3 types of hormone secreting cells which regulate the level of glucose in blood. They are,
 - α Cells - secrete hormone glucagons
 - β Cells - secrete Insulin
 - δ Cells - secrete somatostatin.
- **Insulin:** When glucose levels in blood exceeds Insulin acts on liver, **Muscles & Adipose cells**. Insulin directs the liver to take glucose & let it convert into Glycogen and fat. Somatostatin acts as paracrine to inhibit the secretion of **glucagons and Insulin**.
- Hence, Insulin Lower the blood glucose concentration where as Glucagons Elevates the blood glucose concentration.

Diabetes Mellitus

- It is a metabolic disorder characterized by high glucose blood sugar (Hyperglycaemia), among other signs.
- According to World Health Organization (WTO), there are 3 types of diabetes:



- Type 1, Type 2, and Type 3 (gestational diabetes).
- Type 3, occurring during pregnancy.
- Although these share signs and symptoms but have different causes and population distributions. They are not a single disease or condition.
- Type 1 is generally due to auto-immune destruction of the insulin producing cells - pancreatic beta cells.
- Type 2 is characterized by tissue wide insulin resistance and varies widely. Gestational diabetes is due to a poorly understood interaction between fetal needs and maternal metabolic controls. Type 2 sometimes progresses to loss of beta cell function as well.
- The term diabetes usually refers to diabetes mellitus, there are several other rarer conditions also named diabetes. The most common of these is diabetes insipidus (unquenchable diabetes) in which the urine is not sweet. It can be caused by either kidney (nephrogenic DI) or pituitary gland (central DI) damage.
- It is a group of disorders that lead to an increase in the level of glucose in the blood (hyperglycaemia). This is due to underactivity of β cells in islets of Langerhans. This condition is known as insulin-dependent diabetes.
- Diabetes Mellitus: results from hyposecretion of insulin
- Diabetes Insipidus: results from hyposecretion of antidiuretic hormone.

Ovaries

- Ovaries are egg-producing reproductive organs found in female organisms. They are part of the vertebrate female reproductive system.
- The term gonads refers to the ovaries in females and testes in males.
- Normally, a female will have two ovaries, each performing two major functions viz. producing eggs (exocrine function) and secreting hormones (endocrine function).
- Ovary secretes estrogens which stimulate growth of endometrium, sexual behavior and regulates menstruation.
- It secretes progesterone which stimulates maintenance of uterus and regulates menstruations.
- Menstruation is the periodic shedding of the endometrium of the uterus in some mammals (primates) with bleeding.

- Ovulation occurs usually midway (14th day) during menstrual cycle.
- Sex hormones in female are Estrogen and Progesterone.
- **Estrogen:** It controls the development of secondary sexual characteristics of the female, enlargement of uterus, menstruation.
- Progesterone is the hormone responsible for growth and maintenance of foetus.
- **Corpus luteum:** Progesterone prepares the inner lining of uterus for implementation of young embryo, also maintains it during pregnancy.
- Menopause (climateric) is the period of life when menstruation naturally stops. It usually occurs between the ages of 45 and 50.
- Hormones taken in the form of drugs for contraception are estrogen and progesterone.
- Ovaries in females are homologous to testes in males.

Testes

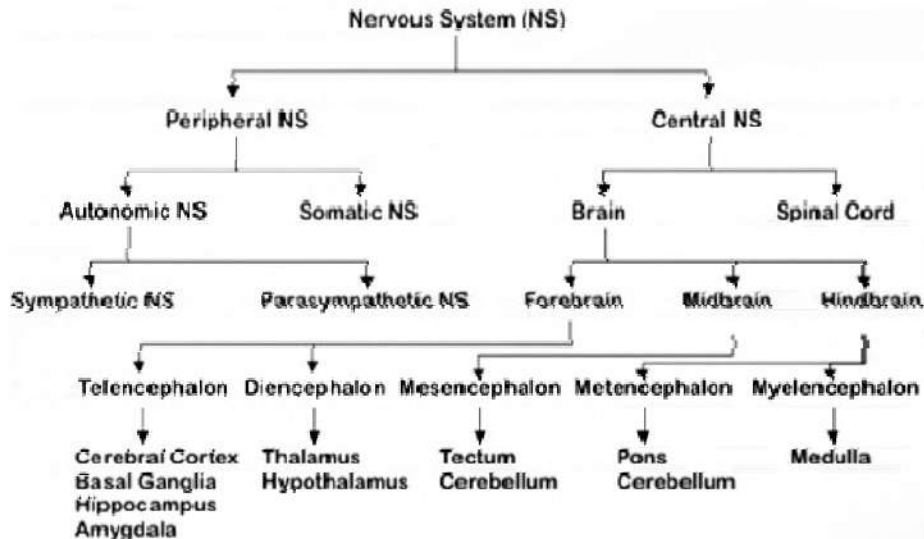
- The testicles, or testes (singular testis), are the male generative glands in animals.
- Interstitial/Leydig cells located in narrow, angular spaces between seminiferous tubules.
- Sex hormones in male are Androgen, Testosterone, Dihydrotestosterone and Androstenedione.
- **Testosterone:** Controls the development of secondary sexual Androgen, Testosterone, Dihydrotestosterone and Androstenedione. characteristics of the male. Sex hormones regulate growth and development, of accessory sex organs, secondary sexual characters and sexual behaviors. The principal male hormone is testosterone. It is responsible for puberty in man.
- **Testosterone:** It is responsible for puberty in man.



CHAPTER-8

HUMAN NERVOUS SYSTEM

Nervous System:



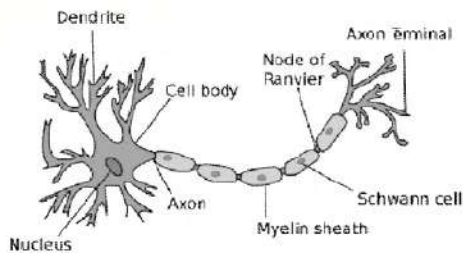
- The study of Nervous system is Neurology.
- Nervous system controls all other systems of the body and command to do certain action at right time. This system is a complex network of neuron cells.
- Neurons may be of 3 types-
 - **Sensory Neurons:** Relay information from body organs to brain.
 - **Motor Neurons:** Carry impulses to brain from body organs.
 - **Association Neurons:** Transmit impulses between sensory and motor neurons

CENTRAL NERVOUS SYSTEM

- The human nervous system consists of two contrasting functional subsystems, the central nervous system and the peripheral nervous system.
- It consume 20% of oxygen inhaled and 20% of glucose intake.
- All nerves are formed before birth and grow up to the age of 5 years and stop by 19-20 years.
- Central nervous system consists of brain and spinal cord, covered by thin membrane meninges which protect them from infection. If it is infected by anyway causes meningitis.
- Peripheral Nervous System includes all the nerve pathways of the body outside the brain and spinal cord.
- These pathways are divided into two:
 - **Sensory pathways:** It transmits information to the central nervous system.
 - **Motor pathways:** It transmits commands from the central nervous system.
- 1) Somatic (voluntary) nervous system, which relay commands to skeletal muscles, and
- 2) Autonomic (involuntary) Nervous System which stimulates the glands and other muscles of the body.
- Along with that, there is the neuro endocrine system, which is a network of endocrine glands whose hormone production is controlled by commands from the central nervous system.



Neuron



- Neuron is the structural and functional unit of nervous system.
- Neuron is the largest cell found in the body.
- A neuron has three components - cell body, dendrites and axon.
- The functional junction between neurons is called synapse.
- Largest number of cell bodies of neurons in our body is found in brain.
- All functions of the body are regulated, integrated and co-ordinated by the neuro endocrine system.
- Nervous system (brain, spinal cord and nerves) develops from ectoderm.
- Information passing through the neurons are in the form of chemical and electrical signals called as nerve impulse.

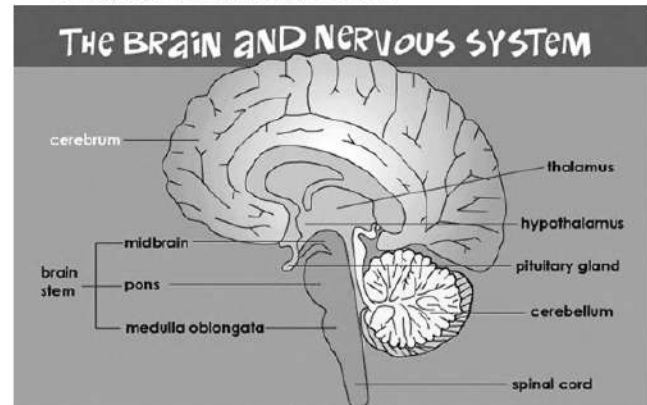
Animals

- Nervous system and endocrine system are two integrative (communication) systems in the animal body.
- Nervous system is found only in animals and absent in plants.

Brain

- The study of brain is called Phrenology.
- Central nervous system consists of brain and spinal cord.
- Brain lies in the cranium of skull.
- Human brain has wrinkled surface and appears as a large, soggy, pinkish grey walnut.
- Human brain is composed of about 30 billion neurons.
- It weighs about 1350grms and forms about 98% of total weight of Central nervous system.
- Adult brain, on average, weighs two per cent of their body weight, growing from a quarter of its size at the time of birth.
- The brain reaches between 75 and 80% of adult size within the first two years, and its full size at the age of 6 years.
- It receives about 15 - 17% of cardiac output.
- It also receives about 20% of the oxygen consumed by the body.

- Roger Sperry (U.S scientist) got Nobel Prize in 1981 for his work on split brain theory.
- The left half of the brain controls the right side of body and vice versa.
- The brain is broadly divided into three regions viz. fore brain, mid brain and hind brain.



- Fore Brain:** It consists, Cerebrum, Diencephalon and Olfactory lobes.
- Mid Brain:** It consists Optic lobes and Crura cerebri, and
- Hind Brain:** It consists Cerebellum and Medulla Oblongata.

Fore Brain

Cerebrum

- It is the largest part of brain which is responsible for higher intellectual activities.
- And also it is the most complex and specialised part of the brain.
- In the true sense it is the seat of mind.
- Cerebrum has 2 outer layer - Cerebral Cortex, Cerebellum.
- Cerebral Cortex which controls sensations like vision, touch, hearing, taste, smell, control of voluntary movements, reasoning, emotion and memory. It is roughly 80% of total brain mass.

Cerebellum;

- It is the small brain or hind brain which maintains the equilibrium of the body and muscles co-ordination during activities like walking, running, cycling.
- In the cerebrum, specific regions for each kind of stimulus and its response are located. (like Occipital lobe is the region for visual reception, Temporal lobe for auditory reception, parietal lobe for touch, smell, temperature and conscious association and frontal lobe for muscular activities).
- Olfactory lobes are well developed in dog. So power of smell is more in dog.

**Diencephalon (hypothalamus)**

- Hypothalamus contains higher nerve centres for temperature
- regulation, hunger, thirst and emotional reactions.
- It secretes neurohormones which control the secretions of anterior pituitary hormones.
- Hypothalamus synthesizes the posterior pituitary hormones and control their release into the blood.

Thalamus

- This small organ is often referred to as the relay station of the human brain because most sensory signals have to go through the thalamus in order to get to the brain for processing

Mid brain

- Midbrain controls the muscle tone and modifies some motor activities initiated by the cortex.

Hind Brain

- It consists of three parts-Cerebellum, Medulla Oblongata, Pons.

Cerebellum

- Cerebellum contains centres for the maintenance of posture and equilibrium of the body and for the muscle tone.
- Coordination and adjustment of movement and posture are under the control of cerebellum.
- Damage to cerebellum is characterized by ataxia. Ataxia means lack of muscle coordination.

Medulla Oblongata

- Medulla oblongata is the regulating centre for swallowing, coughing, sneezing and vomiting.

Pons

- Pons takes part in regulation of respiration.
- The brain stem consists of pons, varolii and medulla oblongata, the latter continuing into the spinal cord.

Spinal Cord

- It is a tubular structure, downward continuation of brain and is housed in the neural canal within the vertebral column.
- It is responsible for simple reflex and connects brain with rest of the body.
- Spinal cord is a cylindrical structure. It begins in continuation with the medulla oblongata and extends downwards, enclosed within vertebral column.
- It is also surrounded by meninges. A total of 31 pairs of spinal nerves arise from the spinal cord.
- Medulla oblongata continues behind as spinal cord. Spinal cord is located in the neural canal of vertebral column.

- In spinal cord, white matter is found outside the grey matter (reverse in brain).
- The cavity of spinal cord is known as central canal.

Meninges

- Brain and spinal cord are surrounded by two meninges-Duramater and Piamater in frog and three meninges-Duramater Arachnoid and Piamater) in mammals.
- Duramater is the outermost, thick and tough non-vascular meninx applied to the inner surface of bone (no epidural space in mammals).
- **Piamater** is the thinnest of all the meninges which is highly vascular and closely adhered to the surface of the brain.

Cerebrospinal Fluid

- Fluid present around brain and spinal cord is known as cerebrospinal fluid (CSF) which is formed by choroid plexuses.
- The main function of cerebrospinal fluid is to absorb any shock or jerk to brain and spinal cord.
- It also helps in the nutrition and excretion of neurons in Central Nervous system.
- An average of some 150cc of cerebrospinal fluid is usually present in the adult nervous system of man.

BRAIN DISEASES**(1) Meningitis**

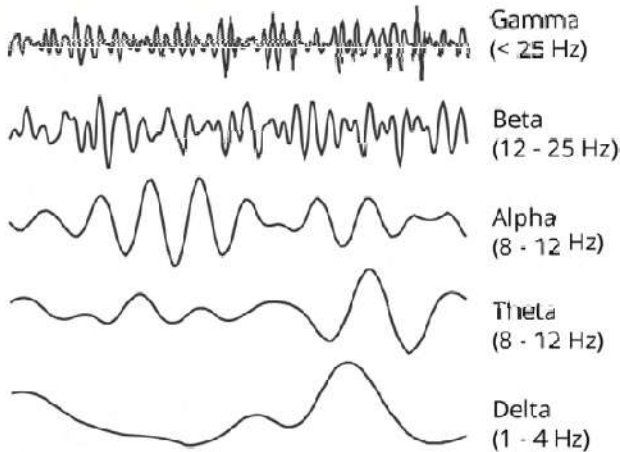
- It refers to infection or inflammation of meninges resulting in severe headache, vomiting and pain and stiffness of the neck.
- The most common cause of meningitis is the infection of bacteria viz. streptococcus pneumoniae, neisseria meningitis and haemophilus influenzae.

(2) Epilepsy**(3) Paralysis****(4) Brain Tumor (Cancer)****(5) Migraine****(6) Encephalitis****(7) Alzheimer's****(8) Parkinson's**

- Parkinson's disease or paralysis agitans is a defect of brain. Parkinson's disease is characterised by tremors and progressive rigidity of limbs caused by a degeneration of brain neurons and a neurotransmitter called dopamine.



Brain Tests



(1) Electro-Encephalo Gram (EEG)

- The instrument, Electro-encephalograph or Cathode Ray Oscilloscope is used to record the electrical activity of the brain and cerebral cortex.
- Electro-encephalic record in normal human subjects consists of four different types of rhythmic waves namely, Alpha, Beta, Theta and Delta waves.

Alpha Waves: Alpha waves are most prominent when the brain is under quiet rest and they disappear entirely during sleep (10-12 cycles/sec.).

Beta Waves: Beta waves appear during tension or during intense activities of Central Nervous System (15-60 cycles/sec.).

Theta Waves: Theta waves are prominent in children between ages 2 to 5 years (5-8 cycles/sec.).

Delta Waves: Delta waves rarely found in normal adults during waking periods, but usually found during deep sleep. Delta waves in EEG also indicate brain damage (1-5 cycles/sec.).

(2) Magneto Encephalo Gram (M.E.G)

(3) Magnetic Resonance Imaging (M.R.I)

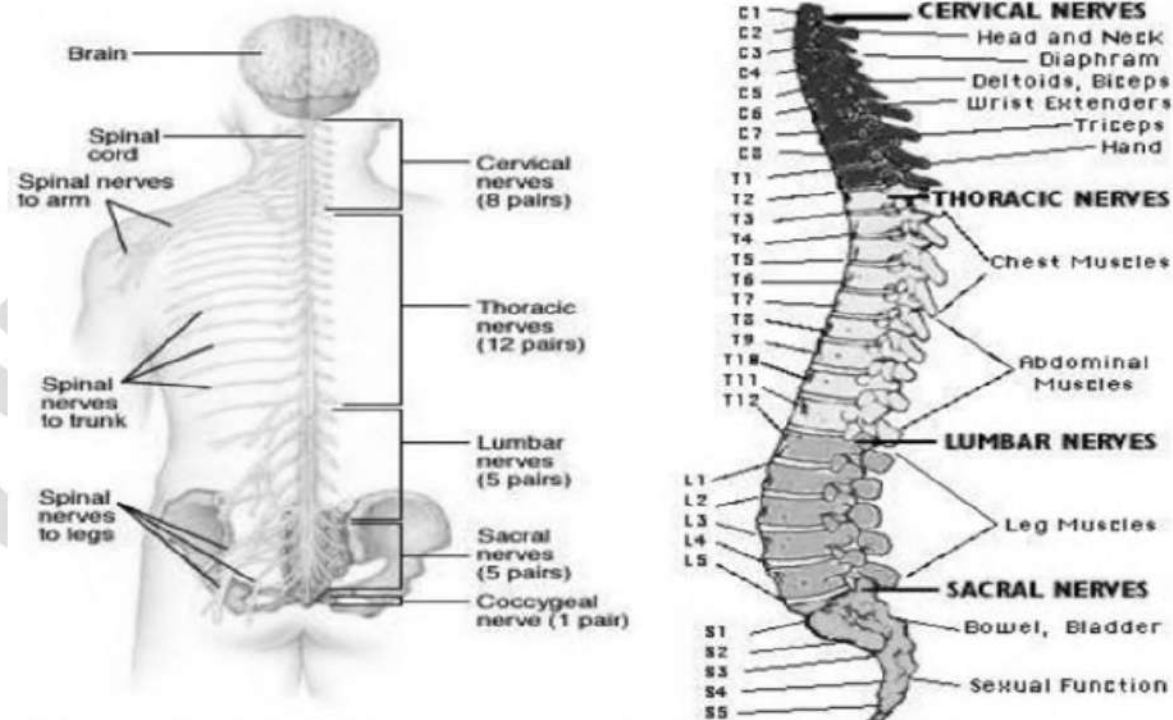
(4) Computed Tomography (C.T.Scan)

NARCO TEST

- It is a process to take out real information from the brain
- Chemicals are used in Narco Test.
 - Sodium Amatoil
 - Sodium Pentothal

PERIPHERAL NERVOUS SYSTEM

The Spinal Nerves



- It is the network of Cranial nerves which emerge from brain and spinal nerve which emerges from spinal cords.



Cranial Nerves

- In human 12 pairs of cranial nerves are present. Smallest cranial nerve is Trochlear, largest cranial nerve is Trigeminal.
- Vagus supplies the regions other than head.
- The sensory cranial nerves are:
 - Olfactory nerves: Responsible for smell
 - Optic nerves: Responsible for vision
 - Auditory nerves: Responsible for hearing and equilibrium

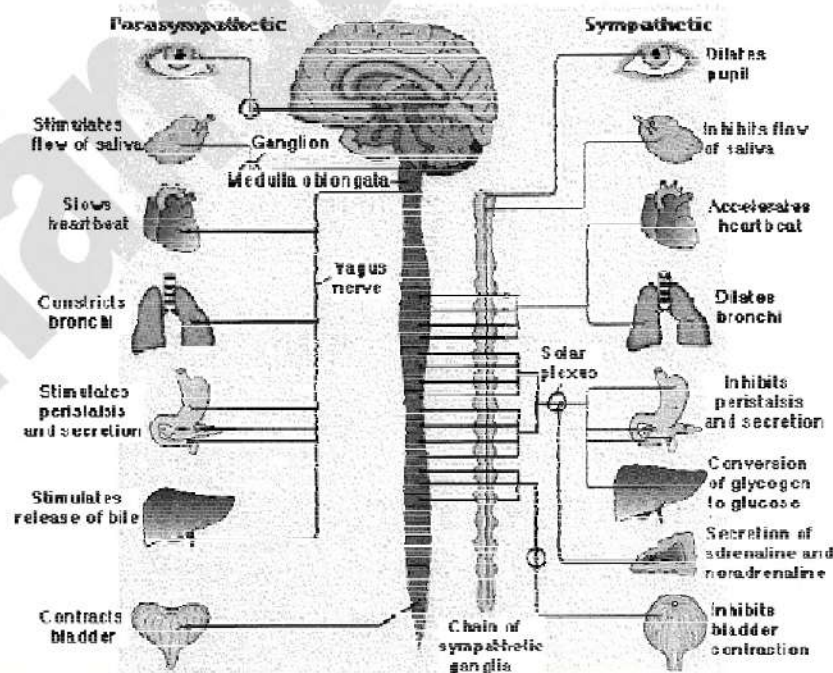
Spinal Nerves

- They are mixed nerves. Each spinal nerve joins the spinal cord at two points called roots and contain both receptor neuron and effector neuron.
- Spinal nerves have 2 roots- Dorsal root (is sensory) and ventral root (is motor).
- 31 pairs of spinal nerves are present in a man.

AUTONOMIC NERVOUS SYSTEM

- Autonomic Nervous System lies outside the central nervous system and works independently of the central nervous system.
- It controls the survival mechanism of the body and all involuntary actions like heart beat, digestion etc. It is usually operates without conscious control.
- It automatically regulates the activities of smooth muscles, cardiac muscles and glands. It is entirely motor.

- All autonomic axons are efferent fibres.
- Autonomic Nervous System consists of two parts- Sympathetic, Parasympathetic.
- Sympathetic preganglionic fibres in man occur only in his thoracic.
- Sympathetic nervous system increases defence system of body against adverse conditions. It is active in stress condition, pain fear and anger.
- Sympathetic nervous system is an accelerator since it releases adrenaline.
- Parasympathetic nervous system provides relaxation, comfort, pleasure at the time of rest. It helps in the restoration and conservation of energy.
- **Parasympathetic Nervous System** is an inhibitor since it releases acetylcholine. Autonomic Nervous System regulates and co-ordinates involuntary activities like heart beating, homeostasis, body temperature, breathing, buteristalsis and secretion of glands. Autonomic Nervous System cannot regulate learning and memory.
- Horner's syndrome results from the damage of sympathetic trunk of one side.
- A patient of Homer's syndrome exhibits lack of sweating on affected side, sunken eyes and constricted pupil.





CHAPTER -9

HUMAN SENSORY SYSTEM

SKIN (Sense of Touch)



- The study of skin is called dermatology.
- Skin is the largest organ of body. It regulates body temperature and acts as the organ of touch. It protects the inner organs from infection & injuries.
- The integumentary system is the largest organ system, comprising skin, hair, feathers, scales, nails, sweat glands and their products (sweat and mucus).
- Skin is known as the largest organ of the human body.
- Synthesis of vitamin D takes place in skin.
- The cutaneous membrane (skin) and its accessory structures (hair, scales, feathers, nails, exocrine glands) make up the integumentary system.
- In humans, the integument or the skin forms the external covering of the body. It is continuous with the mucous- membrane lining the mouth, rectum, nostrils, eyelids, and the opening of urinogenital ducts.
- Colour of skin is mainly due to the pigment melanin.
- Colouration in amphibian and reptiles is due to presence of chromatophores.
- In some animals like whales and other fishes and amphibians, skin folds and flaps helps in movements and locomotion.
- Some amphibians and reptiles can change the intensity of skin colour by contraction and relaxation of pigment cells. This ability is called as 'metachrosis'.
- Skin plays a major role in the regulation of temperature in cold-blooded animals (poikilotherms) like fishes, amphibians and reptiles.

- In warm-blooded animals (homiotherms) like birds and mammals, there is a temperature controlling mechanism. To keep themselves active, warm-blooded animals have to produce and conserve heat.
- Skin plays an important role in heat preservation peripheral vasoconstriction (closure of blood vessels), dilation of cutaneous blood vessels (vasodilation).
- The average body temperature of man is 98.6°F
- An important function of subcutaneous fat in humans is to provide a reserve food depot.
- Skin is an organ of offence and defence in some animals as a reaction to acute danger, cold, fear, etc.
- In some animals skin derivatives are modified as locomotory organs e.g. hoofs of horse.
- There are three layers of skin namely epidermis, dermis and subcutaneous tissue.

Cutaneous Glands Include:

- Sudoriferous glands - sweat glands
- Sebaceous glands - oil producing glands
- Ceruminous glands - glands of the ear canal that produce cerumen (earwax)
- Skin has pigmentation or melanin, provided by melanocytes, which absorb some of the potentially dangerous ultraviolet radiation in sunlight.
- Skin plays the most important role in protecting against pathogens, insulation and temperature regulation, sensation and vitamin D and B synthesis.

Three Primary Layers of Skin

Epidermis - Epidermis is the outermost layer of the skin. It forms the waterproof, protective wrap over the body's surface and is made up of stratified squamous epithelium with an underlying basal lamina.

- It consists no blood vessels, and is nourished by diffusion from the dermis. The main type of cells which make up the epidermis are keratinocytes, melanocytes, Langerhans cells and Merckels cells.
- Epidermis is divided into five sub layers.
- Stratum corneum
- Stratum lucidum
- Stratum granulosum
- Stratum spinosum (also known as-Prickle-cell layer)



- Stratum germinativum (also known as- Stratum Basale)
- These five layers of epidermis are found only in the skin of palm and soles.
- Only four layers are found in the skin in the rest of the body.
- Spinosum is the fifth layer which is absent in rest of the body.

Cornified Layer: Have dead cells to cut the loss of water from evaporation & to prevent from microbes.

Malpighian Layer: Colour, contain malpighian cells which secrete 'Melanin' which protect the inner cells from ultra-violet radiations when more melanin is secreted.

- The skin colour becomes 'tan'. White-man have more possibility of skin cancer.

Skin cancer may be two types:

- Localised Skin Cancer- can be treated, not -fatal, less harmful
- Melanoma- most serious, known as skin cancer, fatal, pigment cell are affected, most common in Southern Hemisphere due to erosion of ozone layer.

Dermis (Inner layer)

- It is more thicker, true skin regulates the body temperature through capillary vessels & sweatglands, have millions of capillary blood vessels, sweatglands, nerve endings, sebaceous glands and hair follicles.
- It is the layer of skin beneath the epidermis that consists of connective tissue and cushions the body from stress and strain.
- Dermis is tightly connected to the epidermis by a basement membrane. It also harbors many nerve endings that provide the sense of touch and heat.
- If temperature increase the capillary vessels expands & sweat glands evaporation; air conditioner of the body; sweat glands also remove the waste products; sebaceous glands-kills bacteria & keeps the skin soft & the body hair lustrous; found at the root of the hair follicle.
- Dermis of large-sized mammals yields leather after tanning.

Hypodermis

- It is also called as hypoderm, subcutaneous tissue or superficial fascia.
- It is the lowermost layer of the integumentary system in vertebrates.
- Cells that are found in the hypodermis are fibroblasts, adipose cells, and macrophages.

- It is derived from the mesoderm, but unlike the dermis, it is not derived from the dermatome region of the mesoderm.

- In arthropods, the hypodermis is an epidermal layer of cells that secretes the chitinous cuticle.

The term also refers to a layer of cells lying immediately below the epidermis of plants.

- This cell is also used to supervise the outer skin from burns and other such articulates. It is concerned with storage of fat.
- In marine animals like whales, seals and sea cows and also in elephants this fatty layer is thick and called blubber.
- Skin of humans does not yield leather due to thin dermis.

SOFT DERIVATIVES OF SKIN

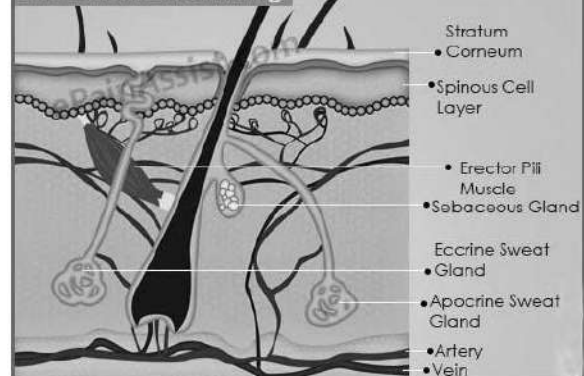
Consist of cutaneous glands. They are derived from stratum germinativum of epidermis.

Types of cutaneous glands - Sweat glands, mammary glands, Sebaceous glands,

Meibomian glands, Zeis glands, Ceruminous glands, penneal glands, Lacrimal glands.

Sweat Glands

Function of Sweat Glands and Benefits of Sweating



- Sweat is like a diluted urine.
- Sweat are simple, coiled, tubular glands which are primarily concerned with regulation of the body temperature.
- These glands maintain homeostasis of water and salts.
- The sweat glands are controlled by sympathetic cholinergic nerves which are controlled by a centre in the hypothalamus.
- Sweat glands are more numerous on palms, soles, forehead and axilla (arm pit).



Sweat glands in Human

- In man, sweat glands are absent in the skin of lips and nail-beds.
- Sweat is an aqueous fluid containing 95% water, 5% of other constituents like chlorides, phosphates, ammonia, urea and uric acid.
- In humans, there are two kinds of sweat glands which differ greatly in both the composition of the sweat and its purpose viz. Eccrine sweat glands and Apocrine sweat glands.

Exocrine Sweat Glands

- They are distributed over the entire body surface but are particularly abundant on the palms of hands, soles of feet, and on the forehead.
- These produce sweat that is composed chiefly of water with various salts.
- These glands are used for body temperature regulation.

Apocrine Sweat Glands

They develop around the age of 15 and release more than normal amounts of sweat for approximately a month and subsequently regulate and release.

Normal amounts of sweat after a certain period of time.

These glands are mainly present in the armpits and around the genital area and their activity is the main cause of sweat odour, due to the bacteria that break down the organic compounds in the sweat from these glands. Apocrine sweat glands essentially serve as scent glands.

Mammary Glands

- Mammary glands exist in both sexes, but they are rudimentary until puberty when in response to ovarian hormones, they begin to develop in the female.
- Development of mammary glands is controlled by hormones. The Estrogen promotes formation, while testosterone inhibits it.
- Almost all instances of breast cancer originate in the lobules or ducts of the mammary glands. These are modified sebaceous glands.
- The ducts of mammary glands open at nipples and teats (nipples are absent in prototheria).

Sebaceous Glands

- Sebaceous glands are simple, branched, alveolar (acinar) glands.
- They secrete oil or sebum which keeps the hair and skin smooth, soft, supple and water proof.
- Sebum is odorless, but its bacterial breakdown can produce odors. Sebaceous glands can usually be found in hair-covered areas where they are connected to hair follicles to deposit sebum on the hairs, and bring it to the skin surface along the hair shaft.

- They are also found in non haired areas of lips, eyelids, penis, labia minora and nipples. Here the sebum reaches the surface through ducts.

Meibomian Glands

- Also known as Tarsal glands.
- They are modified sebaceous glands on the edges of eyelids.
- They keep a thin film of oil over the cornea.

Zeis Glands

- Zeis glands are sebaceous glands associated with follicles of eye lashes.
- Their secretion keeps eye lashes smooth and oily.

Ceruminous Glands

- They are present in the external ear canal.
- They secrete earwax or cerumen which prevents the dust particles from entering the ear passage.

Penneal Glands

- Musk glands are found in certain deers producing attractive smell.
- Scent glands help in the recognition of the members of the same species.

Lacrimal (Lachrymal) Glands

- They are located at the superior lateral portion of each eye orbit.
- They secrete tears.

HARD DERIVATIVES OF SKIN

Consist of Hairs, scales, claws, nails, hooves and horns which are made up of keratin. Two types of keratin are α-keratin and β-keratin.

- α-keratin contains many molecules of the sulphur containing amino acids, cysteine and have abundant sulphur cross-linkages between adjacent polypeptide chains.
- α-keratin are the major constituents of skin horn and nail, wool and hair of mammals.
- β-keratin are found in the skins and scales of birds and reptiles.
- A tiny involuntary smooth muscle called erector Pill or elevator of the hair operates the hair.

EYE (Vision)

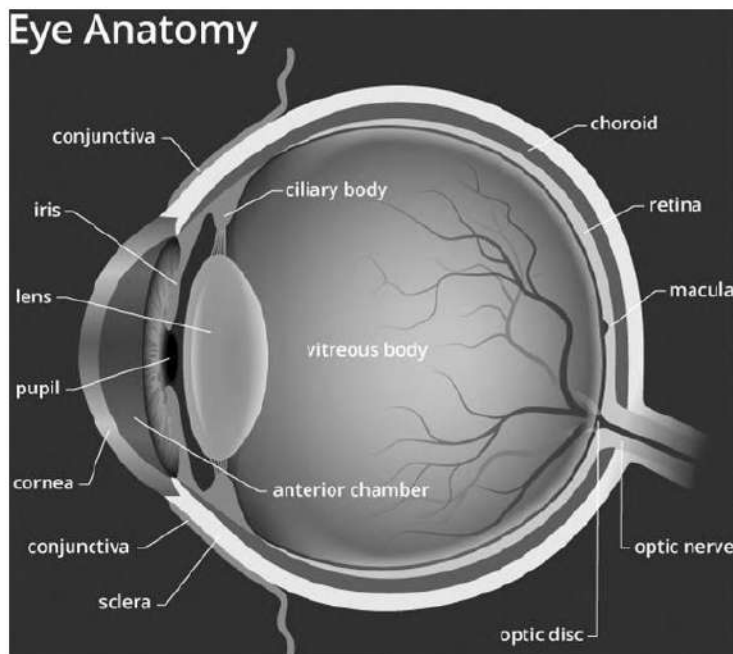
- The study of eye is called **Ophthalmology**.
- With two eyes 1800 objects can be seen with single eye 1500 objects can be seen.
- Eye enhance the functioning of other sense organs.
- Eye is the most specialised and most sensitive organ.
- Simple eye (only one lens) not compound eye.
- The normal eye is known as emmetropic.





- Eyes are the sensitive detectors of light, hence called photo-receptors.
- Each eye weighs roughly 7 grams.
- Eyeball is made up basically of three layers

- Sclera (outermost), Uvea (choroid) and Retina
- Sclera and uvea are Mesodermal and rest of the eye is Ectodermal (Conjunctiva, lens and retina are ectodermal).



Cornea

- Cornea is in curved shape. It is called as window of eye.
- It is transparent and light enters through it.
- Refraction of the light take place here (convergence).
- Cornea has no blood vessels.
- Cornea takes O₂ from atmosphere.
- Behind the cornea we have humour.
- Loss of blindness due to opacity of Cornea.
- Cornea is transplantable, must be removed from within some hours after death.

Sclera

- Fibrous tissues, which helps in maintaining the shape of eye-ball.

Retina

- It is the screen of eye (like film of camera) light sensitive thin and has rods and cones.
- It is the third and inner coat of the eye.
- The primary function of retina is image formation.
- It consists of a nervous tissue layer and a pigmented layer.
- Retina is composed of several layers of cells, each containing a characteristic type of cell. First, there is the photoreceptor layer containing the photosensitive cells, the rods, and cones, partially embedded in the

microvilli of pigmented epithelium cells of the choroids.

- The image falling on retina is real and inverted and reversal.
- If we accidentally focus on intense light sources such as sun, it will damage Macula Lutea. Macula Lutea (Yellow spot) is a tiny circular area, about 6 mm in diameter in the retina where the vision is sharpest.

Rods

- Rods are numerous in number, responsible for vision dim light.
- Rod cells are more sensitive towards light as compared to cone cells.
- Rods are used for vision in dim light (scotopic vision), having no ability to detect colour whereas, cones are used for bright light vision (photopic vision), with the ability to make coloured image of the object.
- The photosensitive chemical substance or the pigment in rods, is called rhodopsin and those in cones are called cone pigments, specific for the basic colours blue, green and red. Thus, cones are of three different types, short wavelength-sensitive (blue) cones, medium wavelength-sensitive (green) cones, long wavelength-sensitive (red) cones.
- These are well developed in dogs and cats.



**Cones**

- Cones are less in number, vision in bright light and colour vision, sensitive to blue, green and red colours.

Choroid

- Choroid is composed of loose connective tissue with abundant blood vessels.
- Choroid is also called the pigmented layer of eyes.
- It absorbs light rays and not reflected back out of the eyeball.

Eyelid

- It protects from dust & injury, by blinking action.
- It helps to maintain the eye moist.
- Tear glands-contains sodium bi-carbonate & sodium chloride washes away & moist the eye.
- Eyelids have enzyme called 'lysozyme'.

Canal of Schlemm

- Through canal of schlemm excess tear is drained into the nose.
- Conjunctiva-covers' inner part of eye-lid & visible part of eye-ball.
- It fight infection & keep the eye healthy conjunctivity.

Aqueous Humour

- Liquid & transparent, helps in maintaining shape of eye.

Iris

- It is comparable to shutter of comma, controlled through pupil also contains pigments cells and decide the colour of eye.
- Iris is the third portion of the vascular tunic.
- The function of iris is to alter the size of pupil.

Pupils

- Pupil is like aperture of camera; opening varies through light. Black, brown, grey eye have pigments respectively. Blue colour absence of pigment.
- Pupil is the black hole in the centre of the iris.
- Pupil is the area through which light enters the eyeball.
- Atropine is a chemical used by doctors to dilate the pupil.

Lens

- Lens cristaline biconvex transparent, no blood converses the light rays before retina, image is real inverted and diminished so brain have straight image.

Ciliary Body/Muscles

- Helps the eye lens to change its shape thin or thicker (accommodation of eye).
- Ciliary muscle alters the shape of the lens for near or far vision.

- The focal length of the eye lens is controlled by contraction and relaxation of ciliary muscles.
- Focusing of image is brought about by altering lens curvature.

Vitreous Humour

- Transparent, help in maintain the shape of inner eye, supplies the nutrients to eye lens.
- Human eye is sensitive to electromagnetic spectrum between 4000A - 7000A wavelength.
- Eyes are most sensitive to yellow-green colour having wave length approximately 5550A.
- Stereoscopic vision (3-dimensional) is found in human beings.
- Only human beings, apes, monkeys, birds, lizards, turtles and some fishes possess colour vision. But most domestic mammals and sharks do not possess colour vision.
- Eyes of carnivores like cat, dog, lion, seal, etc. glow in night. It is due to tapetum lucidu, a reflecting layer next to retina.

Eye Defects**(1) Colour Blindness**

- Colour blindness (also called daltonism) is caused by the absence of one of these types of cones.

(2) Colour Vision

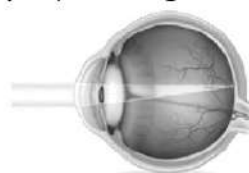
- The visual pigments for colour vision are erythropsin (sensitive to red), chloropsin (sensitive to blue).

(3) Blind Spot

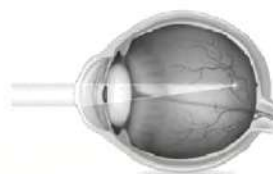
- Blind Spot- region of retina where the optic numbers branch off from retina to brain, have no rods and cones.

(4) Night Blindness

- It is most often caused by Vitamin A deficiency. Vitamin A is required for proper vision.

Eye Diseases**Myopia (Short Sightedness)**

NORMAL VISION
FARAWAY OBJECT IS CLEAR



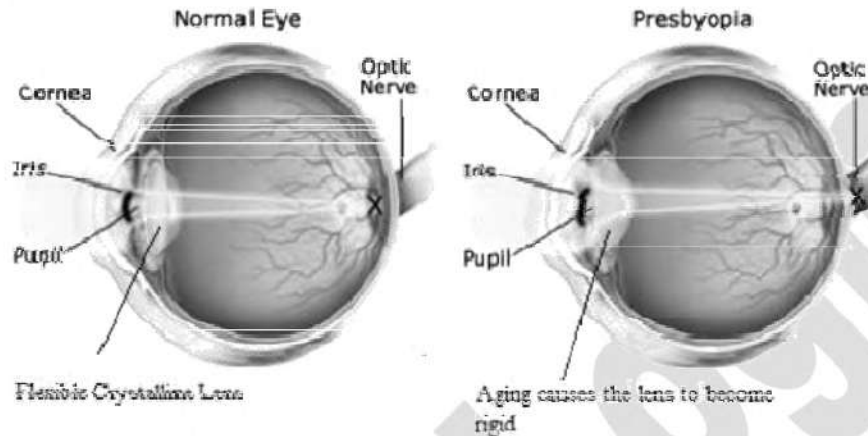
MYOPIA
NEARSIGHTED EYE
THE EYEBALL IS TOO LONG
FARAWAY OBJECT IS BLURRY





- Object close to eye are seen clearly but object of distance are blurred.
- Due to this eyeball becomes longer than normal.
- The focal point is in front of retina.
- Correction of Myopia can be done through concave lens the power of which is denoted by negative (-ve) sign.

Hypermyopia/Hypermertopia (Long Sightedness)

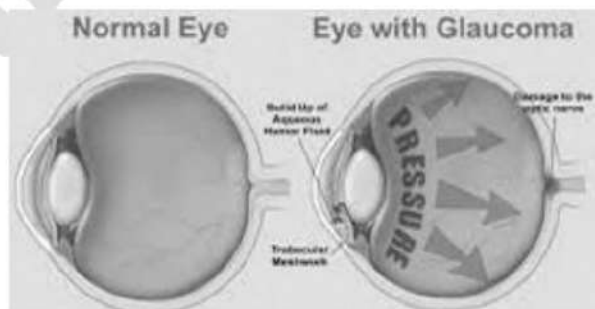


- Image is formed behind the retina, less refraction takes place.
- Convex lens corrects it.
- Eyeball is shorter in anterior-posterior direction.
- A person can clearly see distant objects. The focal point is behind retina.
- Correction of hypermetropia can be done through convex lens the power of which is denoted by positive (+ve) sign.

Presbiopia/Old Sightednes

- Image behind the retina normally after 40 years of age.
- This is an inability to focus on nearby objects due to loss of elasticity of the crystalline lens.
- It is due to excessive fat & lose ciliary muscles.
- Eye lens become more thinner.
- Correction of presbyopia can be done through convex lens. Bifocal lens is prescribed.

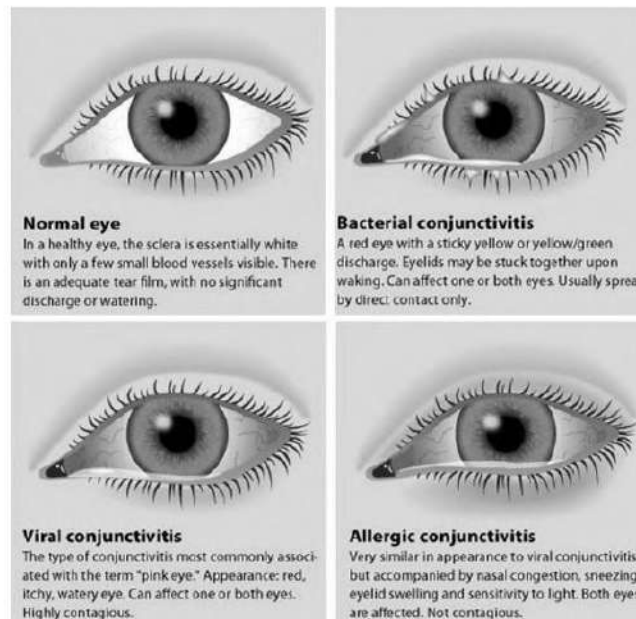
Glaucoma



- Due to Glaucoma retina and optic nerves are affected, due to increase 'P' of fluid in the Inner eye.
- Abnormal high pressure of fluid inside eyeball.
- Aqueous humor does not return into blood stream through canal of schlemm.
- Treatment for this is drugs and surgery.



Conjunctivites



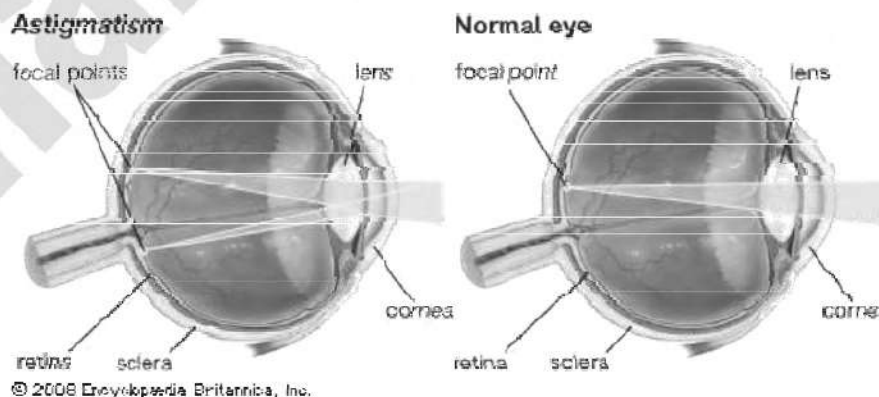
- It is caused due to bacterial.
- This is also called as pinkeye.
- Inflammation of conjunctiva by bacteria like pneumococci or staphylococci or viruses or even due to hypersensitivity reaction.

Trachoma



- Trachoma is caused by bacterial chlamydia trachomatis (also called trachoma agent).
- It is a chronic form of conjunctivitis, thin blood capillary develops over cornea and sand like nodules appears over cornea, leads to blindness.
- Conjunctiva becomes red with secretion of watery fluid.

Astigmatism



- This is a defect due to irregular cornea or lens.
- Correction to this is use of cylindrical lens.

**Presbyopia**

- No clear image in bright light

Ophthalmia Neonaterun

- It also called as conductivity of the new born, due to gonorrhea in mother.
- It is a sex disease.

Cataract

- It is not a disease.
- Degeneration of eye lens due to old age leads to opacity of lens and blindness.

Eye Defects

Types of Defect	Can See	Cannot See	Reason	Ratification
Long-sightedness or hypermetropia	Far objects	Near objects	Eye ball becomes smaller or focal length of the eye lens is large	Convex lens
Short-sightedness or myopia	Near objects	Far objects	Elongation of eye ball or focal length of the eye lens is small	Concave lens
Astigmatism (image are formed at varying distances from their retina)	-	-	Eye ball is not curved equally in all directions	Cylindrical lens

Tongue

- Tasty buds Spread all over the tongue.
- Tastes are in the form of chemical impulses like-Bitter, Sweet, Salty and Sour.

Nose

- Mucous-coated Olfactory region is responsible in the form of chemical impulses.
- Olfactory lobes are well developed in Dogs so that they have sensation of power 40 times greater than man.

Ear

- Ear is an organ of hearing and organ of maintenance of equilibrium of the body.
- Audible range of human -20 to 20000Hz/sec (hertz)
- Audible range in human 85db
- Ear has 3 parts- Outer External Ear, Middle Ear, Inner Ear.





CHAPTER -10

HUMAN REPRODUCTIVE SYSTEM

HUMAN REPRODUCTIVE SYSTEM

- Human reproductive system becomes functional at a definite age called puberty.
- Usually, males attain puberty at the age of 13 to 14 years, while females at the age of 10 to 12 years.
- Testis is the primary sexual organ of male, Ovary is primary sexual organ of female.
- Male gonad produces sperm, female gonad produces ovum at the age of puberty.
- Testes and ovary produce not only viable gametes but also secrete hormones.
- These hormones are testosterone (male hormones from testes) and estrogen (female hormone from ovary).
- In human beings there are 23 pairs of chromosomes. Chromosomes carry the determinants of the body that are genes.
- 22 pairs of chromosomes in human cells called autosomes are alike in men and women, they determine the features like color of skin, hair and eyes, shape of body organs.
- 23rd pair, called the pair of sex chromosomes consists of two similar members in women XX. In men the two members are dissimilar i.e. XY.
- Two microscopic specks of life one from the mother and one from the father constitutes life each called a gamete.
- Female gamete is called ovum or egg cell formed at ovaries. It is largest of human cells. It is roughly about 0.13 mm in diameter and 0.000004 gm in weight.
- Usually one ovum is shed from the ovary every month. During the reproductive age of woman about 400 ova are produced in a life time.
- All animals and plants are made up of cells.

The cells have nucleus which contains a large number of paired thread like structures called chromosomes.

Reproductive System in Male

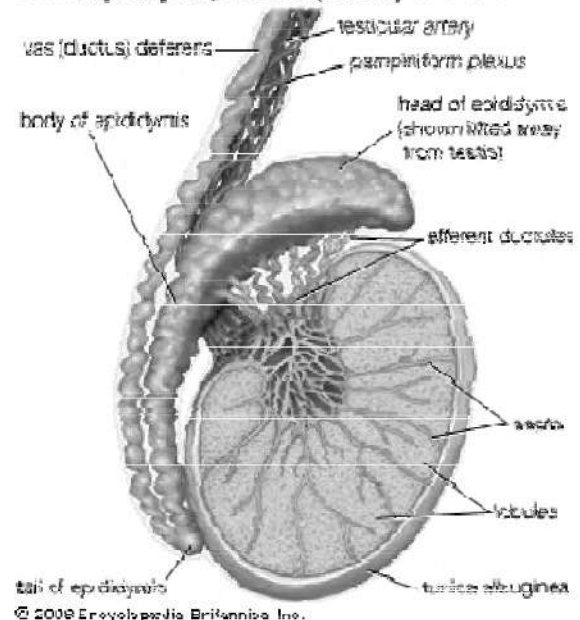
- The study of male reproductive system is called Andrology.
- The human male reproductive system is composed of a pair of testes, genital ducts, several accessory glands and penis.

- Male gamete is called sperm which is roughly about 0.008mm. It is charged with an unimaginable store of energy.
- It swims for 12cm journey up to the ovum which is equivalent of a man's 5 mile swim upstream a river.
- About 5 hundred millions sperms are deposited at a time so that at least one is likely to find the ovum.

Testes

- Testis is the primary male reproductive organ. It lies in small sac-like muscular structure outside the abdominal cavity called scrotum.
- In adult males, each testis is a small, pinkish and oval, but composite structure about 5 cm in length, 2.5 cm in width and 3 cm in thickness.
- Testes produce testosterone (hormone) along with sperms.
- It produces spermatozoa and secretes the male sex hormone testosterone.
- Maturation of sperms needs temperature less than the body.

Testis, epididymis, and vas (ductus) deferens



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- Sperms achieve maturity and motility in epididymis.
- For the production of sperms testes require temperature at least 1°C less than the body temperature.

**Seminal Vesicles**

- It is situated behind the bladder.
- About 70% of the semen in seminal fluid produced by seminal vesicles.

Prostate Gland

- It is situated around the first part of the urethra.
- Enlargement of prostate results in prostatitis i.e. urination is difficult or impossible.

Urethra

- It is a thick walled muscular duct and is a common passage for both urine and semen.
- Urethra traverse and open at the tip of the penis.

Penis

- It is a vascular, spongy and erectile structure.
- It contains three longitudinal columns of spongy erectile tissue which becomes filled with blood during sexual excitement.

Fertilisation

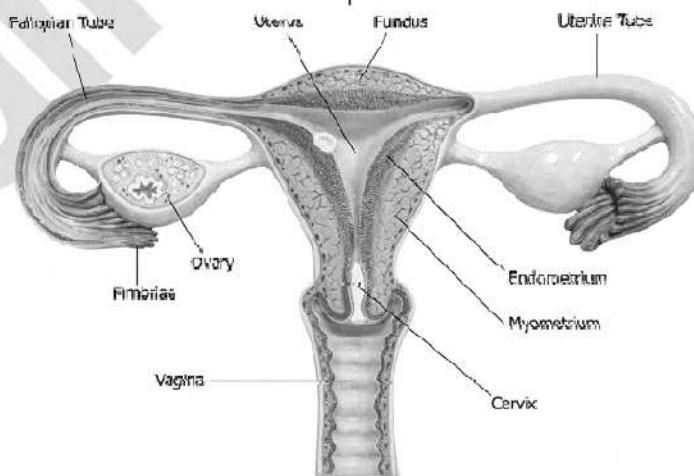
- Fertilisation is internal process in human beings.
- Male gamete (sperm) is introduced inside the female genital tract by the process called copulation (mating). The sperms are highly active and mobile. Millions of sperms are released in the vagina, and they move up through cervix and uterus.
- Finally, in the fallopian tube only one sperm fertilises the ovum.
- After the union of the sperm and the ovum (fertilisation) zygote is formed, if copulation has taken place during ovulatory period (middle of the menstrual cycle).
- Fertilisation is not taken place during the menstruation.

Pregnancy

- The embryonic development of the zygote starts immediately in the fallopian tube and pregnancy begins.
- The embryo moves down to reach the uterus, and gets attached to its thickened inner wall.
The close attachment of the embryo with the uterus is called implantation.
- After implantation, a special tissue develops between uterine wall and the embryo (fetus) called placenta.
- Through the placenta, the developmental needs of the fetus, such as nutritional, respiratory and excretory are met from the maternal body.
- The development of fetus inside the uterus till birth is called gestation. On completion of gestation, the birth of the fully developed fetus is termed as parturition.
- The average duration of human pregnancy is about 280 days or 40 weeks and it is calculated from the first day of the last menstrual cycle.
- The normal new born child, after 40 weeks of gestation, weighs about 3.5 kg.

Reproductive System in Female

- Female reproductive system consists of, Two ovaries, Two oviducts (fallopian tubes), Uterus, Cervix, Vagina, Accessory genital glands, Mammary glands, etc.
- In females, paired ovaries are located in the abdominal cavity near the kidney.
- Ovaries perform dual function of Production of female gamete (ovum) and Secretion of female hormones (estrogen and progesterone).



- Each ovary is composed of ovarian follicles. At puberty, these follicles undergo maturation to produce ova.
- Near the posterior end of ovary, a funnel like structure leads into a long convoluted tube called fallopian tube. The fallopian tube from both sides open into a muscular structure, the uterus. Uterus opens into the vagina.



- Vagina is a tubular structure. It is also called birth canal. In female, urinary opening (urethra) and vaginal openings are separate.
- Ovaries start producing ovum (female gamete) at a definite age called puberty (10-12 years). Ovarian follicles grow into mature follicle. Usually, one mature follicle develops to surround one ovum.
- Maturing ovum is from one of the two ovaries. The ovum is then released from the respective ovary by the process called ovulation. As the ovarian follicle matures, the inner wall of uterus thickens to get prepared for receiving the developing zygote in case fertilisation occurs.
- In case fertilisation does not take place, the thickened inner wall of the uterus breaks down along with its blood vessels, and moves out of the vagina in the form of bleeding, called menstrual flow (menstruation). It lasts for about 4-7 days.
- The cycle of events taking place in the ovaries and uterus every twenty eight days and marked by the menstrual flow, is called menstrual cycle (sexual cycle in human female).
- In a normal healthy woman, ovulation takes place in the mid of the menstrual cycle around 14th day. Menstruation occurs every 28 to 30 days.
- Commencement of menstruation at puberty is termed menarche and marks the beginning of reproductive life of a woman.
- The cycle of events in ovary and uterus stops around the age of 50 years in a human female. It is marked by the absence of menstrual flow. The stoppage of menstrual flow and other events at this age is termed menopause. The period between menarche and menopause, (12 to 50 years) is the normal reproductive life in human female.
- The function of ovaries and uterus change during pregnancy. Both ovulation and menstruation stop till birth of the offspring. The functions of the ovary and other reproductive organs are controlled by hormones released from hypothalamus and pituitary and the female hormones.
- A new born girl contains 2-4 million ova in number but of these only about 400 will mature during her life time. The beginning of this period of life is called puberty.
- The Luteinizing Hormone (LH) causes release of mature egg into fallopian tube follicle.
- The egg is fertilized in fallopian tube.
- Fertilization may occur between 11th to 15th days of the menstrual cycle.

Ovary

- Ovary is primary female sex organ.
- It produces ova and secretes female sex hormones e.g. estrogen & progesterone.
- Discharge of a mature ovum from graafian follicle is known as ovulation.
- After ovulation, a ruptured follicle turns into a yellowish solid mass of cells called corpus luteum.
- Corpus luteum secretes the female hormone called progesterone.
- Progesterone (sex hormone) stimulates the uterus to develop blood capillaries to receive and nourish the fertilized ovum and also suppresses release of further production of ova.
- Progesterone controls the last two weeks of menstruation cycle. Secretion of progesterone is stopped if ovum is not fertilized.
- The destroyed capillaries of uterus is called menstruation blood.
- Estrogen is continuously secreted and controls first two weeks of menstruation cycle.

Fallopian Tubes

- There are 2 Fallopian tubes which are also called oviducts or uterine tubes.
- Normally, fertilization in mammals occurs in fallopian tube.
- Fertilization in uterus is rare.
- If there is blockage in fallopian tube, ova cannot be fertilized.





- Both fallopian tubes open into uterus.
- Uterus is situated above and behind the urinary bladder and remains attached to the body wall by ligaments.
- The wall of uterus is composed of smooth muscle fibres called myometrium.
- Lumen of the uterus is lined by a mucous membrane called endometrium.
- Lower narrow end of uterus is called cervix.

Vagina

- Uterus opens into an elastic muscular tube called vagina.
- The vagina is lined by a stratified squamous epithelium without any glands.
- During reproductive life the vagina contains Lactobacillus acidophils which keeps the vaginal pH between 4.9 and 3.5 by producing lactic acid from glycogen.

Chromosomes

In the nucleus of each cell, the DNA molecule is packaged into thread-like structures called chromosomes. Each chromosome is made up of DNA tightly coiled many times around proteins called histones that support its structure.

Number of Chromosomes: 23 pairs or 46.

Sex chromosomes: 1 pair (2)

Number of autosomes: 22 pairs (44)

Twins

(1) Identical (or) Monozygotic Twins

Monozygotic (MZ) or identical twins occur when a single egg is fertilized to form one zygote (hence, "monozygotic") which then divides into two separate embryos.

(2) Non Identical (or) Dizygotic Twins

Fraternal or dizygotic (DZ) twins (also referred to as "non-identical twins", "dissimilar twins", "biovular twins", and, in cases of females, sororal twins) usually occur when two fertilized eggs are implanted in the uterus wall at the same time. When two eggs are independently fertilized by two different sperm cells.

Siamese Twins

Twins those are physically joined at birth sometimes sharing organs, and sometimes separable by surgery.





CHAPTER -11

HUMAN PATHOLOGY

- Factors like malfunctioning or improper functioning of various body parts, genetic disorder, hormonal imbalance, malfunctioning of immune system of body, causes diseases to the human health.

CONGENITAL DISEASES

- They are present since birth which are caused due to genetic abnormality or due to metabolic disorders or malfunctioning of any organ.
- They are permanent in nature, generally not easily curable.
- They are inherited to the children.
- They are classified as
- Communicable or infectious diseases:
They are caused by different variety of pathogenic viruses, bacteria, protozoa, fungi and worms. The pathogens are generally carried with the help of a vector.
- Non-communicable or non-infectious or degenerative diseases: They occur due to the malfunctioning of some organ or organ system in the body.
- Deficiency diseases: They occur due to the deficiency of some nutrients, minerals or Vitamins.

Disease spread by air

- | | |
|----------------|----------------|
| (1) Cold | (2) Chickenpox |
| (3) Birdflu | (4) Swineflu |
| (5) Smallpox | (6) Measles |
| (7) Influenza | (8) Pneumonia |
| (9) Diphtheria | (10) Pertussis |
| (11) T.B | (12) S.A.R.S |

Disease caused by contaminated food and water

- | | |
|----------------|---------------|
| (1) Cholera | (2) Typhoid |
| (3) Amoebiasis | (4) Polio |
| (5) Diarrhea | (6) Hepatitis |
| (7) Madcow | |

Disease spread by mosquitoes

- | | |
|-----------------|---------------------------|
| (1) Filariasis | (2) Japanese encephalitis |
| (3) Chikungunya | (4) Dengue fever |

DDT (Di-chloro di-phenyl-trichloro ethane)

It is a first insecticide discovered in 1937)

Disease caused by flies

- (1) Tse -Tse fly: Sleeping sickness
- (2) Sand- Fly: Black fever (Kala-azar)
- (3) Rat- Fly: Plague

Vaccines

First vaccine was developed by Edward Jenner in 1798. He was called Father of Immunology and father of Vaccination

Vaccine Name	Disease	Discovered by
Smallpox	Smallpox	Edward Jenner
B.C.G	T.B	Calmette, Guérin
Polio	Polio	Sebein
Measles	Measles	John Enders
Cholera	Cholera	Louis Pasteur
Anthrax	Anthrax	Louis Pasteur
Hydrophobia	Rabies	Louis Pasteur

Cancer

- It is caused by H.P.V virus (Human papilloma virus).
- The study of cancer is called Oncology.
- It is characterized by the uncontrolled growth and division of cells.
- It leads to a mass of cells termed as neoplasm.
- Abnormal and persistent cell division localised in a particular region is called benign tumor.
- Cancers are 2 types
 - (1) Benign Cancer (No harm to the humans)
 - (2) Malignant Cancer (It is dangerous)
- Cancer can be determined by the Biopsy test.
- Always Women suffer with Uterine cancer and Breast cancer.
- Always men suffer with throat cancer and Mouth cancer.
- **Pink ribbon** related to cancer.

Cancer Treatments

- (1) Cobalt -60
- (2) γ-rays
- (3) Laser rays

Cancer treatment is also called **Chemotherapy**.

Gout

- It results from accumulation of uric acid crystals in the synovial joints.



- Gout is a disease associated with an inborn error of uric acid metabolism that increases production or interferes with the excretion of uric acid.

Haemophilia

- It is called bleeder's disease.
- A disorder which is sex-linked, it is a recessive condition.
- In a patient of haemophilia, blood clotting is deficient.
- Haemophilia A is characterized by lack of anti-haemophilia globulin factor VIII. About four-fifths of the cases of haemophilia are of this type.
- Haemophilia B or Christmas disease results from a defect in plasma thromboplastic component (Factor IX).

BLOOD TRANSFUSION DISEASES AIDS

(Acquired Immune Deficiency Syndrome)

- AIDS is the end stage of the disease.
- AIDS destroys immune system of the body.
- Unsafe homosexual or bisexual activities are the most common mode of transmission of AIDS (82.5%).
- AIDS is caused by the Human Immuno deficiency Virus (HIV).
- There are two types of HIV namely HIV -1 and HIV-2.
- The most common virus currently associated with AIDS is HIV-1.
- A virus found in the blood of wild African green monkey called the Simian Immuno deficiency Virus (SIY) is similar to HIV-2.
- HIV is a retrovirus. It can synthesize DNA from RNA.
- The major cell infected by HIV is the helper T-lymphocyte that bears the CD-4 receptor site.
- HIV progressively destroys T -lymphocytes.
- HIV antibodies can be detected by the ELISA test (Enzyme-Linked Immuno Sorbent Assay).
- The patient occasionally will suffer from swollen lymph nodes, mild prolonged fever, diarrhoea or other non-specific symptoms.
- In India AIDS was first reported in 1986.
- World AIDS Day is on 1st December.

VIRUS

- Virus was first discovered by Dimitri of Russia in 1892.
- Viruses are smallest a cellular organism, Obligately parasites and possess characters of both living and non - living. They can be crystallized and stored for many years.
- They live inside the host cell of living body, can't live outside the host or reproduce on their own. Their shape may be - spherical, isometric, cubical, tadpole or elongated shaped.
- They are filterable, non cellular and ultramicroscopic of protein and nucleic acids. They generally lack the energy - containing chemicals, raw materials. They can't live Viruses pass from one host to another by direct or indirect contact.
- Antibiotics and sulfa - drugs are generally used against virus.

Hepatitis

- Hepatitis is a viral disease.
- HA V causes hepatic anorexia resulting in liver damage or jaundice causes liver cancer.
- General mode of transmission is by the faecal-oral route.
- Children and young adults are susceptible to HAV.
- Adults are only susceptible to HBV but vaccines are available.
- Virus can be categorized into Three:
 - Animal Virus** (Consists DNA & RNA)
 - Plant Virus** (Consists RNA)
 - Bacterial Virus** (Consists DNA)

Types of Virus

Virus	Invaded
1.Zoophase	Animals
2.Phytophase	Plants
3.Mycophase	Fungus
4.Zymophase	Yeast
5.Bacteriophage	Bacteria

N.P.V (Nuclear Polyhydrosis Virus)

It is a most useful virus for farmers.

It is used as a Bio Insecticide

**DISEASES CAUSED BY VIRUS**

Disease	Pathogen	Mode of transmission	Main symptoms
Smallpox	Variola Virus (DNA virus)	Direct contact (droplets on skin) indirect by infected articles: 12 days	High fever, headache, backache, skin rashes. later pass to muscles
Chickenpox	Varicella virus	Direct contact (droplets) indirect by infected objects, 12 - 16 days	Fever, cold, skin eruption starts as red spots, vesicles and crusts, scab formation in 36 hours. 0 days
Common cold Influenza / Flu	Rhina virus Orthomixovirus (RNA virus)	Droplets from nose and throat.)	Fever, headache, cough, nasal discharge, muscular pain, chills, sore throat and dry cough.
Measles (Rubella)	Paramyxovirus (RNA virus) Effects Respiratory passages.	Virus transmitted through air by droplets during talking, coughing and sneezing 10 - 14 days	running nose, sneezing, fever, headache, backache and chills, inflammation of mucous membrane.
Mumps	Mumps virus (Paramyxovirus)	Virus in saliva and secretion of nose invades salivary glands; 12 - 21 days	High fever, headache, backache, reddened taste buds, dry throat swelling of salivary glands about ear and jaws for 7 - 10 days
Viral encephalitis	Encephalitis virus (arbovirus)	Domestic animals reservoir of virus; transmitted by mosquito bite to man; 4 - 21 days	Sudden fever, headache, vomiting, pains, drowsiness, stiffness of neck, mental confusion, blindness, deafness, mental defects, paralysis
Poliomyelitis (polio)	Polio virus (Enterovirus) RNA virus	houseflies, fleas, food and water; 7 - 14 days. Effects pharynx, intestines, blood. Paralysis occurs.	headache, stomach upset, vomiting, high fever, sore throat and loose bowels, stiffness of neck and back, paralysis some parts of the body
Rabies (Hydrophobia)	Rabies virus (rhabdovirus)	Bite of a mad (rabid) Dog: 2- 16 weeks or more	Headache, vomiting, fever, insomnia, hoarse voice, (fear of water), paralysis, death
Dengue fever or break bone fever	Dengue virus (arbovirus) herpes virus simplex	Mosquito (Aedes) bite; 4 - 8 days contact, saliva, stools, contaminated articles	Fever, headache, pain in joint back muscles and eye balls, rashes, vesicular virus simplex eruption the mouth of small children
Herpes zoster (Shingles)	Herpes virus zoster	Contact droplets; 7-14 days	Fever, severe pain in the skin or mucous membrane, vesicles on skin
AIDS (Acquired Immuno Deficiency Syndrome)	Human T-cell (HTLV - III) AV (Retro virus) RNA Virus	Via blood and sperm among homo & Hetero sexual, drug- users, individuals & prostitutes	Body loose weight, fever, cancer pneumonia, brain tumors, hemorrhage, diarrhea, swelling of lymph glands, mouth ulcers.





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Yellow Fever	Arthropod-borne virus (arbovirus)	Vector arthropods	Effects blood vessels and liver
Influenza	Myxovirus (RNA virus)	Droplets from Nose and throat. Effects mucous membrane.	Nasal discharge, Sneezing and Cough.

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Disease	Pathogen	Mode of transmission	Main symptoms
Dengue fever	Flavivirus	Mosquitoes	Body pains, high fever, Running nose, red pimples on skin
Chickenguniya	Alfa Virus	Mosquitoes	Body pains, high fever, Running nose, red pimples on skin
Birdflu	H5N1 Virus	Birds	High Fever, low breathing rat
Swine flu	H1N1	Swines	High Fever, low breathing rat

BACTERIA

- **A. Van Leuwenhoek** was the first person to observe bacteria. The chemical reactions like fermentation were first studies by **Luis Pasteur** while the pathogenic properties of bacteria were discovered by Robert Koch.
- Bacteria are present everywhere in the nature. They live in or on bodies of living organism, in dead or non living organic materials like dung, milk etc.
- These are single celled body can float in air & live in icy mountains.
- Bacteria has the capacity of reproduction (also sexually) and can split into two.
- Bacteria categorized into Four:
 - Bacilli - (Cylinder / rod shape)
 - Cocci - (Spheres / ball shape)
 - Vibrios - (curved / comma shape)
 - Spirilla - (Cork Screw curved / spiral shape)

Points to remember

- Bacteria are cosmopolitan in distribution and lead a saprophytic, autotrophic or parasitic mode of life.
- Bacteria cells contain a circular, coiled DNA and ribosome's of 70 types but not membrane bound organelles.
- Bacteria reproduce 'generally by budding and binary fission (asexual reproduction). Some Bacteria can also reproduce sexually by conjugation, transformation and transduction.
- Photosynthetic Bacteria do not use water as raw material in photosynthesis and do not evolve oxygen.
- **Chemosynthetic Bacteria** oxidize inorganic nitrate, nitrite or iron to generate ATP. Bacterial flagellum is made up of one or more molecules of the same protein called flagellin.
- The Bacterial genome is called nucleoid.
- Escherichia coli (E.Coli) is a Bacterium which is common inhabitant of human intestine.
- Clostridium butylicum has been used in the synthesis of Vitamin B.
- Fixation of free nitrogen by Bacteria in the soil is done by Azotobacter.
- **Marsh gas** is produced by methanogens (methane-producing Bacteria).
- Bacteria have incipient nucleus (nucleoid) and hence, they are placed in prokaryota.
- **Chemosynthetic Bacteria** do not need sunlight to grow because they prepare their food without the help of light.
- Bacteria are classified on the basis of Gram positive (+ve) and Gram negative (-ve) staining property.

Diseases Caused by Bacteria

Disease	Pathogen	Mode of transmission	Main symptoms
Septic sore throat	Streptococcus (SP).	Bacteria infect throat and nasal membranes by droplet	Sore throat with fever and Cough
Diphtheria	(Coryne – bacterium diphtheria)	Bacteria infect respiratory tract by carrier, direct contact droplet and food	Sore throat, formation of membranous deposit in throat difficult breathing
Tuberculosis (TB)	Mycobacterium tuberculosis. Effects mainly lungs	Transmitted to lungs, bones and other organs by direct contact, droplet infection.	Cough, fever in evening, fatigue, loss of weight, infection in the lungs



Disease	Pathogen	Mode of transmission	Main symptoms
Plague Bubonic plague	Short rod, yersinia pestis	Rat flea spreads disease from rat to man 2 – 10 days	High fever, vomiting, dry skin, thirst, black spots on skin
Tetanus or (lockjaw typhoid or enteric fever cholera)	Clostridium tetani salmonella typhoid vibrio cholerae. Mainly effects Blood.	Bacteria in soil, enter through wound, food, faeces, stools, water	Spasm of muscles, fever, vomiting, abdominal pain, severe diarrhea, urine stops
Bacillary dysentery / Diarrhoea	Shigella dysenteriae, Salmonella, Escherichia coli, Campylobacter	Flies, food, faeces water and carriers, 1 – 4 days. Effects alimentary canal, ileum, colon	Fever, nausea, vomiting, severe abdominal pain, blood in the stools and diarrhoea
Whooping cough (Pertussis)	(Bordetella pertussis) effects Respiratory tract	Droplets projected during coughing and sneezing 7 – 14 days	Cold & cough, swelling, pus discharge, frequent & burning urination
Syphilis	(Treponema pallidum) effects Reproductive organs	Direct contact, chiefly sexual intercourse, 10 – 90 days	A hard, painless sore or chancre (ulcer) on the genitalia
Leprosy	Mycobacterium leprae	Long & close contact with infected persons	Ulcers, deformities of fingers, toes and wasting of body parts
Typhoid Fever	Salmonella typhi	Due to contamination of food and water	Effects alimentary canal, blood, lungs, spleen, bone marrow. Constant fever.
Gonorrhoea	Neisseria gonorrhoeae	Through sexual contact	Effects Urinogenital mucosa. Burning sensation in micturition
Cholera	Vibrio cholera	Contaminated food and water	Effects mainly small intestine. Severe diarrhea and vomiting.
Botulism	Clostridium Botulism	Food	Nervous system, digestive system

ALGAE

- Chiefly water plants through some grow on stones, the bark of trees. They are multi cellular and aquatic. Algae prepare their own food (autotrophic)
- These primitive organisms are divided into 7 primary divisions - one division of the kingdom Monera and six division of the plant kingdom.
- Algae is not subjected to direct sunlight.
- The remaining are,
 - a) **Blue-Green Algae (Moneran.).** It also known as Cyanobacteria. Commonly available forms of Blue-algae are Nostoc, Anabaena and Oscillatoria. They are being used as fertilizers as they fix atmospheric Nitrogen.
 - b) **Euglena**
 - c) **Green Algae** (has Chlorophyll & Carotene. In addition, it contains red pigment also.)
 - d) **Brown Algae**
 - e) **Diatoms** (Yellowish green to yellowish brown)
 - f) **Red Algae** (Red, Brown and Violet)





Disease	Pathogen	Mode of transmission	Main Symptoms
Ring worm (Tinea)	Microsporum, Trichophyton	Direct contact from unbathed cats & dogs or infected objects.	Sores begin as small, slightly raised reddish areas, enlarge, become redder and contain one or more blistered areas on skin and scalp cause partial and temporary baldness in children
Athlete's foot	Trichophyton	Bad foot hygiene where skin remains warm and moist for long periods, fungi find optimum condition to invade dead outer layer of skin.	Painful itching or burning sensation in the infected areas. Crack appears in the skin at toes. Otherwise reddening, scaling & thickening of skin between toes.

FUNGI

- They lack chlorophyll, non-green. Some exist as parasites (do not produce food of their own & dependants), other feed on dead organic matter (saprophytes) & grow in soil.
- Unable to take in solid foods as animals do, they do this by secreting enzymes into their surroundings, Which break down complex organic materials into simple substances that can be dissolved into water.
- Having no definite tissues. Mushrooms, Toad balls, Moulds, Puff balls, Yeast (Unicellular and saprophytic) are common

Fungi.

Diseases Caused By Fungi





Madura foot	Maaurella mycetoni	Fungi gain entry through some minor injury to the skin	Produce a chronic, granulating infection of the lower extremities affected part becomes enlarged and develop many deep sores, extensive bones destruction leading to crippling deformities.
Dhobi itch	Several different fungi	Direct contact through objects handled by infected person.	A type of ringworm infection usually located in the grain and inner surfaces of thighs, red rash which itches intensely.

Diseases Caused by Protozoan

Disease	Pathogen	Habitat	Mode of transmission	Main Symptoms
Malaria	Plasmodium	Passes through a developmental phase in liver, resides inside RBCs and carried by blood to all organs	Transmitted to man by bite of an infected female anopheline mosquito	Cold, headache, shivering, rising temperature Fever rises maximum, severe headache pain in back & joints, vomiting, sweating fall in temperature
Amoebic dysentery or amoe- basis	Entamoeba histolytica	Large intestine	Transmitted to man to man through ingestion of cysts in drinking water, vegetables & food contaminated with faces	Acute dysentery with blood, mucous in stools, severe stools, severe abdominal pain. Multiple abscesses in liver, lung brain, spleen, ulceration of skin vagina & penis



Sleeping Sickness (trypanosomiasis)	Trypanosome brucei	Reaches lymph nodes via lymphatics blood & infects brain	Transmitted by bite of tsetse fly	Fever severe headache, enlargement of glands at back at neck, rash on the back and chest, joint pains swelling of eyelids ankles & hands, trembling, loss of appetite, mental disturbances, coma and death.
Oriental sore (Delhi boil) Kala-azar or black sickness diarrhea 'giardiasis'	Leishmania tropica leishmania donovani giardia intestinalis	Reticuloendothelial cells reticuloendothelial cells bile tract, duodenum and upper part of jejunum	Transmitted to man by the bite of infected sand fly. transmit from man to man through in drinking water, food. Contaminated with stools	Ulcer forms wherever sand fly bites, ulcers, found on face, arms, legs spleen enlarge, liver fever, jaundice, skin become dark. passage of yellowish, green stools with excess fat, fever anaemia.
Vaginitis	Trichomonas vaginalis	Female genital tracts, urinary tract of both male and female	Transmitted during coitus in females vagina as part of ejaculate & by shared towels and toilet seats	Itchy inflammation of vagina and a frothy or yellow vaginal discharge

LICHEN

It is the combination of Fungus and Algae

- They are grown on tree barks and on rocks.
- Water provided by Fungus, Food manufactured by Algae

Worms Carry Diseases

Disease	Pathogen	Mode of transmission	Main Symptoms
Hook Worm	Ancylostoma duodenale	Habitat-small intestine. Transmit from person to person	Severe anaemia, oedema of feet and ankle, Dermatitis
Filaria	Wuchereria	Habitat-Lymphatic vessels & nodes. Mosquito's larvae deposited on skin & enters lymphatic vessels & nodes	Enlarge the size of legs, penis, breast, forearm, labia etc.
Pin Worm	Enterobius	Habitat-Vermiform appendix & Caecum. Infection due to itching in anus. Scratching anus with fingers, transfer into food.	Inflammation of vermiform appendix, bed wetting at night

Genetical Disease (or) Hereditary Disease**Gregor John Mendel (1822-1884)**

- He opened a new chapter in the history of Modern Genetics by conducting hybridization experiments with garden Pea (Pisum Sativum).
- Though first published in 1866, such valuable contribution of Mendel passed unnoticed.
- It was received in 1900 by the memorable work of Devries, Correns and Tshemark. From then Mendel is regarded as the Father of Genetics.





Some terms used in Genetics

Phenotype: It means external appearance of a plant or an animal.

Genotype: It means what type of genes a plant or an animal has.

Allele: The genes which control a character occur in pairs in the cell. Each gene is called the allele of the other.

Together the genes make up an allelic pair.

Some Genetic abnormalities in man

- 1) Colour blindness
- 2) Down's Syndrome (Mongolism)
- 3) Haemophilia
- 4) Albinism

Diseases and Parts of the body Affected

Disease	Part of the Body Affected
AIDS	Immune system
Arthritis	Inflammation of joints
Cataract	Eyes
Conjunctivitis	Eyes
Diabetes	Pancreas
Diphtheria	Throat
Eczema	Skin
Elephantiasis	Legs
Encephalitis	Brain
Goiter	Thyroid gland
Infantile paralysis	Limbs (arms and legs)
Jaundice	Liver
Meningitis	Spinal cord and brain
Pneumonia	Lungs
Pleurisy	Lining of the chest wall
Pyorrhea	Gums
Rheumatism	Joints
Trachoma	Eye
Tuberculosis	Intensive, whole
Typhoid	Intensive, whole body

Vaccination Schedules

1	Birth -3 months	BCG	1	Intradermal (1 dose)
2	6 weeks – 9 months Polio	DPT	3	Intra muscular (1m) oral (P.O)
3	9 months – 12 months	Measles	1	Subcutaneous (Sc)
4	9 months – 24 months	DPT Polio	1	Intra muscular oral
5	5 years to 6 years	DT Typhoid	1	Oral
			2	Sub-cutaneous
6	10 years	TT	1	Oral
		Typhoid	1	Sub-cutaneous
7	16 years	TT	1	Intra muscular
		Typhoid	1	Sub-cutaneous

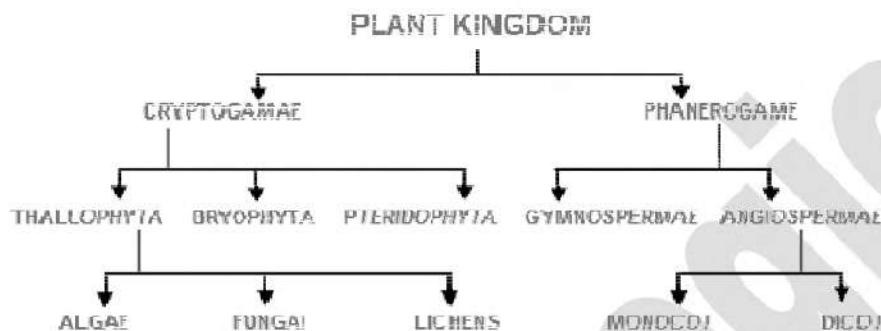




CHAPTER -12

PLANT KINGDOM

Plant Kingdom:



The Plant kingdom comprise multicellular, Photosynthetic producers. They are primarily aquatic red, brown and green algae and the land plants -bryophytes, ferns, gymnosperms and angio sperms.

Except algae & bryophyte all are vascular plants (Vascular system consists of Xylem and phloem Xylem is water conducting tissue and phloem is the food conducting tissue of the Vascular Plant).

Most vascular plants are flowering plants. They form seeds within a fruit. So, they are called angio sperms (Covered Seed). Some Vascular plants produce seeds but no fruit. They are called gymnosperms (unprotected seed). Ferns produce neither flowers nor seeds but only spores in sporangia on their leaves.

PLANTS CLASSIFICATION

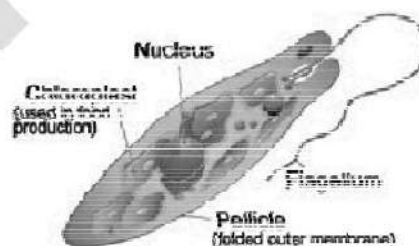
- Entire plant kingdom is divided into Cryptogams and Phanerogams depending upon the presence or absence of flowers and seeds.

(1) CRYPTOGRAMS

All plants without seeds known as cryptogams. All non-flowering plants such as algae, fungi, lichens, mosses and fern are included in Cryptogams (Kryptos: concealed; gamus: marriage).

It is further divided in to Bryophyte, Thallophyte and Pteridophyta.

(i) Thallophyta



- Plants of this group are the simplest and most primitive ones. Their bodies are not differentiated into organs like root, stem and leaf.
- Algae and Fungi both are considered as example of Thallophyta. However, nutrition is autotrophic in algae and fungi exhibits heterotrophic nutrition.

Algae:

- The Study of algae is called Phycology.
- Father of Indian Phycology M.O.P Ayyangar
- These are called Autotrophes.
- They can make their own food through Photosynthesis.



TYPES OF ALGAES

1. Green Algae

Ex: Spirogyra, Sea lettuce, Purple Laver

2. Red Algae

Ex: Irish Moss

3. Brown Algae

- These are called sea weeds and Kelps.
- These are used as a Biogas.
- Iodine is more available in this Algae.

4. Phytoplanktons

The floats on the water. These are called Floating grass lands.

5. Blue Green Algae

Spirulina (it consists of more proteins)

FUNGI

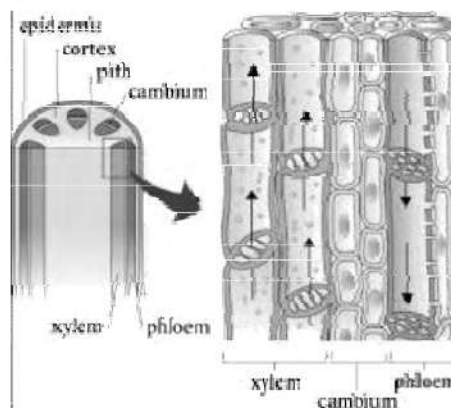
- The study of Fungi is called Mycology. .
- There is no chlorophyll
- There is no photosynthesis in Fungi
- These are called Heterotrophes and Saprophytes. Ex: Yeast, Pencillium
- Pencillin was discovered by Alexander Fleming in 1928.
- It is a wonder drug.
- It is derived from pencillium notatum (Fungi) and it acts on bacterial disease.
- **Mushrooms (Fungi)**
 - **Food Mushrooms:** Button & Penny Bun Mushrooms
 - **Poison Mushrooms:** Toad Stools

(ii) BRYOPHYTES

- It comprises mosses and liverworts. These are small plants that grow densely together in moist shady places. They form green carpets on damp soils, rocks, walls and on barks of trees.
- Bryophytes have no vascular tissues for conduction of water and food. Due to the absence of true roots, their cells absorb moisture directly from the ground (or) the atmosphere. Transportation of materials is done from cell to cell.
- They inhabit land but they need water for their sexual reproduction. they show regular heteromorphic alteration of generations, (i.e) gametophyte (gamete bearing plants) and sporophyte (Spore producing plants).

Eg: Riccia, Funaria

(iii) PTERIDOPHYTA



- Pteridophyta (Pteron means feather, Phyton means Plants) are the most primitive vascular plants. They are known as vascular cryptogams.



- They occur in humid and tropical climates and usually grow on soil, rocks, in ponds and as epiphytes on other plants.
- On the basis of organisation of plant body i.e the nature of leaf, vascular system and location of sporangia, the Pteridophyta is divided into four groups, viz. Psilopsida, Lycopsida, Sphenopsida and Pteropsida.

Xylem:

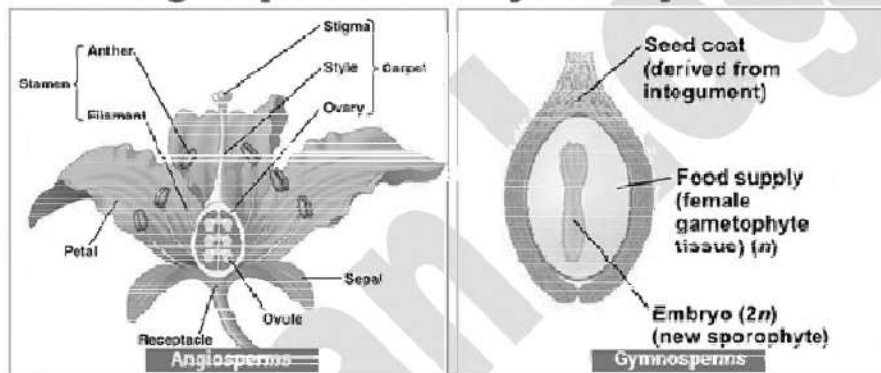
Xylem is one of the two types of transport tissue in vascular plants (phloem is the other). The word xylem is derived from the Greek word (xylon), meaning "wood"; the best-known xylem tissue is wood, though it is found throughout the plant. Its basic function is to transport water, but it also transports some nutrients through the plant.

Phloem:

In vascular plants, phloem is the living tissue that carries organic nutrients (known as photosynthate), in particular, sucrose, a sugar to all parts of the plant where needed. In trees, the phloem is the innermost layer of the bark, hence the name, derived from the Greek word (**Phlios**) meaning "bark". The phloem is concerned mainly with the transport of soluble organic material made during photosynthesis. This is called translocation.

(2) PHANEROGAMS

Angiosperms vs Gymnosperms



- All plants with seeds known as Phanerogamae. These plants bear seeds. (Ex: Phaneros: visible; gamus: marriage).
- Phanerogams are also known as spermatophytes.
- Phanerogams are further divided into two parts - Gymnosperms and Angiosperm.

(i) Gymnosperms (Naked Seeded Plants)

- They are the most ancient seed plants that originated during the late Paleozoic era but, flourished well during the Mesozoic era.
- Gymnosperms are represented by coniferous.
- Gymnosperms are grow in cool climate. They use snow as a source of water.
- But, Cycads and members of Gentiles survive in warm dry climate.
- There is an absence of bisexuality which reduces the chances of selffertilization.
- Gymnosperms are represented by Thuja, pinus, cyads, Abies and cedar.
- Gymnosperms have uncovered (naked) seeds.
- Gymnosperms lack vegetative means of reproduction by means of laying, cutting etc.
- Ex:** Red Wood Tree (World Highest trees)
- Existing in California (U.S.A)
- On the basis of nature of leaves, woods, vascular system and reproductive structures Gymnosperms are again divided into 3 groups **Cycadopsida, Coniferopsida and Gnetopsida**.

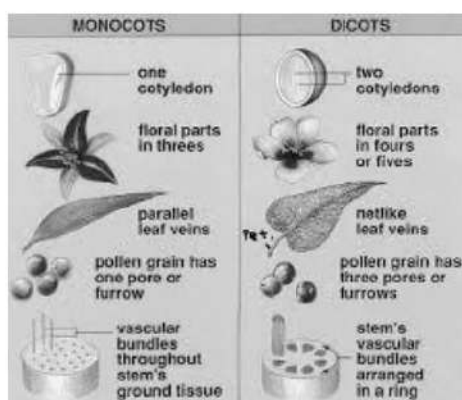
(ii) Angiosperms

- Angiosperms have covered seeds.
- Angiosperms are closed seeded plants(seeds within the fruits).
- They are most highly developed plants which bear flowers having conspicuous accessory and essential whorls.



- d. Each carpel bears an ovary, style and stigma. With the stimulus of fertilization the ovary usually develops into the fruit and the ovules into seeds. Thus, the seeds within the fruit.
- e. From minute floating duckweeds to giant eucalyptus come under Angiosperms. Others trees and plants like silk cotton trees, plants of great variety, water lilies, sunflowers, orchids, pitcher plants, Indian pipe etc.
- f. They are well adapted to the terrestrial life and they occur in diverse habitats like cold tundra to hot tropical and even desert areas. They also survive in aquatic habitats.
- g. Angiosperms evolved from ancient gymnosperms.
- h. Angiosperms plants represented by trees, shrubs and herbs.
- i. Today, many angiosperms are nonwoody herbaceous plants rather than being woody shrubs or trees.
- j. They have a body well differentiated into root, stem and leaves.
- k. The phenomenon of pollination is facilitated by wind, water, insects, birds and other animals including man.
- l. There are two types in angiosperms- Monocotyledons and Dicotyledons.

Monocots



They are almost herbaceous and have parts in trees, parallel-veined leaves, scattered vascular bundles in the stem and one cotyledon or seed leaf.

Dicots

- They are either woody or herbaceous. in members of this class exhibit (net-like) venation and arrangement like alternate, spiral whorled. On the basis age Angiosperms grouped as

Perennials-live for many years (Pipal)

Annuals-live for a year (Tulasi)

Biennialisals- live for two seasons (sugarcane)

(1) Root system

- In vascular plants, the root is the organ of a plant that typically lies below the surface of the soil.
- Roots can also be aerial or aerating (growing up above the ground or especially above water). Furthermore, a stem normally occurring below ground is not exceptional either (see rhizome). Therefore, the root is best defined as the non-leaf, non-nodes bearing parts of the plant's body. However, important internal structural differences between stems and roots exist.
- The first root that comes from a plant is called the radicle. The four major functions of roots are
 - absorption of water and inorganic nutrients
 - anchoring of the plant body to the ground, and supporting it
 - storage of food and nutrients
 - vegetative reproduction.
- In response to the concentration of nutrients, roots also synthesise cytokinin, which acts as a signal as to how fast the shoots can grow.
- Roots often function in storage of food and nutrients. The roots of most vascular plant species enter into symbiosis with certain fungi to form mycorrhizas, and a large range of other organisms including bacteria also closely associate with roots.





ROOT MODIFICATIONS

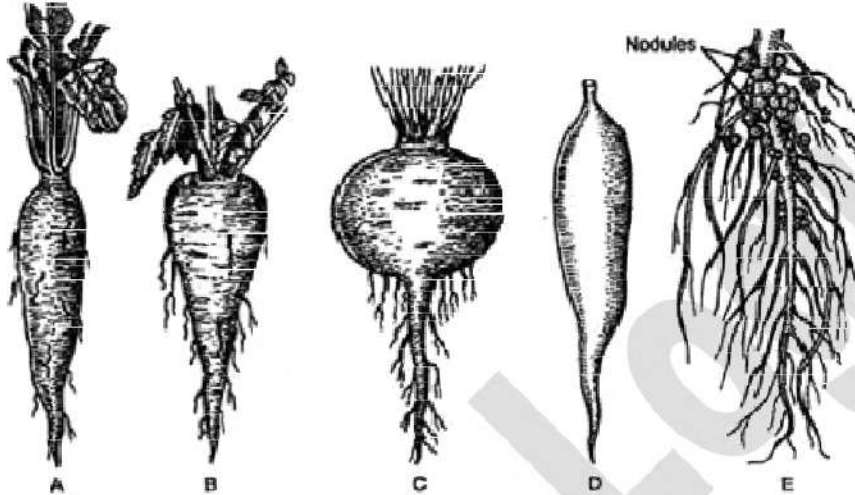
(1) Tap root modifications (For storage of food):

In some plants the tap roots store reserve food for which they become swollen and assume different shapes. There are four types

Ex: Carrot, Beetroot, Raddish

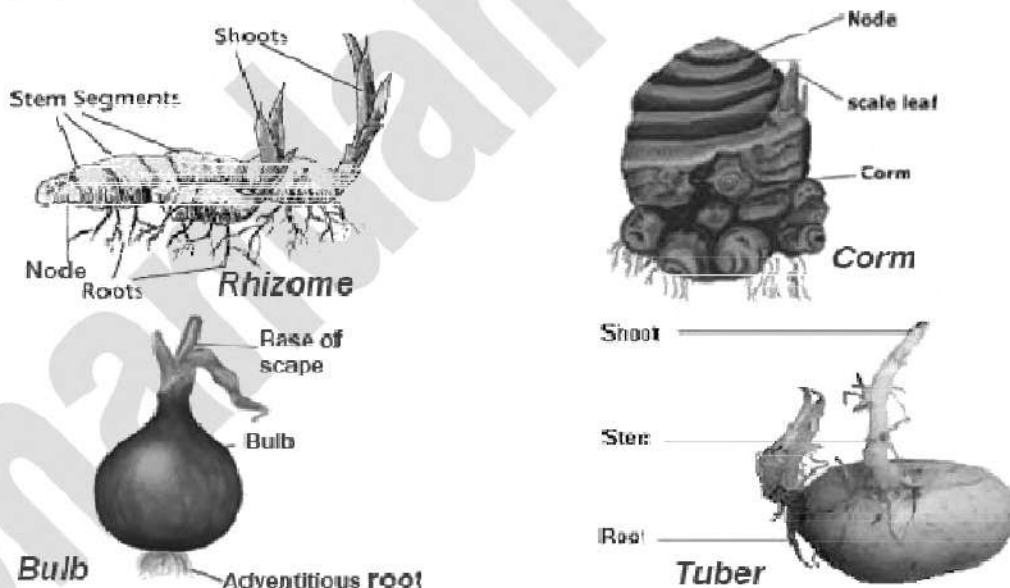
(2) Modified Adventitious Roots (for storage of food)

Ex: Sweet Potato, Ginger



Modified tap roots : A. Fusiform in *Raphanus sativus*. B. Conical in *Daucus carota*. C. Napiform in *Beta vulgaris*. D. Tuberous root of *Mirabilis jalapa*. and E. Nodulated root of *Pisum sativum*

(2) Stems



- A stem is one of two main structural axes of a vascular plant.
- The stem is normally divided into nodes and internodes.
- The nodes hold buds which grow into one or more leaves, conifer cones, roots, other stems, or flowers (inflorescence).
- The internodes distance one node from another. The term shoots is often confused with stems; shoots generally refer to new fresh plant growth and does include stems but also to other structures like leaves or flowers.
- The other main structural axis of plants is the root. In most plants stems are located above the soil surface but some plants have underground stems. A stem develops buds and shoots and usually grows above the ground. Inside the stem, materials move up and down the tissues of the transport system.



Stems have four main functions which are:

- Support for and the elevation of leaves, flowers and fruits. The stems keep the leaves in the light and provide a place for the plant to keep its flowers and fruits.
- Transport of fluids between the roots and the shoots in the xylem and phloem.
- Storage of nutrients.
- The production of new living tissue. The normal life span of plant cells is one to three years. Stems have cells called meristems that annually generate new living tissue.

Underground Stem Modifications

Rhizome

- A rhizome is a thick horizontally growing stem which usually stores food material.
- It has nodes and internodes, scale leaves, axillary buds, adventitious roots and a terminal bud.
- Scale leaves enclosing the axillary buds are seen arising from the nodal points of the stem.
- Some of the axillary buds develop into branches which grow upwards into the air and then produce normal green foliage leaves.
- Usually the growing points of the rhizome continue to remain underground causing an elongation of the rhizome.
- Roots develop from the lower surface of the rhizome. Eg. Ginger, Turmeric.

Tuber

- Tuber is a swollen end of an underground branch which arises from the axil of a lower leaf.
- These underground branches grow horizontally outwards in the soil. Each tuber is irregular in shape due to the deposition of food materials (starch).
- On the surface of each tuber many leaf scars are seen.
- These leaf scars are the impressions of fallen scale leaves.
- Each such leaf scar encloses an axillary bud.
- A leaf scar with an axillary bud is called an eye.
- These eyes of potato are capable of producing new plants by vegetative propagation. E.g., Potato.

Bulb

Here, the stem is reduced and represented by a short disc. The lower surface of the stem produces many adventitious roots. E.g., Onion, Garlic.

- In bulbs of onion, garlic, etc. the inner leaves are fleshy while the outer ones are dry.

- This is called as tunicated bulb since the concentric leaf bases form a complete covering or tunic.
- The apical bud of the bulb produces the shoot.
- The axillary buds sometimes produce daughter bulbs, as in garlic.

Corm

- A corm is a greatly swollen underground basal portion of an erect stem.
- The swelling is due to the storage of reserve food material.
- It bears scale leaves and axillary buds. At the end of the growing season, the aerial parts die.
- With the return of favorable conditions usually one axillary bud (rarely more than one) near the apex develops into a new shoot utilising the food reserve material in the old corm.
- The new plant produces a new corm at its base.
- The earlier corm shrivels off. E.g. Amorphophallus, Colocasia.

Leaf

- The leaves perform three main functions such as manufacture of food, interchange of gases between the atmosphere and the plant body and evaporation of water.
- Some of the functions are described below.
- The functions of modified leaves are not dealt as they carry on specialised functions as discussed earlier.

A. Manufacture of Food

- It is the primary function of green leaves.
- The process is called photosynthesis by which leaf manufactures food material in the presence of sunlight and green pigment-chlorophyll present in the leaf.
- The preparation of food material takes place out of water and carbon dioxide obtained from the soil and the air respectively.

B. Interchange of Gases

- It takes place both for respiration and photosynthesis.
- There are numerous, minute openings present on the leaves, called stomata, through which gaseous exchange takes place between the atmosphere and the plant body.

C. Evaporation of Water

- Water absorbed by the root hairs of the plant is evaporated from the leaf surface during the daytime.

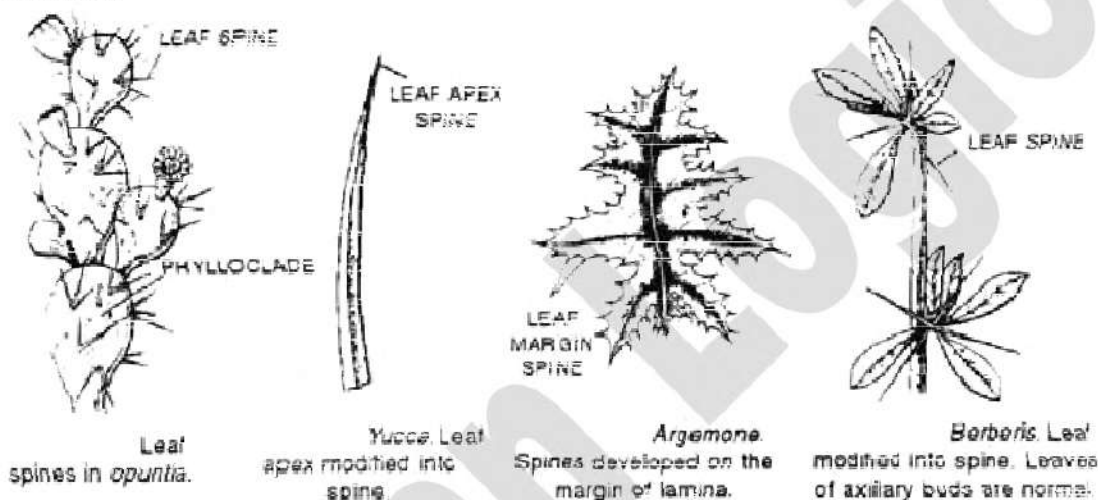


- It generally takes place through the stomatal openings.
- The phenomenon is known as transpiration.
- This process allows the plant to absorb water and minerals from the soil and conduct the same to the top of the plant.

D. Storage of Food

- Fleshy leaves of Indian aloe, Portulaca and fleshy scale leaves of onion store food materials and water for the future of the plants.

Leaf Modifications



Tendrils

- Tendrils are slender, spirally coiled spring like structures.
- They are highly sensitive to contact and when they come in contact with any support, tendrils coil around the support like the stem twiners.
- In glory lily (*Gloriosa superba*) the leaf apex is modified into a tendril.
- In pea (*Pisum sativum*) the terminal leaflets of an unipinnately compound leaf are modified into tendrils.
- In Lathyrus or wild pea, the entire leaf is modified into a tendril. In Clematis and Smilax, the petiole and stipules respectively, are modified into tendrils.

Phyllode

- A phyllode is the petiole or rachis of a leaf which is modified into a green flat structure for the purpose of photosynthesis.
- In such a leaf the lamina is poorly developed.
- In *Acacia melanoxylon*, the petiole is flattened, green and becomes a phyllode.
- The leaflets and secondary rachis drop off.

- Fleshy and succulent leaves of xerophytic plants growing in desert area always store a huge quantity of water, mucilage and food materials.

E. Vegetative propagation

- Leaves of Bryophyllum, Begonia and Kalanchoe produce buds by means of which they give rise to new plants by vegetative propagation.
- In these cases when leaf lamina touches the ground, the leaf margin at point produces roots and form a bud that grows into a new plant.

- In *Parkinsonia aculeata*, the secondary rachis are modified into phylodes which are photosynthetic.
- The primary rachis is modified into a spine.

Spines

- In some plants, leaves or parts of leaves may be modified into spines.
- In *Opuntia* (prickly pear) leaves are poorly developed and fall off very early, but the minute leaves of the axillary bud are modified into spines. In *Argemone* (prickly poppy), the leaf margin is modified into small spines. In *Zizyphus* the stipules are modified into spines. The spines act as defensive structures.

Scale Leaves

In many desert plants, the leaves are highly reduced and appear as scales. The scale leaves are thin, membranous, dry, stalkless and brownish or colourless. In plants where the leaves are reduced to scales in order to minimise transpiration, the function of "photosynthesis is relegated to the stems (cladodes).

Flowers

- Flowers is the metamorphosed part of the shoot meant for the reproduction of plant.



- There are four whorls in a flower

Accessory whorl:

- (i) Calyx
- (ii) Corolla Essential whorl
- (iii) Androecium
- (iv) Gynoecium

Pollination

Pollination is the process by which pollen is transferred in the reproduction of plants, thereby enabling fertilization and sexual reproduction.

In spite of a common perception that pollen grains are gametes, like the sperm cells of animals, this is incorrect; pollination is a phase in the alternation of generations. Each pollen grain is a male haploid plant, a gametophyte, adapted to being transported to the female gametophyte, where it can achieve fertilization by producing the male gamete (or gametes, in the process of double fertilization).

Abiotic

Abiotic pollination refers to situations where pollination is mediated without the involvement of other organisms. Only 10% of flowering plants are pollinated without animal assistance. The most common form of abiotic pollination, anemophily, is pollination by wind. This form of pollination is predominant in grasses, most conifers, and many deciduous trees. Hydrophily is pollination by water, and occurs in aquatic plants which release their pollen directly into the surrounding water. About 80% of all plant pollination is biotic. In gymnosperms, biotic pollination is generally incidental when it occurs, though some gymnosperms and their pollinators are mutually adapted for pollination. The best-known examples probably are members of the order Cycadales and associated species of beetles. Most conifers are anemophilous; they depend on wind pollination. Of the abiotically pollinated species, 98% are anemophilous and 2% hydrophilous, being pollinated by water.

Biotic

More commonly, the process of pollination requires pollinators: organisms that carry or move the pollen grains from the anther to the receptive part of the carpel or pistil. This is biotic pollination. The various flower traits (and combinations thereof) that differentially attract one type of pollinator or another are known as pollination syndromes. Roughly 200,000 varieties of animal pollinators are in the wild, most of which are insects.

Flower- Fertilization

When ripe pollen from another of the same kind of flower catches on the stigma, each pollen grain sends out a tiny threadlike tube. The tube grows down through the style and pierces one of the ovules in the ovary. This process is called fertilization.

Flower Inflorescence:

An inflorescence is a group or cluster of flowers arranged on a stem that is composed of a main branch or a complicated arrangement of branches. Morphologically, it is the part of the shoot of seed plants where flowers are formed and which is accordingly modified.

Fruits

A fruit is a part of a flowering plant that derives from specific tissues of the flower, one or more ovaries, and in some cases accessory tissues. Fruits are the means by which these plants disseminate seeds. Many of them that bear edible fruits, in particular, have propagated with the movements of humans and animals in a symbiotic relationship as a means for seed dispersal and nutrition, respectively; in fact, humans and many animals have become dependent on fruits as a source of food. Fruits account for a substantial fraction of the world's agricultural output, and some (such as the apple and the pomegranate) have acquired extensive cultural and symbolic meanings.

False Fruits

An **accessory fruit** (sometimes called false fruit, spurious fruit, pseudofruit, or pseudocarp) is a fruit in which some of the flesh is derived not from the ovary but from some adjacent tissue exterior to the carpel. Examples of accessory tissue are the receptacle of strawberries, figs, or mulberries, and the calyx of *Gaultheria procumbens* or *Syzygium jambos*. Pomes, such as apples and pears, are also accessory fruits, with much of the fruit flesh derived from a hypanthium.

True Fruits

A fruit in which all tissues are derived from a ripened ovary and its contents. See more at fruit.

Compare accessory fruit.

Ex: Tomato, Guava, Brinjal, Papaya, Chillis, Banana, Coconut, Lady fingers, cucumber, Mango.

Plant Photosynthesis

A primary difference between plants and animals is the plant's ability to manufacture its own food. In photosynthesis, carbon dioxide from the air and water from the soil react with the sun's energy to form photosynthates (sugars, starches, carbohydrates, and proteins) and release oxygen as a byproduct.



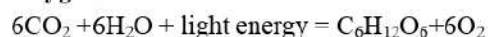


In photosynthesis, the plant uses water and nutrients from the soil, and carbon dioxide from the air with the sun's energy to create photosynthates. Oxygen is released as a byproduct.

Photosynthesis literally means to put together with light. It occurs only in the **chloroplasts**, tiny sub-cellular structures contained in the cells of leaves and green stems. A simple chemical equation for photosynthesis is given

Photosynthesis:

Carbon dioxide + water + light energy = glucose + Oxygen



Simple chemical equation for photosynthesis.

This process is directly dependent on the supply of water, light, and carbon dioxide. Limiting any one of the factors on the left side of the equation (carbon dioxide, water, or light) can limit photosynthesis regardless of the availability of the other factors. An implication of drought or severe restrictions on landscape irrigation is a reduction in photosynthesis and thus a decrease in plant vigor and growth.

In a tightly closed greenhouse there can be very little fresh air infiltration and carbon dioxide levels can become limiting, thus limiting plant growth. In the winter, many large commercial greenhouses provide supplemental carbon dioxide to stimulate plant growth.

The rate of photosynthesis is somewhat temperature dependent. For example, with tomatoes, when temperatures rise above 96°F the rate of food used by respiration rises above the rate of which food is manufactured by photosynthesis. Plant growth comes to a stop and produce loses its sweetness. Most other plants are similar.

TRANSPIRATION

Water in the roots is pulled through the plant by transpiration (loss of water vapor through the stomata of the leaves). Transpiration uses about 90% of the water that enters the plant. The other 10% is an ingredient in photosynthesis and cell growth.

Transpiration serves three essential roles:

- **Movement of minerals up from the root (in the xylem) and sugars (products of photosynthesis) throughout the plant (in the phloem).** Water serves as both the solvent and the avenue of transport.
- **Cooling** - 80% of the cooling effect of a shade tree is from the evaporative cooling effects of transpiration. This benefits both plants and humans.

- **Turgor pressure** - Water maintains the turgor pressure in cells much like air inflates a balloon, giving the non-woody plant parts form. Turgidity is important so the plant can remain stiff and upright and gain a competitive advantage when it comes to light. Turgidity is also important for the functioning of the guard cells, which surround the stomata and regulate water loss and carbon dioxide uptake. Turgidity also is the force that pushes roots through the soil.

OSMOSIS

Osmosis is a process in which a fluid passes through a semipermeable membrane, moving from an area in which a solute such as salt is present in low concentrations to an area in which the solute is present in high concentrations. The end result of osmosis, barring external factors, will be equal amounts of fluid on either side of the barrier, creating a state which is known as "isotonic." The fluid most commonly used in demonstrations of osmosis is water, and osmosis with a wide variety of fluid solutions is key for every living organism on Earth, from humans to plants.

DIFFUSION

Diffusion is a spontaneous movement of particles from an area of high concentration to an area of low concentration. (Ex: tea flavoring moving from an area of high to low concentration in hot water.)

PHYTO HORMONES

Plant hormones (also known as **phytohormones**) are chemicals that regulate plant growth, which, in the UK, are termed 'plant growth substances'.

Plant hormones are signal molecules produced within the plant, and occur in extremely low concentrations. Hormones regulate cellular processes in targeted cells locally and, when moved to other locations, in other locations of the plant. Hormones also determine the formation of flowers, stems, leaves, the shedding of leaves, and the development and ripening of fruit.

Abscisic Acid

Abscisic acid (also called ABA) is one of the most important plant growth regulator.

It accumulates within seeds during fruit maturation, preventing seed germination within the fruit, or seed germination before winter. Abscisic acid's effects are degraded within plant tissues during cold temperatures or by its removal by water washing in out of the tissues, releasing the seeds and buds from dormancy.



Auxins

Auxins are compounds that positively influence cell enlargement, bud formation and root initiation. They also promote the production of other hormones and in conjunction with cytokinins, they control the growth of stems, roots, and fruits, and convert stems into flowers.

Cytokinins

Cytokinins or CKs are a group of chemicals that influence cell division and shoot formation. They were called kinins in the past when the first cytokinins were isolated from yeast cells. They also help delay senescence or the aging of tissues, are responsible for mediating auxin transport throughout the plant, and affect internodal length and leaf growth.

Gibberellin

Gibberellins, or GAs, include a large range of chemicals that are produced naturally within plants and by fungi.

Ethylene

It is only gaseous hormone which is used for fruit ripening.

Ethylene is a gas that forms through the breakdown of methionine, which is in all cells. Ethylene has very limited solubility in water and does not accumulate within the cell but diffuses out of the cell and escapes out of the plant. Its effectiveness as a plant hormone is dependent on its rate of production versus its rate of escaping into the atmosphere. Ethylene is produced at a faster rate in rapidly growing and dividing cells, especially in darkness. New growth and newly germinated seedlings produce more ethylene than can escape the plant, which leads to elevated amounts of ethylene, inhibiting leaf expansion.

MEDICINAL PLANTS

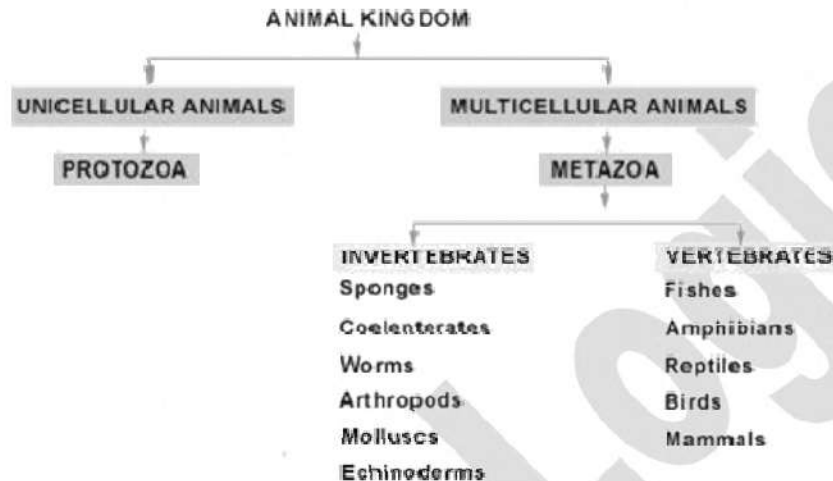
Name of the plant	Part	Alkaloid	Medicinal Use
Amla	Fruit	Embllica officinalis	Vitamin – C, Cough, Diabetes, cold, Laxativ, hyper acidity
Neem (It is called Indian Herbal doctor)	Rhizome	Nimbin	Sdedative, analgesic, epilepsy, hypertensive
Opium	Latex	Morphine	Anaesthesia, Pain killer, Brain stimulates
Cinchona Aficinalis	Bark	Quinine	Anti-malarial drug
Cocoa	Leaves	Cocaine	Heart beat rate, B.P controller
Fox Glove	Leaves	Digitalis	Mycardial stimulant, improve blood supply to kidneys
Rauwofia Serpentina	Root	Respirin	Treatment of epilepsy, hypertension, insanity, snake-bites.2
Belladonna	Leaves and fruits	Atropine	Athama and whooping cough, Encephalitis
Nicotiyana, Tobaccomm	Leaves	Nicotin	Brain stimulator, Muscle stimulator
Coffeia, Arabica	Seeds	Caffein	Used in soft drinks



CHAPTER -13

ANIMAL KINGDOM

Animal Kingdom

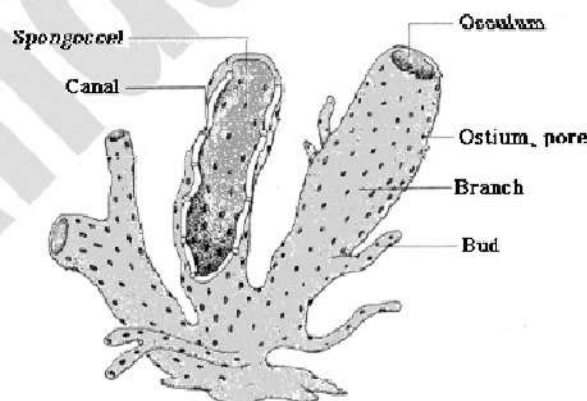


Protozoa (Proto = first, Zoa = animals)

They are microscopic organisms in which a single cell performs all the vital activities.

- Many of which live as parasites on plants and animals.
- They are aquatic mostly.
- Different types of locomotory organs are found in protozoans locomotory organs are absent in parasitic forms of Protozoans. Some Protozoans are holophytic i.e they prepare their own food by photosynthesis (e.g Euglena)
- The parasitic protozoans feed on materials obtained from the hosts. (e.g. Monocystis).

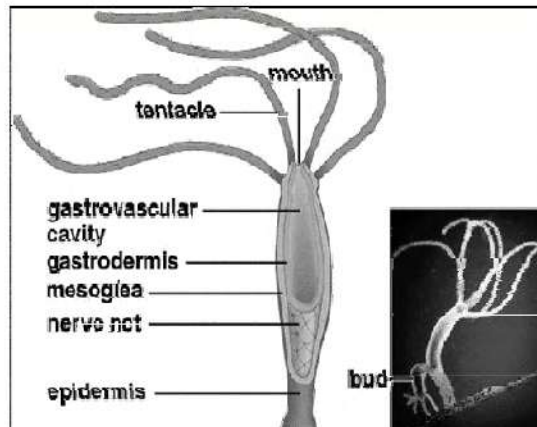
Porifera (Pore = Holes fera = have)



- Members of this phylum are commonly known as sponges.
- These are primitive multi-cellular animals and have cellular level of organization.
- Sponges have a water transport or canal system.
- Sponges reproduce asexually by fragmentation. Sponges sexes are not separate (hermaphrodite). That means eggs and sperms are produced by the same individual.
- Most of them are marine and remain attached to rocks. A few live in fresh water. Ex: Sycon (Scypha), SpongiUa (Fresh water sponge), Euspongia (Bath sponge).

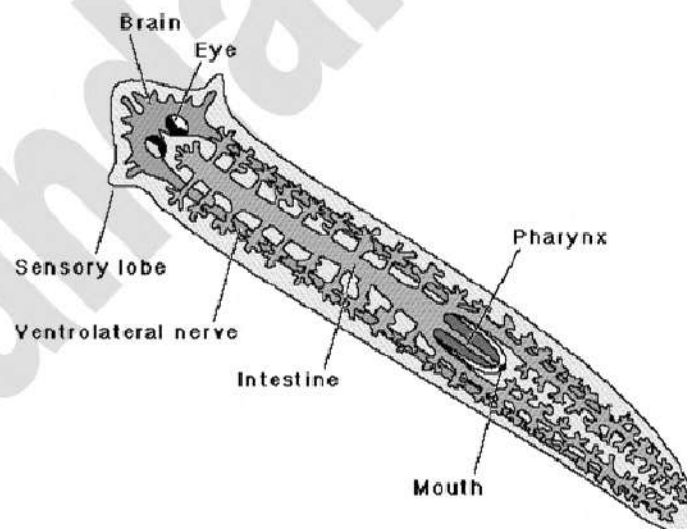


Coelenterata (Cnidaria)



- They are aquatic mostly marine, sessile or free-swimming, radially symmetrical animals.
- The name cnidaria is derived from the cnidoblasts or cnidocytes (which contain the stinging capsules or nematocytes) present on the ectoderm of tentacles and the body of the carnivorous animals.
- Some of the cnidarians e.g. corals have a skeleton composed of calcium carbonate.
- Cnidarians exhibit two basic body forms called polyp and medusa.
- **Polyp** is sessile and cylindrical form like Hydra, Adamsia etc.
- **Laksha dives, Andaman Nicobar Islands** are formed by coral, polyps
- World largest coral polyp island great barrier (Near Australia)
- Examples: Hydra, Obelia, Porpita, Vellela, Physalia (Portuguese man of war), Aurelia (Jellyfish), Adamsia (Sea anemone), Pennatula (Sea-pen) and Gorgonia (Sea-fan) Meandrina (Brain Coaral).

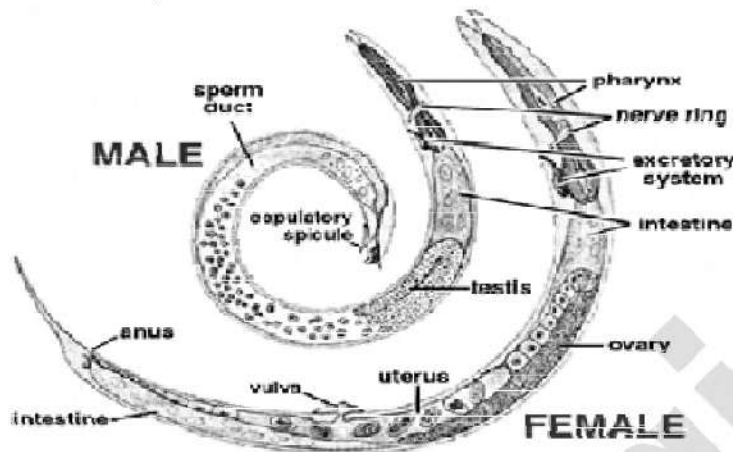
Platyhelminthes (Flatworms)



- They have dorso-ventrally flattened body. That's why, they are called as flatworms.
- These are mostly endoparasites found in animals including human beings.
- Platyhelminthes are triploblastic and unsegmented animals exhibiting bilateral symmetry.
- Platyhelminthes are hermaphrodites are bisexual
- Some of them absorb nutrients from the host directly through their body surface.
- **Ex: Taenia (Tapeworm), Fasciola (Liver Fluke), Schistosoma (Blood fluke) etc.**



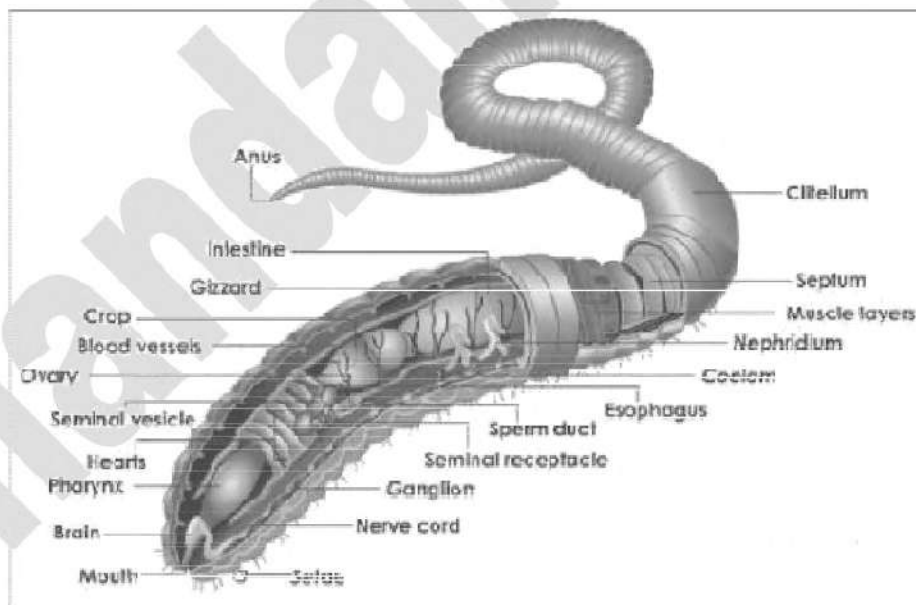
Nematehelmenthes (Round Worms)



The bodies of Aschelminthes is circular in cross-section, hence, they are called as roundworm or nematodes. They are bilaterally symmetrical, triploblastic, pseudocoelomate animals

- They may be free living, aquatic and terrestrial or parasitic in plants and animals.
- They have organ system level of body organization.
- Alimentary canal is complete with a well developed muscular pharynx.
- Sexes are separate (dioecious) i.e males and females are distinct.
- They have a tube within a tube body plan with mouth, pharinges, intestine and anus.
- **Ex:** Ascaris (round worm), Wuchereria (Filaria worm), Ancylostoma (Hook worm), Enterobius (Pin worm) and Rhaditis.

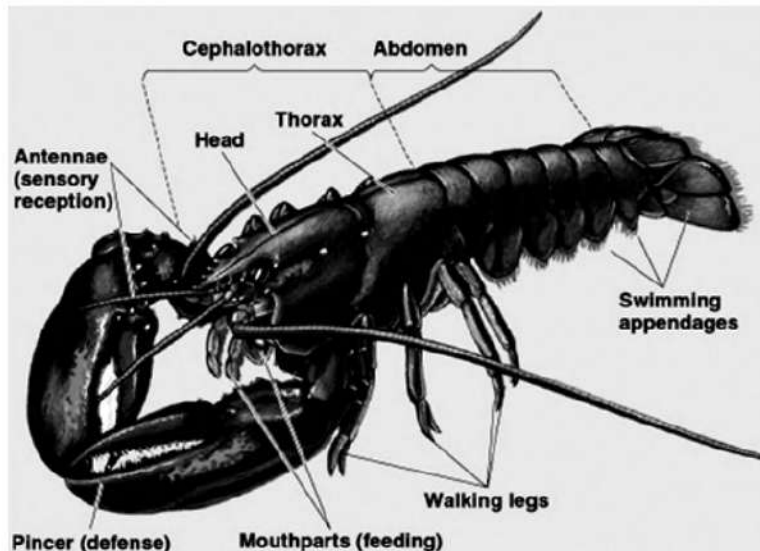
Annelida



- They are aquatic and terrestrial free living and sometimes parasitic.
- Their body surface is distinctly marked out into segments or metameres (little rings), hence named as Annelida. (i.e. metamerically segmented)
- They possess longitudinal and circular muscles which help in locomotion. Posses true coelom.
- **Ex:** Nereis, Aphrodite (sea mouse), Pheretima (Earthworm), Tubifex, Hirudinaria (Leech), Chaetopterus, Erebella, and Bonnellia.

**Earth Worm:**

- Cultivation of earth worm is called **vermiculture**.
- A fertilizer is made by earth worms is called **vermicompost**.
- Earth worm is called Natural tractor and friend of the farmer.

Arthropoda (Jointed Legs)

- This is largest phylum of animalia which includes insects. Over 2/3 (900,000 species) of all named species on the earth are arthropods.
- Arthropods have a segmented body covered by chitinous exoskeleton. Each segment consists of a pair jointed appendages.
- Body consists of head, thorax and abdomen.
- Respiratory organs are gills, book gills, book lungs or tracheal system.
- Excretion takes place through malpighian tubules.
- They are unisexual.
- **Examples:** Araneus (garden spider), Limulus (king crab), Buthus (scorpion), Eupagurus (hermit crab), Cancer (common crab), Macrobrachium (prawn), Lepisma (silverfish), Periplaneta (cockroach), Apis (bee), Daphnia (waterflea), Anopheles (mosquito), Musca (housefly), Leptocorisa (paddy pest: gandhi poka), Triops (tadpole fish), Cyclops, Squilla, Astacus (crayfish), Lepas and Balanus (Barnacle), Silk worm.

Silk Worm:

- It is one type of **Math**.
- Cultivation of silk worm is called **Sericulture**.
- Food for silk worms Mulberry leaves
- Artificial silk is called Rayon.

Honey Bee:

- Cultivation of Honey bee Apiculture.
- These are social animals

Ants:

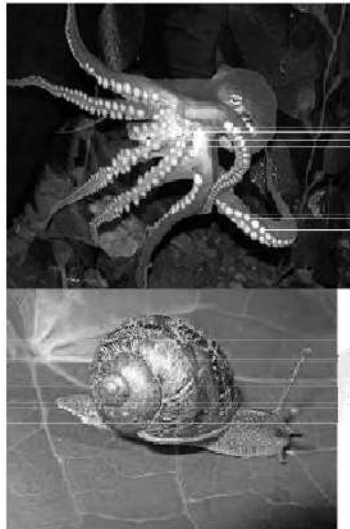
- By Birth deaf animals
- Ants have no ears

Social Animals:

They produce formic acid.



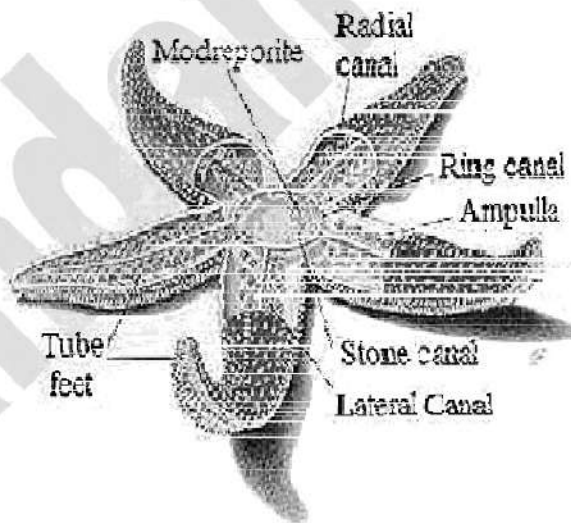
Mollusca (Soft Bodied Animals)



- Mollusca are aquatic & terrestrial inhabitants.
- This is second largest animal phylum.
- They have organ system level of body organization.
- Their body covered by a calcareous shell and unsegmented with a distinct head, muscular foot and visceral hump.

Ex: *Pilla* (apple snail), *Achatina* (land snail), *Lamellidens* (mussel), *Pinctada* (pearl oyster), *Sepia* (cuttlefish), *Loligo* (squid), *Octopus* (devilfish), *Doris* (sea-lemon), *Aplysia* (sea-hare) and *Teredo* (shipworm).

Echinodermate (Spiny Skinned Animals)



- These animals have an endoskeleton of calcareous ossicles and, hence, the name Echinodermate (spiny bodies).
- Most distinctive feature of these animals is the presence water vascular system which helps to capture food and locomotion.
- All are marine with organ system level of organisation.
- All echinoderms are oviparous.

Ex: *Asterias* (star fish), *Echinus* (sea urchin), *Antedon* (sea lilly), *Cucumaria* (sea cucumber), *Ophiura* (Brittle star).

Fish which are not real fish:

(1) Jelly fish, (2) Silver fish, (3) Devil Fish, (4) Cuttle fish, (5) Star fish

**CHORDATE ANIMALS (VERTEBRATES)**

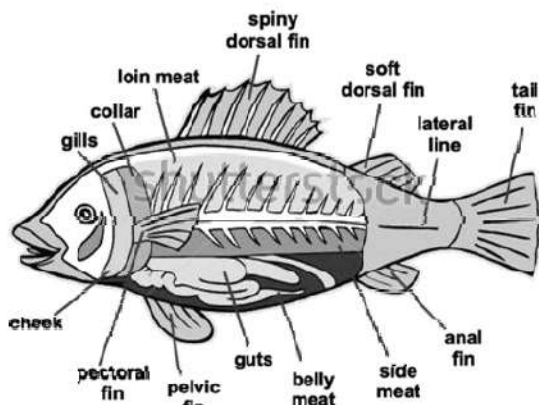
- (1) Fish
- (2) Amphibians
- (3) Reptiles
- (4) Birds
- (5) Mammals

Cold Blood Animals:

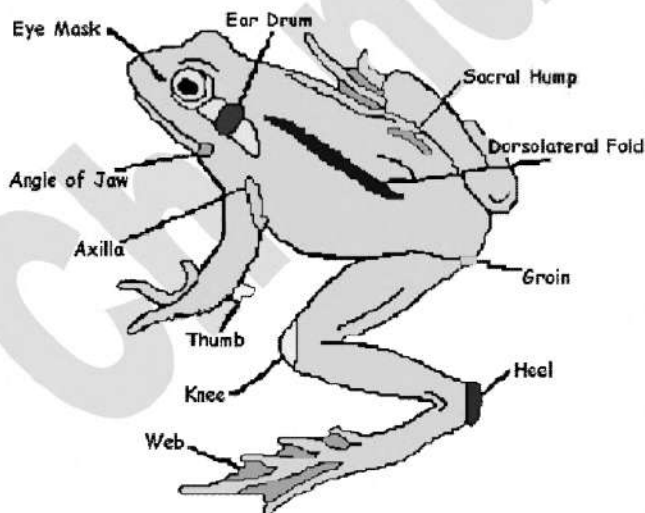
Ex: Fish, amphibians, reptiles

Warm Blood Animals:

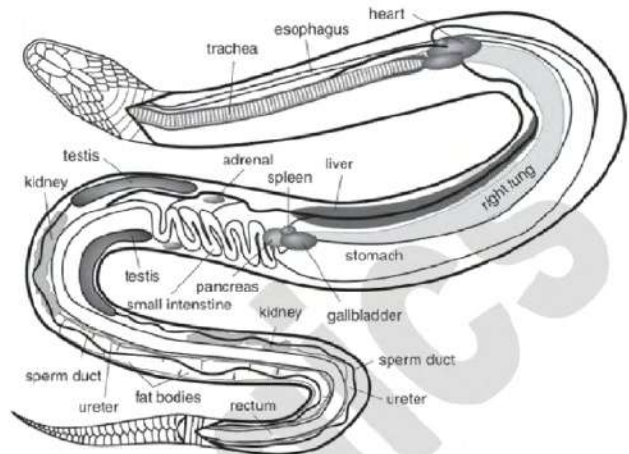
Birds, mammals

Fish:

- Study of fish is called Ichthyology.
- Cultivation of fish is called Pisciculture
- Fins are used for movement in the fish.
- Gills are used for respiratory system in the fish.
- Cat Fish has no scales so that it is dangerous to eat.
- Largest fish blue whale.

Amphibians:**Frog:**

- Flying frog - ranoferous
- Poison frog - blue dart mundukam
- Largest Frog-- Rana goliath

Reptiles:**Snakes:**

- No legs and ears
- Largest snake - American python
- Longest snake - Anaconda
- Longest and largest poison snake - King cobra
- Snake national park in India - Guindy national park (T.N)

Lizards:

- Largest lizard - Commido
- Largest Lizard in India- Veranous
- Poison lizard in India- Gila Monster
- Flying lizard - Draco

Birds:

- Study of birds Ornithology
- Father of Indian Ornithologist - Saleem Ali.
- Largest Bird sanctuary in India - Barthpur or Ghana Bird sanctuary (Rajasthan)
- Highest body temperature animals (40-46°C)
- No urinal bladder.

Flight less birds

- (1) Ostrich (Largest Bird)
- (2) Kiwi (Newzeland national Bird)
- (3) Emu
- (4) Penguin (Home place Antarctica)

Flying Birds:

- 1) Hamming bird (smallest bird)
- 2) Albatross bird (largest wings bird)
- 3) Parrot (It is only bird which moves upper jaw)
- 4) Arctic tern (Longest journey bird 40000 km)
- 5) Godwit (without' breaking journey longest traveling bird)
- 6) Wood coack (Longest journey bird in India)
- 7) Swift (Japan Bird and fastest bird)
- 8) Great Indian bustard (Endangered bird and largest bird in India)

**Mammals:**

- **Blue Whale:** (Largest animal)
- **African elephant:** (Largest land animal)
- **Cheetah** (Fastest running animal)
- **Bat** (flying mammal)

ApesChimpanzee - **Wisest ape**Gorilla - **largest ape**Gibbon - **small ape**Hillock gibbon - **Indian ape**Orangutan - **largest ape in India.**

Animal	Gestation period (days)
Rat	21
Rabbit	30
Cat	60
Dog	60
Sheep	149
Goat	149
Women	270

Animal	Gestation period (days)
Cow	280
Buffalo	280
Horse	336
Elephant	600 (More)
America Opossum	12-13 (less)
Golden Hamster	15-17
Fox	51-63
Coatis	76-78
Leopards	94
Lions	108
Pigs	115
Beavers	122
Baboons	187
Moose	245
Cattle	283
Tapirs	380-400
Giraffes	420-450
Sperm whales	480-590
Rhinos	487



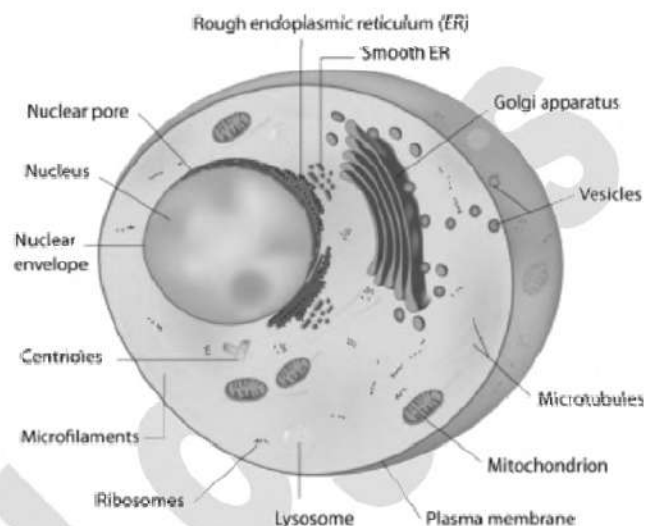


CHAPTER -14

CELL BIOLOGY

- Study of structure and composition of cells is known as **Cytology**.
- All organism in the universe are made up of cell.
- Cell is the basic unit of life. A cell always differs from other cells in shape, size, volume and weight. **Plasma membrane** is the living limit of the cell and the coat present outside it is called the cell wall.
- In plant, the cell-wall is composed of 3 layers - middle lamella, primary wall and the secondary wall. In animal no such cell wall is present.
- Cell is made up of life giving substance, **protoplasm**.
- Cell was discovered by **Robert Hooke** in 1665.
- All living organisms are composed of one or more cells & all cells came from pre-existing cells.
- Cell Theory was propounded for the first time by **M. Schleiden (1838)** and **T. Schwann (1839)** of **Germany**.
- **Rudolf Virchow (1855)** was the first to explain that the cells divide and new cells are formed from the pre-existing cells.
- If the organisms are made up of a single cell, they are called **Unicellular Organisms** (Ex: Amoeba, chlamydomonas, bacteria).
- If organisms are made up of many cells, they are called **multicellular organisms**. The multicellular organisms may be made up of a few cells (Ex: some algal and fungal forms) to several million cells (Ex: human beings, Tree, whale, etc.)
- Each Cell acts as an autonomous unit.
- Cell independently carries out of all fundamental biological processes.
- It oxidizes food molecules to produce energy and utilizes that energy and some nutrient molecules to synthesize complex molecule.
- The general fluid mass of the cell outside the nucleus makes cytoplasm. The cytoplasm contains several organelles.

CELL



Prokaryotic Cell

- Prokaryotic Cell are represented by bacteria, blue-green algae, mycoplasma / pleuro pneumonia-like organism, spirochete and rickettsiae.
- Prokaryotic cells are far smaller than the eukaryotic cells. They multiply more rapidly as compared to most of the eukaryotic cells.
- The Prokaryotic cells are nuclear material i.e. DNA, RNA, Proteins, etc. is not bound by a definite nuclear membrane.
- Prokaryotic cells vary in shape and size.
- The cytoplasm lacks well defined cytoplasmic organelles such as endoplasmic reticulum, mitochondria, centrioles etc.

Eukaryotic Cell

- Organism in whose cell or cells chromosomal genetic material is contained within one or more nuclei and so. Separated from the cytoplasm by two nuclear membranes known as Eukaryotic cell.
- They have a nucleus which contains nuclear material enclosed by a double layered membrane. All plant and animal cells come under this category.
- Components of Eukaryotic cells- Cell wall which is absent in animal cells and some protists, plasma membrane, cytoplasm and organelles.



- Important cytoplasmic organelles are mitochondria, plastids, ribosomes, endoplasmic reticulum, lysosomes, vacuoles, golgi complex, centrioles.

CELL STRUCTURE

Nucleus

- In cells, nucleus lies in the centre.
- It was discovered by Robert Brown in 1831.
- It also be in the periphery of the cell as in the case of plant cells.
- Liquid protoplasm in the nucleus is called nucleoplasm which is bound by a membrane called nuclear membrane.
- Nucleus controls the activities of the cells. It contains a network of fibrous material called chromatin.
- The condensed chromatin is in the form of fibre like structures called chromosomes.
- Main components in Nucleus are

Chromatin and RNA (Ribonucleic acid)

Chromatin

- Chromatin material consists of DNA (deoxy ribonucleic acid). It is in the form of fibre like structures called chromosomes.
- It helps in inheritance or transfer of characters from the parents to next generation.
- It condenses into compact rod-like bodies called chromosomes at the time of cell division.

Ribosomes

- Ribosomes were discovered by G.E. Palade.
- They are small granular bodies found in the wall of ER (Endoplasmic Reticulum) or freely suspended in the cytoplasm.
- Ribosomes are made up of Ribonucleic acid (RNA) and protein.
- Ribosomes are the site of protein Synthesis.

Endoplasmic Reticulum (ER)

- ERs are of two types - **Rough Endoplasmic Reticulum** (with ribosomes attached on its surface for synthesizing proteins) and **Smooth Endoplasmic Reticulum** (without ribosome's for secreting lipid).
- ER gives mechanical support to cytoplasmic matrix,
- ER involve in the vital process of metabolism.
- ER also perform - Synthesis of secretory (serum proteins) lysosomal / membrane proteins on the ER membrane, Synthesis of lipids, Detoxification of drugs and deputed with muscle contraction by release and uptake of Ca^{+2} ions,

Gene

- Genes are in chromosomes. Genes are segments of DNA.
- Gene is a functional unit of chromosome.
- Genes are arranged in a single linear order along the chromosomes,
- A single gene is responsible for one or several cell functions, or a single function may be carried out by a set of genes.

Protoplasm

- It is called as the physical basis of life.
- Protoplasm Theory is given by Schultze.
- 99% of the protoplasm by its weight is made up of four elements – carbon, hydrogen, nitrogen, and Oxygen.
- Between the nucleus and the plasma membrane is known as Cytoplasm.

Lysosomes

Lysosomes are called as digestive bags as they secrete enzymes to digest food substances, hence called digestive bags.

They even digest the entire damaged or dead cell containing them. That's why they are also known as **suicide bags** of the cell.

Golgi apparatus

- Golgi complex was first discovered in the cell of cat. In plants it is known as Dictyosome.
- Its main function the cells is collecting protein secretions from the ER and passing them to outside the cell.
- It also involved in the formation of lysosomes and peroxisomes.

Vacuoles

- There are droplets of substances dissolved in water. Under the microscope, these droplets appear as empty spaces and are termed as vacuoles.
- In **amoeba**, the vacuoles contain food particles and are known as food vacuoles.
- Most of the plant cells have larger vacuoles as compared to animal cells.

Plastids

- They are the characteristic feature of plant cell and found in the plant cells only.
- Plastids almost resemblance to mitochondria and contains DNA.
- Plastids synthesize fats, proteins and starch.
- Plastids contain pigments. On the basis of pigments, they are 3 types- Leucoplast, Chromoplast and Chloroplast.



- Leucoplasts are colourless and occur in sex cells and in storage cells. They form and store starch grains and oil drops.
- Chromoplasts carry pigments that colour the parts of plants.
- Chloroplast is the site of photosynthesis having chlorophyll
- The green substance convert CO_2 and water into sugar in the presence of sunlight.

Mitochondria

- It contains DNA. Mitochondria was discovered by **Altman**.
- Mitochondria are 2 membrane bound structures having rod shaped or spherical organelles.
- It contains enzymes for cellular respiration, through which energy is released. That's why they are also called '**power houses**' of the cell.
- Here, Energy in the food molecules is converted into usable energy and stored in the form of Adenosine triphosphate (ATP) molecules.

CELL DIVISION

The process by which a cell divides to form two daughter cells. Upon completion of the process, each daughter cell contains the same genetic material as the original cell and roughly half of its **cytoplasm**.

Cell Division is divided into 3 types

(1) Binary Fission:

- Binary fission, asexual reproduction by a separation of the body into two new bodies.

- In the process of binary fission, an organism duplicates its genetic material, organism deoxyribonucleic acid (DNA), and then divides into two parts (cytokinesis), with each new organism receiving one copy of DNA.

(2) Mitosis

- It takes place as a result of mitotic division. One cell divides into 2 daughter cells which are quantitatively and qualitatively identical to the mother cell. No change in the Chromosome number.
- In between two successive mitotic divisions there is a rest period called the interphase.
- Mitosis has 4 phases- Prophase, Metaphase, Anaphase & Telophase.

(3) Meiosis

- Also called Reproductive cell division occurs in the germ cells, because it is associated with all sexual reproduction. As a result of division, In the daughter cells chromosomes number become half to that of the mother cell.
- Each division results into 4 daughter cell in contrast to 2 daughter cells in each mitotic divisions.
- In meiosis a cell is permanently transformed.
- Divisions include 2 phases i.e. Reduction division and Equational division (Metosis).

Difference between Prokaryotic cells and Eukaryotic Cells

Prokaryotic cells		Eukaryotic Cells	
1	Has one envelope organization	1	Has two envelope organization
2	Cell wall, if present, having muramic acid	2	Cell not having muramic acid
3	No organized nucleus.	3	Well organized nucleus present
4	Robosomes are of 70 types	4	Ribosomes are of 80 types
5	Lysosomes and microbodies are present	5	Lysosomes and microbodies are absent
6	DNA is naked	6	DNA associated with histone proteins.
7	DNA is circular.	7	DNA is linear, Extranuclear DNA is commonly circular
8	Plasmids are occur	8	Plasmids are absent
9	Cell organelles like endoplasmic reticulum, mitochondria, Golgi apparatus and centrioles are absent	9	All organelles are always present.



Difference between Meiosis and Mitosis

Meiosis		Mitosis	
1	There are two successive cell divisions	1	The cell divides only once
2	Meiosis takes place in the germ cells	2	Mitosis takes place in the somatic cells
3	The chromosomal number remains constant at the diploid to the haploid	3	The chromosomal number is reduced from the end of mitosis
4	The DNA replicates once for two cell divisions	4	DNA replicates once for one cell division
5	There are two cell divisions but the chromosomes also divide only once	5	The cell divides only once and the chromosome divide only once.
6	It requires longer time.	6	Mitosis requires less time.
7	chromosome of daughter cells contains a mixture of maternal and paternal genes.	7	The genetic constitution of daughter(2) cells is identical to that of Parent cell
8	Crossing over of genes takes place.	8	No crossing over of genes
9	It occurs only in sexually reproducing Organisms	9	Occurs in both sexually as well as asexually reproducing organisms

Difference between Plant Cell and Animal Cell

Plant Cell		Animal Cell	
1	All plant cells have a rigid cell wall	1	They do not have cell wall.
2	They lack centrioles.	2	They have Centrioles
3	Photosynthetic pigment chlorophyll is present to prepare their food	3	They lack chloroplast
4	Plastids are absent	4	Plastids are present.
5	Large in size	5	Small in Size
6	Mitochondria are less	6	Generally more
7	Glyoxysomes may be present	7	They are absent
8	Reserved food is generally starch & Fat	8	Reserved food generally glycogen and fat.

CHROMOSOMES

- The rod like bodies found in nuclear region during cell division are called chromosomes.
- Man has 23 pairs of Chromosomes or 2 sets of genomes or 46 chromosomes.
- In Eucaryotic cells 2 types of chromosomes are present. Sex chromosomes which determine sex of the organism and second set of the organism and second one is autosomes or somatic chromosomes.
- Each chromosome consists of 2 Chromatids which remain attached to a point called centromere. Each chromosome basically consists of DNA and basic proteins called histones.
- Chromosomes are the physical carriers of heredity. They bear genes in a linear order.
- Mutations in chromosomes led to variations which in turn lead to evolution of the organisms.



DNA (Deoxy Ribonucleic Acid)

- DNA is mainly found in nucleus and it also occurs in **chloroplast and mitochondria**.
- It is the genetic material and contains all the information needed for the development and existence of an organism.
- It is located by means of Feulgen staining technique which is specific for DNA. **"Watson and Crick"** were able to co-ordinate all the known facts of DNA into a model and proposed a right handed double helix model.
- DNA consists of two poly nucleotide strands which are coiled helically in clock-wise direction. DNA is a linear polymer of **"PURINE"** and **"PYRAMIDINE"** Nucleotides.
- A Nucleotide is nothing but a nucleoside attached to a nitrogen base. A nucleoside is in turn a linkage of sugar and phosphate radical.
- In DNA, the sugar molecule is dextro ribose sugar. It is different from RNA in not having an oxygen molecule (in RNA, the sugar molecule is ribose sugar).

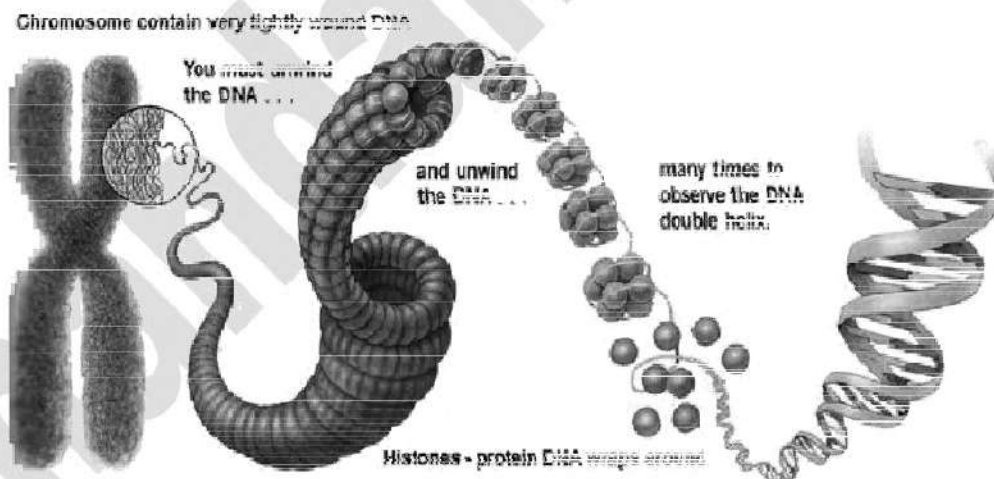
Sugar Molecule + Phosphate Radical → Nucleoside

Nucleoside + Nitrogen base → Nucleotide

- Such nucleosides in large numbers form a poly nucleotide strand. Such two strands which are anti parallel coil together to form DNA.

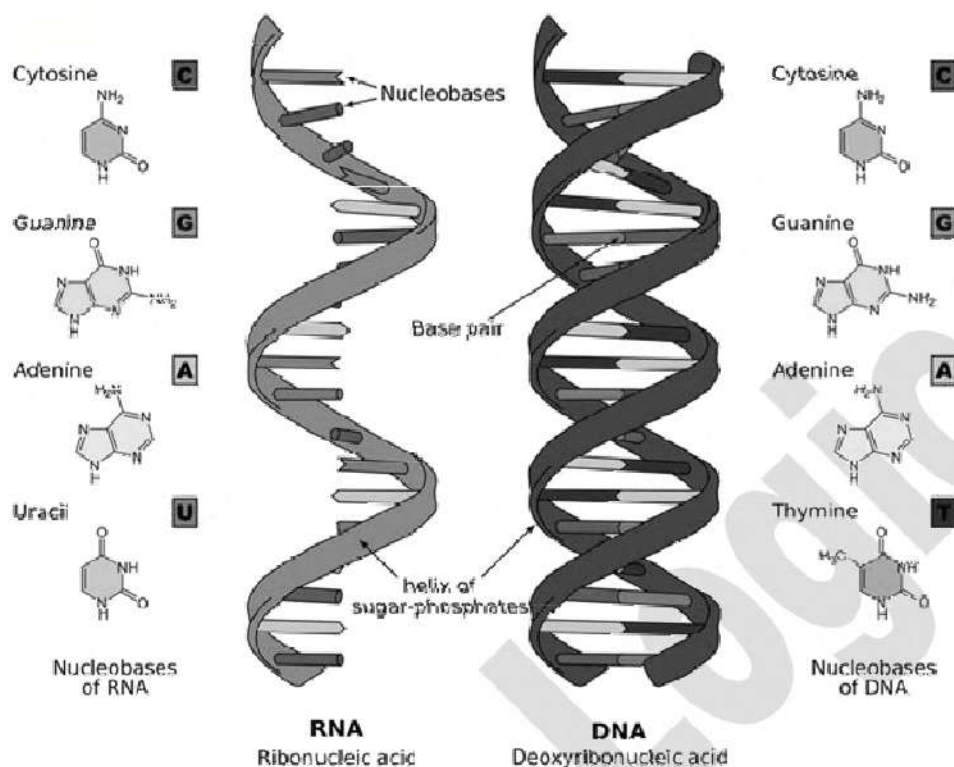
There are 4 nitrogen bases in DNA.

- Adenine (A)
- Thymine (T)
- Guanine (G)
- Cytosine (C)



- Adenine and guanine are purines and Thymine and cytosine are pyrimidines. A purine is always paired with a pyrimidine. That is Adenine always pairs with Thymine and Guanine is always paired with Cytosine. This complementary is known as Base Pairing rule.
- A polynucleotide strand of sugar and phosphate radicals are linked alternately to form external back bones. In between the sugar molecules of two back bones, nitrogen bases are arranged in the form of steps. The nitrogen bases are interconnected by weak hydrogen bonds.
- The DNA molecule is the most unique molecule in the living world. It is the only molecule that can replicate itself.





- It is found in all living cells.
- RNA is synthesized mostly in nucleus but moves out into the cytoplasm.
- RNA is responsible for transmitting the information from the nucleus to the ribosomes where protein - synthesis occurs.
- RNA is single stranded, consists of only one polynucleotide strand.

There are 3 types of RNA

- a) Ribosomal RNA or 'r' RNA found in the Ribosomes where protein synthesis occurs.
- b) Messenger RNA or 'm' RNA which is produced in the nucleus and carries the information for the synthesis of proteins.
For each protein there is specific 'm' RNA.
- c) Transfer RNA or 't' RNA. Its role is to collect amino acids from the cytoplasm for protein synthesis.
 - In general RNA is non-genetic and helps in protein synthesis but in certain viruses the RNA acts as genetic material.
 - RNA contains a single polynucleoside strand where in a linear polymer of ribonucleotides are present.
 - A Nucleotide is nothing but Ribose sugar, Phosphate radical and a nitrogen base.
 - Here also there are 4 nitrogen bases but here Uracil is present instead of Thymine. The rest of the nitrogen bases are same as DNA.